

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	1 of 68	

☒ Confidential

☐ Non-Confidential

☒ 使用說明 ☐ 需求手冊 ☐ 設計文件 ☐ 測試報告

分發：

文件編號

生效日期

3 月 20 日 2026 年

文件名稱： oToCAM_NvidiaDriver_UserGuide.docx

歷史資料

版本	制修訂人/核准人	生效日期	變更項目
0.1	YukiChen	2021/01/11	Initial Version
0.5	PierreHsieh	2021/05/06	Modify for Maxim SerDes.
0.8	GoldenLin	2021/09/06	Add ISP type camera full-size preview command.
0.9	GoldenLin	2022/12/02	Modify for JetPack 5.0 changes.
1.0	GoldenLin	2023/7/31	Add DTS description, Linux Kernel Patch
1.1	GoldenLin	2024/08/30	Add DTS new config: MIPI datarate Add JetPack6.0 support
2.0	JohnTu	2025/06/20	Modify for new camera driver
2.1	JohnTu	2025/06/25	Fine tune string
2.2	JohnTu	2025/06/30	Add for device tree programming
2.3	JohnTu	2025/09/26	Add for device tree programming Dynamic device tree example
2.4	JohnTu	2025/09/26	Fix for otocam driver VER_1.04.04/VER_1.03.03
2.5	JohnTu	2025/11/06	Add for dynamic dts generator tool Add for JP7.0
2.6	JohnTu	2025/11/13	Add oToAdapter for dynamic dts generator tool

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	2 of 68	

			Add for static device tree & tools Add for oToCAM274isp
2.7	JohnTu	2025/11/25	Remove session of "Configure Camera Nodes and Properties for dynamic-overlay device tree creation" Add installation of device tree For driver v1.5.3(JP7.x) v1.4.9(JP6.x)v1.3.8(JP5.x)
2.8	JohnTu	2025/12/10	For driver v2.5.0(JP7.x) v2.4.0(JP6.x)v2.3.0(JP5.x) oToAdapter OrinNX/Nano new HW fix
2.9	JohnTu	2026/1/22	For driver v2.5.2(JP7.x) v2.4.3(JP6.x)v2.3.2(JP5.x) For Frame sync (section 2.1.7 & 2.1.2)
2.10	JohnTu	2026/1/22	For driver v2.6.0(JP7.1) v2.5.3(JP7.0) v2.4.4(JP6.x)v2.3.3(JP5.x) Fix For Frame sync (section 2.1.7 & 2.1.2)
2.11	JohnTu	2026/2/9	Remove JP5.x related Add for JP7.1 add For Frame sync (section 2.1.7 & 2.1.2) add fsync_start tag
2.12	JohnTu	2026/3/2	Add for frame drop and link error check(section 3.3)
2.13	JohnTu	2026/3/17	Add for JP6.2.2
2.14	JohnTu	2026/3/20	Fix section 2.1.4 for JP6.2.2

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	3 of 68	

文件名稱	oToCAM_nVidiaDriver_UserGuide-v2.14.docx
------	--

目錄

1. INTRODUCTION	4
1.1. FEATURES	4
1.2. BLOCK DIAGRAM	5
1.3. DRIVER PACKAGE	8
2. INSTALLATION AND INTEGRATION	10
2.1. INTEGRATE DEVICE-TREE (DTS).....	10
2.2. RECONFIGURE AND PATCH LINUX KERNEL.....	47
2.3. UPDATE DTB.....	47
3. DRIVER TESTING	48
3.1. LOAD DRIVER KERNEL MODULE.....	48
3.2. RUN GSTREAMER TO DISPLAY	48
3.3. RUN V4L2-UTILS OR OTHER TOOL	49
4. FEEDBACK.....	53
5. REFERENCE	54
5.1. NVIDIA DEVELOPMENT DOCUMENTS.....	54
5.2. NVIDIA CARRIER BOARD SPECIFICATION.....	54
5.3. NVIDIA XAVIER PINMUX TABLE	54
APPENDIX A. DEVICE TREE PROGRAMING	55

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	4 of 68	

1. Introduction

oToCAM Nvidia Driver package supports oToBrite's oToCAM series cameras on Jetson Linux platform. This User Guide documents the package contents, and driver installation/integration/testing procedures.

1.1. Features

Catalog	Feature
Platform	➤ Supported Jetson DevKits: AGX-Orin, Orin NX(Nano), AGX-Xavier, Xavier NX ➤ Supported Cameras: ar0823atisp(oToCAM276isp), imx728isp(oToCAM271isp), imx623isp(oToCAM274isp), imx390isp(oToCAM264isp), imx490isp(oToCAM260isp), imx290isp, isx021/031, ar0144/233, svb56g, and more.
MIPI CSI-2	➤ MIPI CSI-2: 1.5Gbps (1.0~2.5Gbps), x2/4 lanes per port ➤ GMSL SerDes: Maxim MAX9296/96712/96724/96714, MAX9295/96717 ■ Each Dser can connect 1/2/4 cameras. Max. 4 Dsers on a system. ➤ FPD L SerDes: TI UB954/960, UB953/933/913 ■ Each Dser can connect 2/4 cameras. Max. 4 Dsers on a system.
JetPack SDK	➤ JetPack-7.1/L4T-38.4 (oToCAM Driver V2.06.XX) (oToCAM.ko...) ➤ JetPack-7.0/L4T-38.2 (oToCAM Driver V2.05.XX) (oToCAM.ko...) ➤ JetPack-6.2.2/L4T-36.5 (oToCAM Driver V2.14.XX) (oToCAM.ko...) ➤ JetPack-6.2.1/L4T-36.4.4 (oToCAM Driver V2.04.XX) (oToCAM.ko...) ➤ JetPack-6.2/L4T-36.4.3 (oToCAM Driver V2.04.XX) (oToCAM.ko...) ➤ JetPack-5.1/L4T-35.6 (oToCAM Driver V2.03.XX) (oToCAM.ko...) ➤ JetPack-7.0/L4T-38.2 (oToCAM Driver V1.05.XX) (oToCAM.ko...) ➤ JetPack-6.2.1/L4T-36.4.4 (oToCAM Driver V1.04.XX) (oToCAM.ko...) ➤ JetPack-6.2/L4T-36.4.3 (oToCAM Driver V1.04.XX) (oToCAM.ko...) ➤ JetPack-5.1/L4T-35.6 (oToCAM Driver V1.03.XX) (oToCAM.ko...) ➤ JetPack-6.1/L4T-36.4 (oToCAM Driver V0.04.XX) ➤ JetPack-5.1/L4T-35.6 (oToCAM Driver V0.03.XX) ➤ JetPack-4.6/L4T-32.6 (oToCAM Driver V0.02.XX) ➤ JetPack-4.5/L4T-32.5 (oToCAM Driver V0.01.XX)

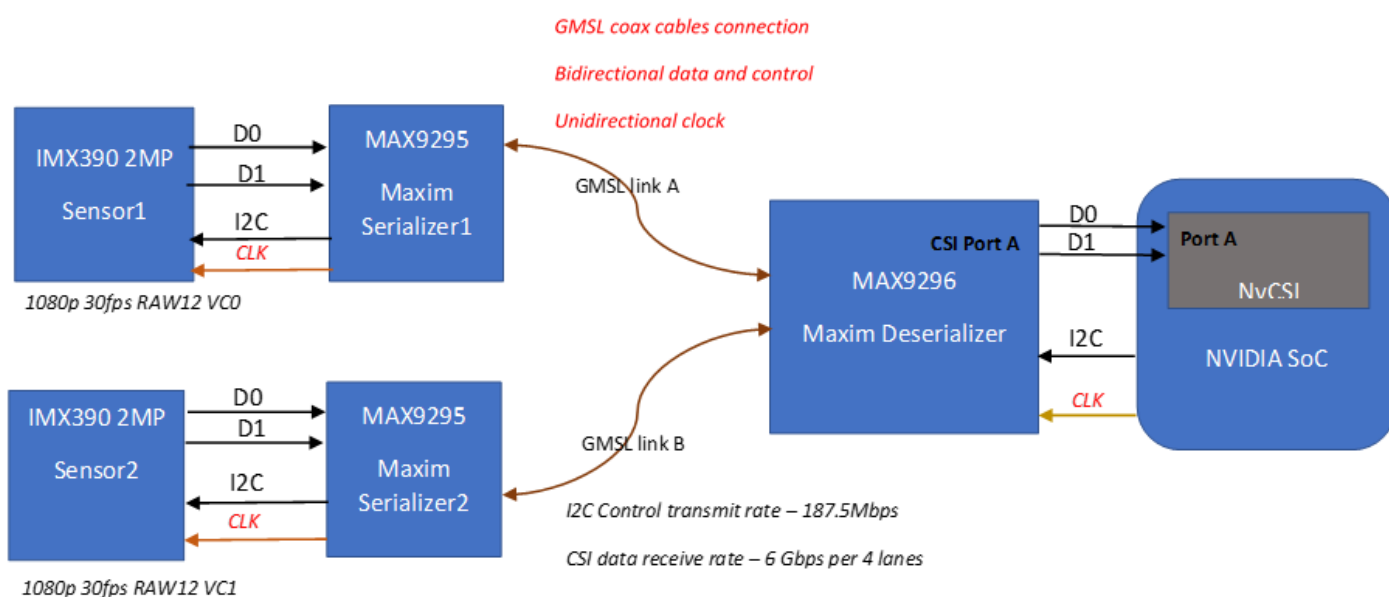
文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	5 of 68	

1.2. Block Diagram

The following diagram shows a GMSL connection with 1:2 Deserializer, which can link two Serializer-Sensor devices. (Ref:

<https://docs.nvidia.com/jetson/archives/r35.4.1/DeveloperGuide/text/SD/CameraDevelopment/JetsonVirtualChannelWithGmslCameraFramework.html>)

Reference GMSL setup with 2x aggregator



The following diagram shows a GMSL connection with 1:4 Deserializer, which can link four Serializer-Sensor devices.

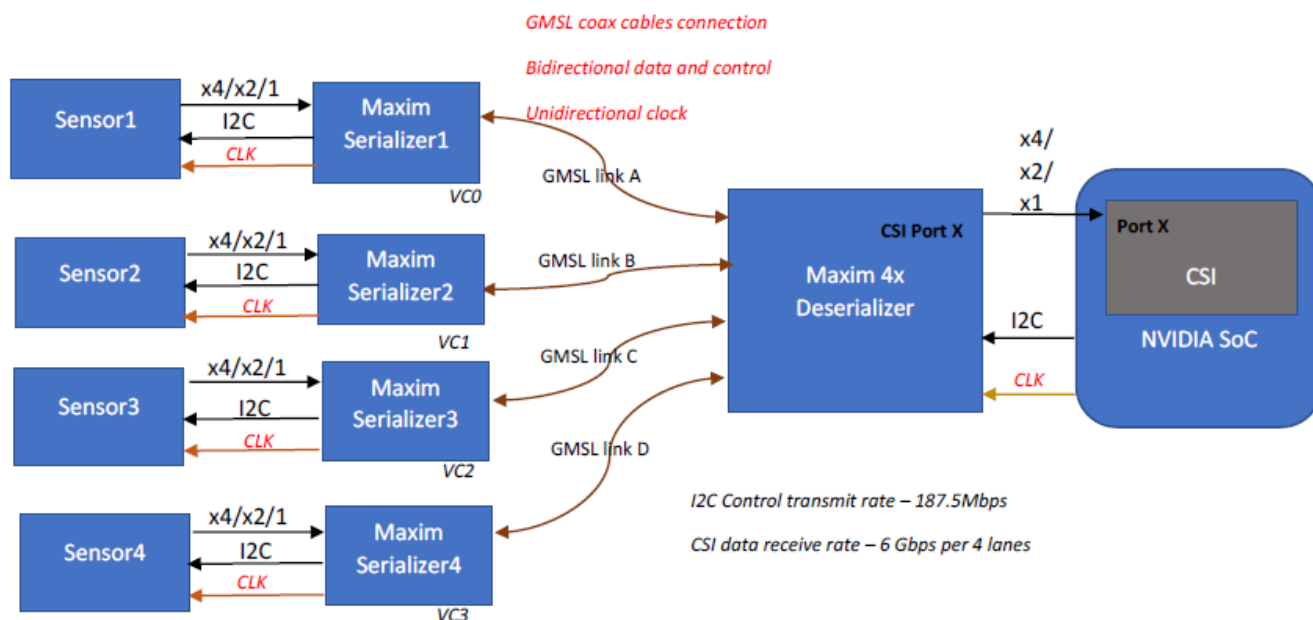


歐特明電子股份有限公司

oToBrite Electronics, Inc.

文件名稱	oToCAM_nVidiaDriver_UserGuide-v2.14.docx	文件編號		版本
		生效日期	2026/3/20	V2.14
		制定部門	System-SW	
		頁數	6 of 68	

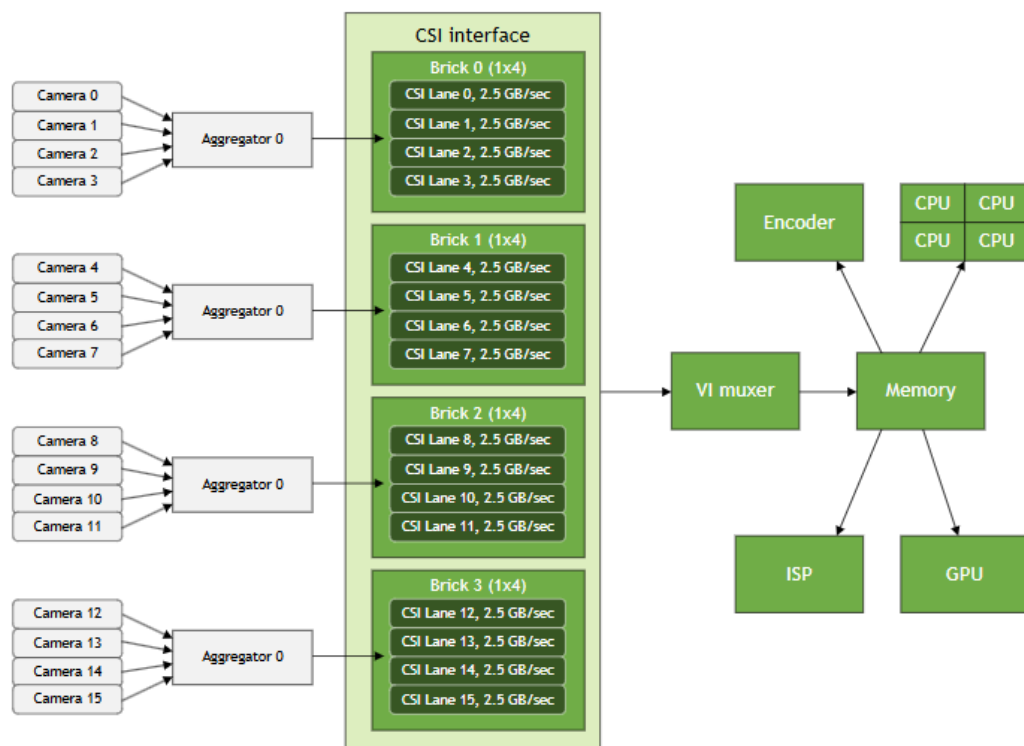
Proposed GMSL setup with 4x aggregator



文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	7 of 68	

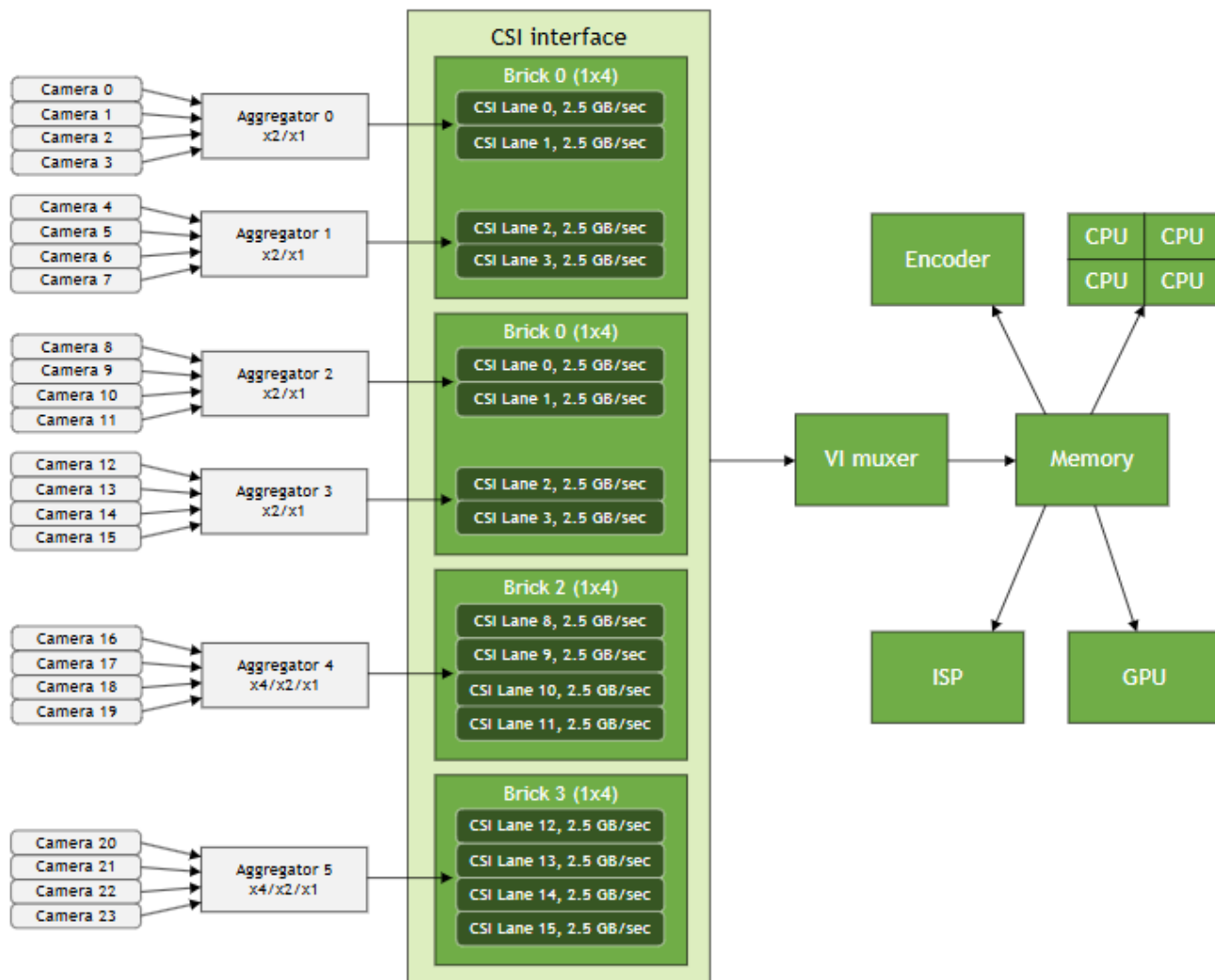
Jetson AGX Orin Series

Jetson AGX Orin series supports a maximum of 16 virtual channels with ISP, or 24 virtual channels without ISP.



The diagram above shows sensor connections to each of the NVIDIA® Orin™ CSI bricks (a total of 4 CSI bricks) in x4 or /x2 or /x1 possible lane configuration.

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	8 of 68	



The diagram above shows four sensors connected to each port of CSI bricks AB and CD in x2 or x1 lane configuration, and four sensors connected to each of the remaining CSI bricks EF and GH in x4, x2, or x1 lane configuration.

1.3. Driver Package

The driver package can be extracted as follows:

```
$ unzip otocam-driver.zip
```

It has two folders inside:

- bin/: contains kernel modules (oToSerDes.ko, oToCAM.ko).
- dts/: contains sample device-tree source files.

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	9 of 68	

Kernel modules are built for specific JetPack/Linux version. Use below command to check module's **vermagic**, and see whether it can match the JetPack/Linux version on your Jetson DevKit.

modinfo oToCAM.ko

```
version:      1.04.02
license:      Dual BSD/GPL
author:       oToBrite Corporation
description:   Media Controller driver for oToCAM
description:   support oToCAM222/oToCAM223/oToCAM264ISP/oToCAM260ISP/oToCAM271ISP/oToCAM272ISP/oToCAM251
srcversion:   5E00BEC365CF244A377DF11
alias:        of:N*T*oToBrite,otocamC*
alias:        of:N*T*oToBrite,otocam
alias:        i2c:otocam
depends:       oToSerDes
name:         oToCAM
vermagic:     5.15.148-tegra SMP preempt mod_unload modversions aarch64
parm:         sensor_model:array of charp
parm:         test_mode:int
parm:         deser_notset_poweroff:int
```

modinfo oToSerDes.ko

```
version:      1.04.02
license:      Dual BSD/GPL
author:       oToBrite Corporation
description:   GMSL/FPD-Link Serializer/Deserializer driver for oToCAM
description:   support max9295/max96717/TI-UB953
description:   support max96724/max96712/max9296/TI-UB960
srcversion:   E23FD517D6575FAF750D862
alias:        i2c:deserializer
alias:        of:N*T*Cti,ub953C*
alias:        of:N*T*Cti,ub953
alias:        of:N*T*oToBrite,serializerC*
alias:        of:N*T*oToBrite,serializer
alias:        of:N*T*Cti,ub960C*
alias:        of:N*T*Cti,ub960
alias:        of:N*T*oToBrite,deserializerC*
alias:        of:N*T*oToBrite,deserializer
alias:        i2c:serializer
depends:
name:         oToSerDes
vermagic:     5.15.148-tegra SMP preempt mod_unload modversions aarch64
parm:         test_mode:int
parm:         clk_en:int
parm:         prdbg:int
```

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	10 of 68	

2. Installation and Integration

2.1. Integrate Device-Tree (DTS)

Driver package provides the following sample DTS files:

- (1) Dynamic (for make device tree include camera module)
- (2) Dynamic by dts generator tool
- (3) Static (for overlay example of Orin-NX evk)

OToBrite Confidential

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	11 of 68	

dynamic
agx
agx-orin
JP6_example
device-tree-out
AGXOrin_EVK
OrinNX_EVK
nx
nx-orin
otocam
readme.txt
t23x
tegra234-camera-config.h
tegra234-p3737-0000+p3701-0004-nv-otocam.dts
tegra234-p3737-camera-imx390-overlay-otocam.dts
tegra234-p3737-camera-modules-otocam.dtsi
tegra234-p3767-camera-imx390-overlay-otocam.dts
tegra234-p3768-0000+p3767-0000-nv-otocam.dts
tegra234-p3768-camera-modules-otocam.dtsi
dynamic_dts_generator
linux 22.04
make_overlay_dts_orin-devkit-r36
windows10
make_overlay_dts_orin-devkit-r36.exe
oToBriteEVK DTS Gen.xlsm
nvthon310.dll
dynamic_overlay
static
connection-example.jpg
JP514-EVK-orin-nx
JP61-EVK-orin-nx
JP62-EVK-orin-agx
JP62-EVK-orin-nx
overlay-note.txt
README.docx
tools
check_IQ
example.png
exp_time_float_to_hex.py
oToCAM-gw5300-IQ.sh
readme.txt
sensor_gain_float_to_hex.py
gst-showall2.sh
gst-showall.sh
insmod-otocam-rename.sh
rmmod-rename.sh
run_replace.sh
set-dtb-to-dts.sh
set-dts-to-dtb.sh
show-debug.sh
show-parse-tree.sh
test-nv16.sh
test-otocam276isp.sh

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	12 of 68	

2.1.1 Configure Camera Nodes and Properties for dynamic device tree creation

(JetPack 6.x)

```
dynamic/
├── t23x/
│   ├── tegra234-camera-config.h
│   ├── tegra234-p3737-0000+p3701-0004-nv-otocam.dts (AGX-Orin)
│   ├── tegra234-p3737-camera-modules-otocam.dtsi (AGX-Orin)
│   ├── tegra234-p3768-0000+p3767-0000-nv-otocam.dts (NX-Orin)
│   └── tegra234-p3768-camera-modules-otocam.dtsi (NX-Orin)
└── otocam/
    ├── tegra234-camera-imx390-i2c.dtsi
    ├── tegra234-camera-imx390-a00.dtsi
    ├── tegra194-camera-all-mode.dtsi
    ├── tegra194-camera-imx390isp-mode.dtsi
    ├── tegra194-camera-imx490isp-mode.dtsi
    ├── tegra194-camera-isx021-mode.dtsi
    ├── tegra194-camera-isx031-mode.dtsi
    └── etc.
```

These files are modified/extended from Jetson Linux SDK below files:

(JetPack 6.x, in **hardware/nvidia/t23x/nv-public/**)

```
nv-platform/
├── tegra234-p3737-0000+p3701-0004-nv.dts
├── tegra234-p3768-0000+p3767-0000-nv.dts
└── ...
overlay/
├── tegra234-p3737-camera-modules.dtsi
├── tegra234-p3737-camera-imx390-overlay.dts
├── tegra234-camera-imx390-a00.dtsi
└── ...
```

To integrate, place Driver package's DTS files into Jetson Linux SDK source tree as below, and modify Makefile accordingly.

(JetPack 6.x)

```
hardware/nvidia/t23x/nv-public/
├── nv-platform/ (<= t23x/*)
├── otocam/*
└── ..
```

2.1.1.1 Configure Camera Nodes and Properties

DTS camera nodes and properties can be configured via the parameters defined in **tegra234-camera-config.h**:

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	13 of 68	

I2C_BUS, I2C_MUX, I2C_MUX_TCA9546	Used in otocam/tegra234-camera-imx390-i2c.dtsi, to configure the path of dser/ser/sensor I2C device nodes
I2C_BUS_CAM0-7	Used in otocam/tegra234-camera-imx390-a00.dtsi, to compose tegra-camera-platform's sysfs-device-tree property
CSI_NUM_CHANS	Maximum number of cameras can be connected, eg. 2 4 8
CSI_NUM_PORTS	Number of (aggregated) ports on CSI Host Controller
CSI_PORT_LANES	Bus-width of each CSI Port, eg. 2 4
CSI_SERDES_TYPE CSI_SERDES, CSI_SERDES_GMSL FPDL	0:m9296, 1:m96712, 2:m96724, 3:m96714 define the maximum number of ser devices that can be linked to a dser, eg. 1 2 4 (Ref: 1.2 Block Diagram)
CAM_MODE	Used in otocam/tegra194-camera-all-mode.dtsi, to select a camera-mode .dtsi file, eg. imx390isp isx021 .. (Ref: 1.1 Features)

Beside modifying tegra234-camera-config.h, check otocam/tegra234-camera-imx390-i2c.dtsi for configured I2C bus/address, dser reset-gpios, etc. If they still can't match your board design, please directly modify the nodes and properties.

NOTE: If user want to assign different camera models for each SerDes port, modify in otocam/tegra234-camera-imx390-a00.dtsi to directly include a specific camera-mode .dtsi (instead of otocam/tegra194-camera-all-mode.dtsi). As below example:

```
&cam_0 {
    // #include "otocam/tegra194-camera-all-mode.dtsi" // -
    #include "otocam/tegra194-camera-imx390isp-mode.dtsi" // +
    mode0 {
        ...
    };
    ...
};
```

NOTE: If user want to provide an external FSYNC signal to fsync-mode enabled cameras, assign Dser's GPIO/MFP pin number via fsync-gpio property in otocam/tegra234-camera-imx390-i2c.dtsi. As below example:

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	14 of 68	

```
dser_a: max9296@48 {
    compatible = "maxim,max9296";
    ...
    fsync-gpio = <0>;    // +
};
```

- For MAX9296, MFP0/7-10 pins are suitable for FSYNC signal input, because they are in GPI mode by default.
- For pulse-fsync enabled cameras (eg. imx, isx series), driver will provide initial FSYNC pulse to start streaming. External FSYNC signal can be provided optionally.
- For trigger-fsync enabled cameras (eg. isx series), frames are output only when External FSYNC signal are provided.

NOTE: Dser's MIPI bus data rate can be configured by property `max_lane_speed`, as below:

```
tegra-camera-platform {
    ...
    max_lane_speed = <1500000>;    // in unit of 1Kbps
    ...
};
```

Camera property `pix_clk_hz` (in camera-mode .dtsi) is calculated based on MIPI bus data rate.

Therefore, it should be updated according to modified MIPI bus speed.

<https://docs.nvidia.com/jetson/archives/r36.4.3/DeveloperGuide/SD/CameraDevelopment/SensorSoftwareDevelopment/riverProgramming.html#sensor-pixel-clock>

2.1.1.2 FPD-Link SerDes Support

TI SerDes devices can be declared in the same way as Maxim SerDes. Below is the comparison between them.

GMSL/Maxim SerDes	FPD-Link/TI SerDes
<pre>dser_a: max9296@48 { compatible = "maxim,max9296"; reg = <0x48>; csi-mode = "2x4"; max-src = <2>; reset-gpios = <&tegra_main_gpio CAM0_RST_L ...>; }; ser_a: max9295_a@40 { compatible = "maxim,max9295"; reg = <0x40>;</pre>	<pre>dser_a: ub960@30 { compatible = "ti,ub960"; reg = <0x30>; csi-mode = "2x4"; max-src = <4>; reset-gpios = <&tegra_main_gpio CAM0_RST_L ...>; }; ser_a: ub953@18 { compatible = "ti,ub953"; reg = <0x18>;</pre>

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	15 of 68	

```
nvidia,gmsl-dser-device = <&dser_a>;
};
```

```
nvidia,gmsl-dser-device = <&dser_a>;
};
```

Camera nodes use same properties to describe SerDes link, regardless of GMSL or FPD-Link. Like below:

tegra234-camera-imx390-i2c.dtsi	tegra234-camera-imx390-a00.dtsi
<pre>cam_0: imx390@10 { reg = <0x10>; ... nvidia,gmsl-ser-device = <&ser_a>; nvidia,gmsl-dser-device = <&dser_a>; };</pre>	<pre>&cam_0 { ... gmsl-link { src-csi-port = "b"; dst-csi-port = "a"; serdes-csi-link = "a"; csi-mode = "1x4"; st-vc = <0>; vc-id = <0>; num-lanes = <2>; }; };</pre>

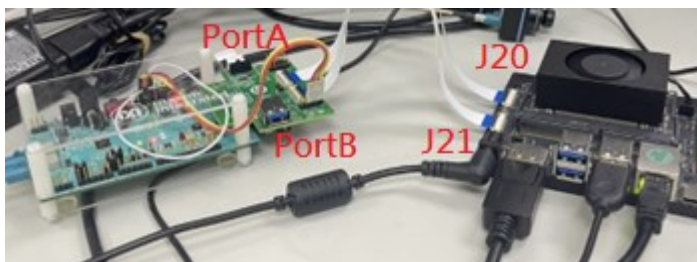
2.1.1.3 JP6.2 dynamic dts example



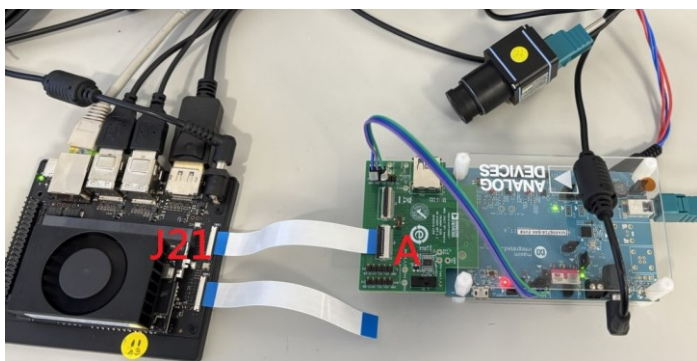
Example :

Orin NX(Nano) :

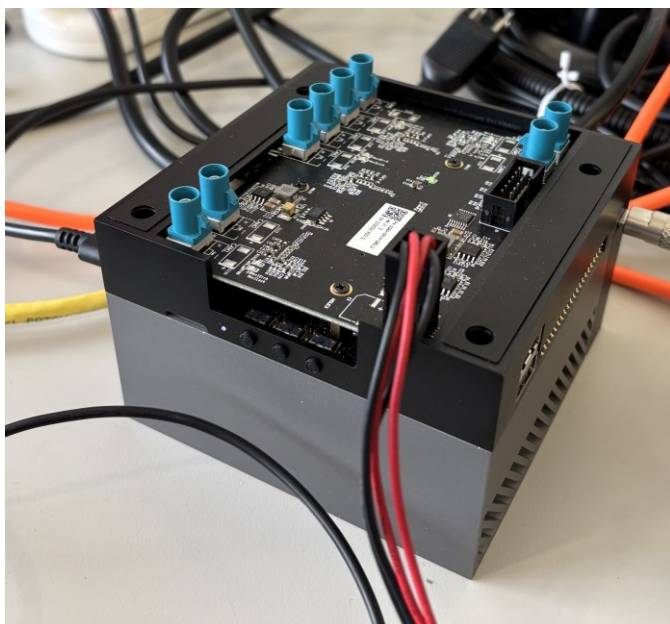
文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	16 of 68	



Orin NX(Nano) A port-> J21 only :



AGX Orin :



文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	17 of 68	

tegra234-camera-config.h
tegra234-camera-config-m9296-2lane_2otocam222.h
tegra234-camera-config-m9296-2lane_8otocam222.h (AGX Orin)
tegra234-camera-config-m9296-4lane_2otocam222.h
tegra234-camera-config-m9296-4lane_8otocam222.h (AGX Orin)
tegra234-camera-config-m96712-2lane_4otocam222.h
tegra234-camera-config-m96712-4lane_4otocam222.h
tegra234-camera-config-m96714-2lane_1otocam222.h
tegra234-camera-config-m96714-4lane_i2cmux_1otocam222.h (A port-> J21)
tegra234-camera-config-m96724-2lane_4otocam222.h
tegra234-camera-config-m96724-4lane_4otocam222.h

oToAdapter for Orin NX/Nano & AGX Orin :

-  tegra234-camera-config-otoadapter-2lane_4otocam222.h
-  tegra234-camera-config-otoadapter-4lane_4otocam222.h
-  tegra234-camera-config-otoadapter-4lane_8otocam222.h

tegra234-camera-config-m9296-2lane_2otocam222.h :

oToBrite Confidential

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	18 of 68	

```
#define I2C_BUS»i2c@3180000// cam_i2c
//#define I2C_BUS»cam_i2cmux
#if 0
#define I2C_MUX»//tca9546@70
//#define I2C_MUX_TCA9546
#define I2C_BUS_CAM0»/bus@0/I2C_BUS/i2c@0
#define I2C_BUS_CAM1»/bus@0/I2C_BUS/i2c@0
#define I2C_BUS_CAM2»/bus@0/I2C_BUS/i2c@1
#define I2C_BUS_CAM3»/bus@0/I2C_BUS/i2c@1
#define I2C_BUS_CAM4»/bus@0/I2C_BUS/i2c@1
#define I2C_BUS_CAM5»/bus@0/I2C_BUS/i2c@1
#define I2C_BUS_CAM6»/bus@0/I2C_BUS/i2c@1
#define I2C_BUS_CAM7»/bus@0/I2C_BUS/i2c@1
#else
#define I2C_BUS_CAM0»/bus@0/I2C_BUS
#define I2C_BUS_CAM1»/bus@0/I2C_BUS
#define I2C_BUS_CAM2»/bus@0/I2C_BUS
#define I2C_BUS_CAM3»/bus@0/I2C_BUS
#define I2C_BUS_CAM4»/bus@0/I2C_BUS
#define I2C_BUS_CAM5»/bus@0/I2C_BUS
#define I2C_BUS_CAM6»/bus@0/I2C_BUS
#define I2C_BUS_CAM7»/bus@0/I2C_BUS
#endif

#define CSI_NUM_CHANS»2 //support max cams
#define CSI_NUM_PORTS»1 //Number of aggregated ports
#define CSI_PORT_LANES»2 //4lane or 2lane
#define CSI_SERDES_TYPE 0 //0:m9296, 1:m96712, 2:m96724
#if (CSI_SERDES_TYPE==0)
#define CSI_SERDES»2 // max cams connected to a dser
#elif (CSI_SERDES_TYPE==1)
#define CSI_SERDES»4 // max cams connected to a dser
#elif (CSI_SERDES_TYPE==2)
#define CSI_SERDES»4 // max cams connected to a dser
#else
#define CSI_SERDES»2 // max cams connected to a dser
#endif
#define CSI_SERDES_GMSL

#define CAM_MODE»isx021 //isx021, isx031, imx390isp, imx490isp, imx728isp, ar0823isp
```

M9296 2lane (2*otocam -> a -> J20) v.s. 4lane (2*otocam -> b -> J21) :

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	19 of 68	

tegra234-camera-config-m9296-2lane_2otocam222.h	tegra234-camera-config-m9296-4lane_2otocam222.h
<pre> 1 #define I2C_BUS i2c@3180000 // cam_i2c 2 //define I2C_BUS cam_i2cmux 3 #if 0 4 #define I2C_MUX //tca9546@70 5 //define I2C_MUX_TCA9546 6 #define I2C_BUS_CAM0 /bus@0/I2C_BUS/i2c@0 7 #define I2C_BUS_CAM1 /bus@0/I2C_BUS/i2c@0 8 #define I2C_BUS_CAM2 /bus@0/I2C_BUS/i2c@1 9 #define I2C_BUS_CAM3 /bus@0/I2C_BUS/i2c@1 10 #define I2C_BUS_CAM4 /bus@0/I2C_BUS/i2c@1 11 #define I2C_BUS_CAM5 /bus@0/I2C_BUS/i2c@1 12 #define I2C_BUS_CAM6 /bus@0/I2C_BUS/i2c@1 13 #define I2C_BUS_CAM7 /bus@0/I2C_BUS/i2c@1 14 #else 15 #define I2C_BUS_CAM0 /bus@0/I2C_BUS 16 #define I2C_BUS_CAM1 /bus@0/I2C_BUS 17 #define I2C_BUS_CAM2 /bus@0/I2C_BUS 18 #define I2C_BUS_CAM3 /bus@0/I2C_BUS 19 #define I2C_BUS_CAM4 /bus@0/I2C_BUS 20 #define I2C_BUS_CAM5 /bus@0/I2C_BUS 21 #define I2C_BUS_CAM6 /bus@0/I2C_BUS 22 #define I2C_BUS_CAM7 /bus@0/I2C_BUS 23 #endif 24 25 #define CSI_NUM_CHANS 2 //support max cams 26 #define CSI_NUM_PORTS 1 //Number of aggregated ports 27 #define CSI_PORT_LANES 2 //4lane or 2lane 28 #define CSI_SERDES_TYPE 0 //0:m9296, 1:m9672, 2:m96724 29 #if (CSI_SERDES_TYPE==0) 30 #define CSI_SERDES 2 // max cams connected to a dser 31 #elif (CSI_SERDES_TYPE==1) 32 #define CSI_SERDES 4 // max cams connected to a dser 33 #elif (CSI_SERDES_TYPE==2) 34 #define CSI_SERDES 4 // max cams connected to a dser 35 #else 36 #define CSI_SERDES 2 // max cams connected to a dser 37 #endif 38 #define CSI_SERDES_GMSL 39 40 #define CAM_MODE isx021 //isx021, isx031, imx390isp </pre>	<pre> 1 #define I2C_BUS i2c@3180000 // cam_i2c 2 //define I2C_BUS cam_i2cmux 3 #if 0 4 #define I2C_MUX //tca9546@70 5 //define I2C_MUX_TCA9546 6 #define I2C_BUS_CAM0 /bus@0/I2C_BUS/i2c@0 7 #define I2C_BUS_CAM1 /bus@0/I2C_BUS/i2c@0 8 #define I2C_BUS_CAM2 /bus@0/I2C_BUS/i2c@1 9 #define I2C_BUS_CAM3 /bus@0/I2C_BUS/i2c@1 10 #define I2C_BUS_CAM4 /bus@0/I2C_BUS/i2c@1 11 #define I2C_BUS_CAM5 /bus@0/I2C_BUS/i2c@1 12 #define I2C_BUS_CAM6 /bus@0/I2C_BUS/i2c@1 13 #define I2C_BUS_CAM7 /bus@0/I2C_BUS/i2c@1 14 #else 15 #define I2C_BUS_CAM0 /bus@0/I2C_BUS 16 #define I2C_BUS_CAM1 /bus@0/I2C_BUS 17 #define I2C_BUS_CAM2 /bus@0/I2C_BUS 18 #define I2C_BUS_CAM3 /bus@0/I2C_BUS 19 #define I2C_BUS_CAM4 /bus@0/I2C_BUS 20 #define I2C_BUS_CAM5 /bus@0/I2C_BUS 21 #define I2C_BUS_CAM6 /bus@0/I2C_BUS 22 #define I2C_BUS_CAM7 /bus@0/I2C_BUS 23 #endif 24 25 #define CSI_NUM_CHANS 2 //support max cams 26 #define CSI_NUM_PORTS 1 //Number of aggregated ports 27 #define CSI_PORT_LANES 4 //4lane or 2lane 28 #define CSI_SERDES_TYPE 0 //0:m9296, 1:m9672, 2:m96724 29 #if (CSI_SERDES_TYPE==0) 30 #define CSI_SERDES 2 // max cams connected to a dser 31 #elif (CSI_SERDES_TYPE==1) 32 #define CSI_SERDES 4 // max cams connected to a dser 33 #elif (CSI_SERDES_TYPE==2) 34 #define CSI_SERDES 4 // max cams connected to a dser 35 #else 36 #define CSI_SERDES 2 // max cams connected to a dser 37 #endif 38 #define CSI_SERDES_GMSL 39 40 #define CAM_MODE isx021 //isx021, isx031, imx390isp, </pre>

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	20 of 68	

tegra234-camera-config-m96724-4lane_4otocam222.h	tegra234-camera-config-m9296-4lane_2otocam222.h
<pre> 1 #define I2C_BUS i2c@3180000 // cam_i2c 2 //define I2C_BUS cam_i2cmux 3 #if 0 4 #define I2C_MUX //tca9546@70 5 //define I2C_MUX_TCA9546 6 #define I2C_BUS_CAM0 /bus@0/I2C_BUS/i2c@0 7 #define I2C_BUS_CAM1 /bus@0/I2C_BUS/i2c@0 8 #define I2C_BUS_CAM2 /bus@0/I2C_BUS/i2c@1 9 #define I2C_BUS_CAM3 /bus@0/I2C_BUS/i2c@1 10 #define I2C_BUS_CAM4 /bus@0/I2C_BUS/i2c@1 11 #define I2C_BUS_CAM5 /bus@0/I2C_BUS/i2c@1 12 #define I2C_BUS_CAM6 /bus@0/I2C_BUS/i2c@1 13 #define I2C_BUS_CAM7 /bus@0/I2C_BUS/i2c@1 14 #else 15 #define I2C_BUS_CAM0 /bus@0/I2C_BUS 16 #define I2C_BUS_CAM1 /bus@0/I2C_BUS 17 #define I2C_BUS_CAM2 /bus@0/I2C_BUS 18 #define I2C_BUS_CAM3 /bus@0/I2C_BUS 19 #define I2C_BUS_CAM4 /bus@0/I2C_BUS 20 #define I2C_BUS_CAM5 /bus@0/I2C_BUS 21 #define I2C_BUS_CAM6 /bus@0/I2C_BUS 22 #define I2C_BUS_CAM7 /bus@0/I2C_BUS 23 #endif 24 25 #define CSI_NUM_CHANS 4 //support max cams 26 #define CSI_NUM_PORTS 1 //number of aggregated ports 27 #define CSI_PORT_LANES 4 //4lane or 2lane 28 #define CSI_SERDES_TYPE 2 //0:m9296, 1:m96712, 2:m96724 29 #if (CSI_SERDES_TYPE==0) 30 #define CSI_SERDES 2 // max cams connected to a dser 31 #elif (CSI_SERDES_TYPE==1) 32 #define CSI_SERDES 4 // max cams connected to a dser 33 #elif (CSI_SERDES_TYPE==2) 34 #define CSI_SERDES 4 // max cams connected to a dser 35 #else 36 #define CSI_SERDES 2 // max cams connected to a dser 37 #endif 38 #define CSI_SERDES_GMSL 39 40 #define CAM_MODE isx021 //isx021, isx031, imx390isp, </pre>	<pre> 1 #define I2C_BUS i2c@3180000 // cam_i2c 2 //define I2C_BUS cam_i2cmux 3 #if 0 4 #define I2C_MUX //tca9546@70 5 //define I2C_MUX_TCA9546 6 #define I2C_BUS_CAM0 /bus@0/I2C_BUS/i2c@0 7 #define I2C_BUS_CAM1 /bus@0/I2C_BUS/i2c@0 8 #define I2C_BUS_CAM2 /bus@0/I2C_BUS/i2c@1 9 #define I2C_BUS_CAM3 /bus@0/I2C_BUS/i2c@1 10 #define I2C_BUS_CAM4 /bus@0/I2C_BUS/i2c@1 11 #define I2C_BUS_CAM5 /bus@0/I2C_BUS/i2c@1 12 #define I2C_BUS_CAM6 /bus@0/I2C_BUS/i2c@1 13 #define I2C_BUS_CAM7 /bus@0/I2C_BUS/i2c@1 14 #else 15 #define I2C_BUS_CAM0 /bus@0/I2C_BUS 16 #define I2C_BUS_CAM1 /bus@0/I2C_BUS 17 #define I2C_BUS_CAM2 /bus@0/I2C_BUS 18 #define I2C_BUS_CAM3 /bus@0/I2C_BUS 19 #define I2C_BUS_CAM4 /bus@0/I2C_BUS 20 #define I2C_BUS_CAM5 /bus@0/I2C_BUS 21 #define I2C_BUS_CAM6 /bus@0/I2C_BUS 22 #define I2C_BUS_CAM7 /bus@0/I2C_BUS 23 #endif 24 25 #define CSI_NUM_CHANS 2 //support max cams 26 #define CSI_NUM_PORTS 1 //number of aggregated ports 27 #define CSI_PORT_LANES 4 //4lane or 2lane 28 #define CSI_SERDES_TYPE 0 //0:m9296, 1:m96712, 2:m96724 29 #if (CSI_SERDES_TYPE==0) 30 #define CSI_SERDES 2 // max cams connected to a dser 31 #elif (CSI_SERDES_TYPE==1) 32 #define CSI_SERDES 4 // max cams connected to a dser 33 #elif (CSI_SERDES_TYPE==2) 34 #define CSI_SERDES 4 // max cams connected to a dser 35 #else 36 #define CSI_SERDES 2 // max cams connected to a dser 37 #endif 38 #define CSI_SERDES_GMSL 39 40 #define CAM_MODE isx021 //isx021, isx031, imx390isp, </pre>

M96724 2lane (2*otocam -> a -> J20 + 2*otocam -> b -> J21) v.s. 4lane (4*otocam -> b -> J21) :

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	21 of 68	

tegra234-camera-config-m96724-2lane_4otocam222.h	tegra234-camera-config-m96724-4lane_4otocam222.h
<pre> 1 #define I2C_BUS i2c@3180000 // cam_i2c 2 //define I2C_BUS cam_i2cmux 3 #if 0 4 #define I2C_MUX //tca9546@70 5 //define I2C_MUX_TCA9546 6 #define I2C_BUS_CAM0 /bus@0/I2C_BUS/i2c@0 7 #define I2C_BUS_CAM1 /bus@0/I2C_BUS/i2c@0 8 #define I2C_BUS_CAM2 /bus@0/I2C_BUS/i2c@1 9 #define I2C_BUS_CAM3 /bus@0/I2C_BUS/i2c@1 10 #define I2C_BUS_CAM4 /bus@0/I2C_BUS/i2c@1 11 #define I2C_BUS_CAM5 /bus@0/I2C_BUS/i2c@1 12 #define I2C_BUS_CAM6 /bus@0/I2C_BUS/i2c@1 13 #define I2C_BUS_CAM7 /bus@0/I2C_BUS/i2c@1 14 #else 15 #define I2C_BUS_CAM0 /bus@0/I2C_BUS 16 #define I2C_BUS_CAM1 /bus@0/I2C_BUS 17 #define I2C_BUS_CAM2 /bus@0/I2C_BUS 18 #define I2C_BUS_CAM3 /bus@0/I2C_BUS 19 #define I2C_BUS_CAM4 /bus@0/I2C_BUS 20 #define I2C_BUS_CAM5 /bus@0/I2C_BUS 21 #define I2C_BUS_CAM6 /bus@0/I2C_BUS 22 #define I2C_BUS_CAM7 /bus@0/I2C_BUS 23 #endif 24 25 #define CSI_NUM_CHANS 4 //support max cams 26 #define CSI_NUM_PORTS 2 //Number of aggregated ports 27 #define CSI_PORT_LANES 2 //4lane or 2lane 28 #define CSI_SERDES_TYPE 2 //0:m9296, 1:m9672, 2:m9674 29 #if (CSI_SERDES_TYPE==0) 30 #define CSI_SERDES 2 // max cams connected to a dser 31 #elif (CSI_SERDES_TYPE==1) 32 #define CSI_SERDES 4 // max cams connected to a dser 33 #elif (CSI_SERDES_TYPE==2) 34 #define CSI_SERDES 4 // max cams connected to a dser 35 #else 36 #define CSI_SERDES 2 // max cams connected to a dser 37 #endif 38 #define CSI_SERDES_GMSL 39 40 #define CAM_MODE isx021 //isx021, isx031, imx390isp, imx </pre>	<pre> 1 #define I2C_BUS i2c@3180000 // cam_i2c 2 //define I2C_BUS cam_i2cmux 3 #if 0 4 #define I2C_MUX //tca9546@70 5 //define I2C_MUX_TCA9546 6 #define I2C_BUS_CAM0 /bus@0/I2C_BUS/i2c@0 7 #define I2C_BUS_CAM1 /bus@0/I2C_BUS/i2c@0 8 #define I2C_BUS_CAM2 /bus@0/I2C_BUS/i2c@1 9 #define I2C_BUS_CAM3 /bus@0/I2C_BUS/i2c@1 10 #define I2C_BUS_CAM4 /bus@0/I2C_BUS/i2c@1 11 #define I2C_BUS_CAM5 /bus@0/I2C_BUS/i2c@1 12 #define I2C_BUS_CAM6 /bus@0/I2C_BUS/i2c@1 13 #define I2C_BUS_CAM7 /bus@0/I2C_BUS/i2c@1 14 #else 15 #define I2C_BUS_CAM0 /bus@0/I2C_BUS 16 #define I2C_BUS_CAM1 /bus@0/I2C_BUS 17 #define I2C_BUS_CAM2 /bus@0/I2C_BUS 18 #define I2C_BUS_CAM3 /bus@0/I2C_BUS 19 #define I2C_BUS_CAM4 /bus@0/I2C_BUS 20 #define I2C_BUS_CAM5 /bus@0/I2C_BUS 21 #define I2C_BUS_CAM6 /bus@0/I2C_BUS 22 #define I2C_BUS_CAM7 /bus@0/I2C_BUS 23 #endif 24 25 #define CSI_NUM_CHANS 4 //support max cams 26 #define CSI_NUM_PORTS 1 //Number of aggregated ports 27 #define CSI_PORT_LANES 4 //4lane or 2lane 28 #define CSI_SERDES_TYPE 2 //0:m9296, 1:m9672, 2:m9674 29 #if (CSI_SERDES_TYPE==0) 30 #define CSI_SERDES 2 // max cams connected to a dser 31 #elif (CSI_SERDES_TYPE==1) 32 #define CSI_SERDES 4 // max cams connected to a dser 33 #elif (CSI_SERDES_TYPE==2) 34 #define CSI_SERDES 4 // max cams connected to a dser 35 #else 36 #define CSI_SERDES 2 // max cams connected to a dser 37 #endif 38 #define CSI_SERDES_GMSL 39 40 #define CAM_MODE isx021 //isx021, isx031, imx390isp, imx </pre>

M96712 v.s. M96724 :

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	22 of 68	

tegra234-camera-config-m96712-4lane_4otocam222.h	tegra234-camera-config-m96724-4lane_4otocam222.h
<pre> 1 #define I2C_BUS i2c@3180000 // cam_i2c 2 //define I2C_BUS cam_i2cmux 3 #if 0 4 #define I2C_MUX //tca9546@70 5 //define I2C_MUX_TCA9546 6 #define I2C_BUS_CAM0 /bus@0/I2C_BUS/i2c@0 7 #define I2C_BUS_CAM1 /bus@0/I2C_BUS/i2c@0 8 #define I2C_BUS_CAM2 /bus@0/I2C_BUS/i2c@1 9 #define I2C_BUS_CAM3 /bus@0/I2C_BUS/i2c@1 10 #define I2C_BUS_CAM4 /bus@0/I2C_BUS/i2c@1 11 #define I2C_BUS_CAM5 /bus@0/I2C_BUS/i2c@1 12 #define I2C_BUS_CAM6 /bus@0/I2C_BUS/i2c@1 13 #define I2C_BUS_CAM7 /bus@0/I2C_BUS/i2c@1 14 #else 15 #define I2C_BUS_CAM0 /bus@0/I2C_BUS 16 #define I2C_BUS_CAM1 /bus@0/I2C_BUS 17 #define I2C_BUS_CAM2 /bus@0/I2C_BUS 18 #define I2C_BUS_CAM3 /bus@0/I2C_BUS 19 #define I2C_BUS_CAM4 /bus@0/I2C_BUS 20 #define I2C_BUS_CAM5 /bus@0/I2C_BUS 21 #define I2C_BUS_CAM6 /bus@0/I2C_BUS 22 #define I2C_BUS_CAM7 /bus@0/I2C_BUS 23 #endif 24 25 #define CSI_NUM_CHANS 4 //support max cams 26 #define CSI_NUM_PORTS 1 //Number of aggregated ports 27 #define CSI_PORT_LANES 4 //4lane or 2lane 28 #define CSI_SERDES_TYPE 1 //0:m9296, 1:m96712, 2:m96724 29 #if (CSI_SERDES_TYPE==0) 30 #define CSI_SERDES 2 // max cams connected to a dser 31 #elif (CSI_SERDES_TYPE==1) 32 #define CSI_SERDES 4 // max cams connected to a dser 33 #elif (CSI_SERDES_TYPE==2) 34 #define CSI_SERDES 4 // max cams connected to a dser 35 #else 36 #define CSI_SERDES 2 // max cams connected to a dser 37 #endif 38 #define CSI_SERDES_GMSL 39 40 #define CAM_MODE isx021 //isx021, isx031, imx390isp </pre>	<pre> 1 #define I2C_BUS i2c@3180000 // cam_i2c 2 //define I2C_BUS cam_i2cmux 3 #if 0 4 #define I2C_MUX //tca9546@70 5 //define I2C_MUX_TCA9546 6 #define I2C_BUS_CAM0 /bus@0/I2C_BUS/i2c@0 7 #define I2C_BUS_CAM1 /bus@0/I2C_BUS/i2c@0 8 #define I2C_BUS_CAM2 /bus@0/I2C_BUS/i2c@1 9 #define I2C_BUS_CAM3 /bus@0/I2C_BUS/i2c@1 10 #define I2C_BUS_CAM4 /bus@0/I2C_BUS/i2c@1 11 #define I2C_BUS_CAM5 /bus@0/I2C_BUS/i2c@1 12 #define I2C_BUS_CAM6 /bus@0/I2C_BUS/i2c@1 13 #define I2C_BUS_CAM7 /bus@0/I2C_BUS/i2c@1 14 #else 15 #define I2C_BUS_CAM0 /bus@0/I2C_BUS 16 #define I2C_BUS_CAM1 /bus@0/I2C_BUS 17 #define I2C_BUS_CAM2 /bus@0/I2C_BUS 18 #define I2C_BUS_CAM3 /bus@0/I2C_BUS 19 #define I2C_BUS_CAM4 /bus@0/I2C_BUS 20 #define I2C_BUS_CAM5 /bus@0/I2C_BUS 21 #define I2C_BUS_CAM6 /bus@0/I2C_BUS 22 #define I2C_BUS_CAM7 /bus@0/I2C_BUS 23 #endif 24 25 #define CSI_NUM_CHANS 4 //support max cams 26 #define CSI_NUM_PORTS 1 //Number of aggregated ports 27 #define CSI_PORT_LANES 4 //4lane or 2lane 28 #define CSI_SERDES_TYPE 2 //0:m9296, 1:m96712, 2:m96724 29 #if (CSI_SERDES_TYPE==0) 30 #define CSI_SERDES 2 // max cams connected to a dser 31 #elif (CSI_SERDES_TYPE==1) 32 #define CSI_SERDES 4 // max cams connected to a dser 33 #elif (CSI_SERDES_TYPE==2) 34 #define CSI_SERDES 4 // max cams connected to a dser 35 #else 36 #define CSI_SERDES 2 // max cams connected to a dser 37 #endif 38 #define CSI_SERDES_GMSL 39 40 #define CAM_MODE isx021 //isx021, isx031, imx390isp, : </pre>

M9296 2lane (2*otocam -> a -> J20) v.s. M96714 2lane (1*otocam -> a -> J20):

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	23 of 68	

tegra234-camera-config-m9296-2lane_2otocam222.h	tegra234-camera-config-m96714-2lane_1otocam222.h
<pre> 1 #define I2C_BUS i2c@3180000 // cam_i2c 2 //define I2C_BUS cam_i2cmux 3 #if 0 4 #define I2C_MUX //tca9546@70 5 //define I2C_MUX_TCA9546 6 #define I2C_BUS_CAM0 /bus@0/I2C_BUS/i2c@0 7 #define I2C_BUS_CAM1 /bus@0/I2C_BUS/i2c@0 8 #define I2C_BUS_CAM2 /bus@0/I2C_BUS/i2c@1 9 #define I2C_BUS_CAM3 /bus@0/I2C_BUS/i2c@1 10 #define I2C_BUS_CAM4 /bus@0/I2C_BUS/i2c@1 11 #define I2C_BUS_CAM5 /bus@0/I2C_BUS/i2c@1 12 #define I2C_BUS_CAM6 /bus@0/I2C_BUS/i2c@1 13 #define I2C_BUS_CAM7 /bus@0/I2C_BUS/i2c@1 14 #else 15 #define I2C_BUS_CAM0 /bus@0/I2C_BUS 16 #define I2C_BUS_CAM1 /bus@0/I2C_BUS 17 #define I2C_BUS_CAM2 /bus@0/I2C_BUS 18 #define I2C_BUS_CAM3 /bus@0/I2C_BUS 19 #define I2C_BUS_CAM4 /bus@0/I2C_BUS 20 #define I2C_BUS_CAM5 /bus@0/I2C_BUS 21 #define I2C_BUS_CAM6 /bus@0/I2C_BUS 22 #define I2C_BUS_CAM7 /bus@0/I2C_BUS 23 #endif 24 25 #define CSI_NUM_CHANS 2 //support max cams 26 #define CSI_NUM_PORTS 1 //number of aggregated ports 27 #define CSI_PORT_LANES 2 //4lane or 2lane 28 #define CSI_SERDES_TYPE 0 //0:m9296, 1:m96712, 2:m96724 29 #if (CSI_SERDES_TYPE==0) 30 #define CSI_SERDES 2 // max cams connected to a dser 31 #elif (CSI_SERDES_TYPE==1) 32 #define CSI_SERDES 4 // max cams connected to a dser 33 #elif (CSI_SERDES_TYPE==2) 34 #define CSI_SERDES 4 // max cams connected to a dser 35 #elif (CSI_SERDES_TYPE==3) 36 #define CSI_SERDES 1 // max cams connected to a dser 37 #else 38 #define CSI_SERDES 2 // max cams connected to a dser 39 #endif 40 #define CSI_SERDES_GMSL 41 42 #define CAM_MODE isx021 //isx021, isx031, imx390isp, </pre>	<pre> 1 #define I2C_BUS i2c@3180000 // cam_i2c 2 //define I2C_BUS cam_i2cmux 3 #if 0 4 #define I2C_MUX //tca9546@70 5 //define I2C_MUX_TCA9546 6 #define I2C_BUS_CAM0 /bus@0/I2C_BUS/i2c@0 7 #define I2C_BUS_CAM1 /bus@0/I2C_BUS/i2c@0 8 #define I2C_BUS_CAM2 /bus@0/I2C_BUS/i2c@1 9 #define I2C_BUS_CAM3 /bus@0/I2C_BUS/i2c@1 10 #define I2C_BUS_CAM4 /bus@0/I2C_BUS/i2c@1 11 #define I2C_BUS_CAM5 /bus@0/I2C_BUS/i2c@1 12 #define I2C_BUS_CAM6 /bus@0/I2C_BUS/i2c@1 13 #define I2C_BUS_CAM7 /bus@0/I2C_BUS/i2c@1 14 #else 15 #define I2C_BUS_CAM0 /bus@0/I2C_BUS 16 #define I2C_BUS_CAM1 /bus@0/I2C_BUS 17 #define I2C_BUS_CAM2 /bus@0/I2C_BUS 18 #define I2C_BUS_CAM3 /bus@0/I2C_BUS 19 #define I2C_BUS_CAM4 /bus@0/I2C_BUS 20 #define I2C_BUS_CAM5 /bus@0/I2C_BUS 21 #define I2C_BUS_CAM6 /bus@0/I2C_BUS 22 #define I2C_BUS_CAM7 /bus@0/I2C_BUS 23 #endif 24 25 #define CSI_NUM_CHANS 1 //support max cams 26 #define CSI_NUM_PORTS 1 //number of aggregated ports 27 #define CSI_PORT_LANES 2 //4lane or 2lane 28 #define CSI_SERDES_TYPE 3 //0:m9296, 1:m96712, 2:m96724, 3:m96714 29 #if (CSI_SERDES_TYPE==0) 30 #define CSI_SERDES 2 // max cams connected to a dser 31 #elif (CSI_SERDES_TYPE==1) 32 #define CSI_SERDES 4 // max cams connected to a dser 33 #elif (CSI_SERDES_TYPE==2) 34 #define CSI_SERDES 4 // max cams connected to a dser 35 #elif (CSI_SERDES_TYPE==3) 36 #define CSI_SERDES 1 // max cams connected to a dser 37 #else 38 #define CSI_SERDES 2 // max cams connected to a dser 39 #endif 40 #define CSI_SERDES_GMSL 41 42 #define CAM_MODE isx021 //isx021, isx031, imx390isp, imx490isp, imx </pre>

M96724 v.s. oToAdapter :

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	24 of 68	

tegra234-camera-config-m96724-4lane_4otocam222.h

```

12 #define I2C_BUS_CAM0 /bus@0/I2C_BUS/i2c@1
13 #define I2C_BUS_CAM7 /bus@0/I2C_BUS/i2c@1
14 #else
15 #define I2C_BUS_CAM0 /bus@0/I2C_BUS
16 #define I2C_BUS_CAM1 /bus@0/I2C_BUS
17 #define I2C_BUS_CAM2 /bus@0/I2C_BUS
18 #define I2C_BUS_CAM3 /bus@0/I2C_BUS
19 #define I2C_BUS_CAM4 /bus@0/I2C_BUS
20 #define I2C_BUS_CAM5 /bus@0/I2C_BUS
21 #define I2C_BUS_CAM6 /bus@0/I2C_BUS
22 #define I2C_BUS_CAM7 /bus@0/I2C_BUS
23 #endif
24
25 #define CSI_NUM_CHANS 4 //support max cams
26 #define CSI_NUM_PORTS 1 //Number of aggregated ports
27 #define CSI_PORT_LANES 4 //4lane or 2lane
28 #define CSI_SERDES_TYPE 2 //0:m9296, 1:m96712, 2:m96724
29 #if (CSI_SERDES_TYPE==0)
30 #define CSI_SERDES 2 // max cams connected to a dser
31 #elif (CSI_SERDES_TYPE==1)
32 #define CSI_SERDES 4 // max cams connected to a dser
33 #elif (CSI_SERDES_TYPE==2)
34 #define CSI_SERDES 4 // max cams connected to a dser
35 #else
36 #define CSI_SERDES 2 // max cams connected to a dser
37 #endif
38 #define CSI_SERDES_GMSL
39
40 #define CAM_MODE isx021 //isx021, isx031, imx390isp, i
mx490isp, imx728isp, imx623isp, ar0823isp
41

```

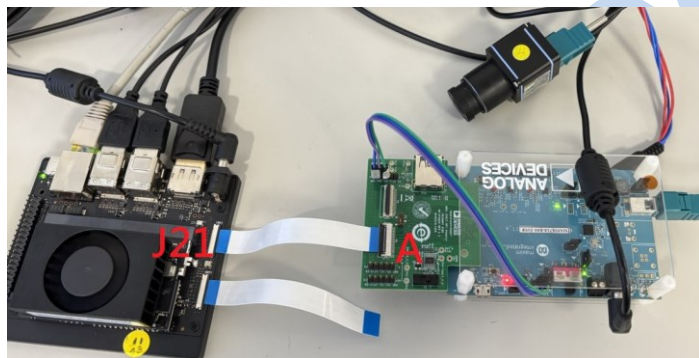
tegra234-camera-config-otoadapter-4lane_4otocam222.h

```

13 #define I2C_BUS_CAM0 /bus@0/I2C_BUS/i2c@1
14 #else
15 #define I2C_BUS_CAM0 /bus@0/I2C_BUS
16 #define I2C_BUS_CAM1 /bus@0/I2C_BUS
17 #define I2C_BUS_CAM2 /bus@0/I2C_BUS
18 #define I2C_BUS_CAM3 /bus@0/I2C_BUS
19 #define I2C_BUS_CAM4 /bus@0/I2C_BUS
20 #define I2C_BUS_CAM5 /bus@0/I2C_BUS
21 #define I2C_BUS_CAM6 /bus@0/I2C_BUS
22 #define I2C_BUS_CAM7 /bus@0/I2C_BUS
23 #endif
24
25 #define CSI_NUM_CHANS 4 //support max cams
26 #define CSI_NUM_PORTS 1 //Number of aggregated ports
27 #define CSI_PORT_LANES 4 //4lane or 2lane
28 #define CSI_SERDES_TYPE 2 //0:m9296, 1:m96712, 2
29 #if (CSI_SERDES_TYPE==0)
30 #define CSI_SERDES 2 // max cams connected
31 #elif (CSI_SERDES_TYPE==1)
32 #define CSI_SERDES 4 // max cams connected
33 #elif (CSI_SERDES_TYPE==2)
34 #define CSI_SERDES 4 // max cams connected
35 #else
36 #define CSI_SERDES 2 // max cams connected
37 #endif
38 #define CSI_SERDES_GMSL
39
40 #define ORIN_NX_OTOADAPTER
41
42 #define CAM_MODE isx021 //isx021, isx031,
43

```

M96714 2lane (1*otocam -> a -> J20) v.s. 4lane (1*otocam -> a -> J21, i2cmux) :



文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	25 of 68	

tegra234-camera-config-m96714-4lane_i2cmux_totocam222.h
tegra234-camera-config-m96714-2lane_totocam222.h

```

1 // #define I2C_BUS i2c@3180000 // cam_i2c
2 #define I2C_BUS cam_i2cmux
3 #if 1
4 #define I2C_MUX //tca9546@70
5 // #define I2C_MUX_TCA9546
6 #define I2C_BUS_CAM0 /bus@0/I2C_BUS/i2c@0
7 #define I2C_BUS_CAM1 /bus@0/I2C_BUS/i2c@0
8 #define I2C_BUS_CAM2 /bus@0/I2C_BUS/i2c@1
9 #define I2C_BUS_CAM3 /bus@0/I2C_BUS/i2c@1
10 #define I2C_BUS_CAM4 /bus@0/I2C_BUS/i2c@1
11 #define I2C_BUS_CAM5 /bus@0/I2C_BUS/i2c@1
12 #define I2C_BUS_CAM6 /bus@0/I2C_BUS/i2c@1
13 #define I2C_BUS_CAM7 /bus@0/I2C_BUS/i2c@1
14 #else
15 #define I2C_BUS_CAM0 /bus@0/I2C_BUS
16 #define I2C_BUS_CAM1 /bus@0/I2C_BUS
17 #define I2C_BUS_CAM2 /bus@0/I2C_BUS
18 #define I2C_BUS_CAM3 /bus@0/I2C_BUS
19 #define I2C_BUS_CAM4 /bus@0/I2C_BUS
20 #define I2C_BUS_CAM5 /bus@0/I2C_BUS
21 #define I2C_BUS_CAM6 /bus@0/I2C_BUS
22 #define I2C_BUS_CAM7 /bus@0/I2C_BUS
23 #endif
24
25 #define CSI_NUM_CHANS 1 //support max cams
26 #define CSI_NUM_PORTS 1 //Number of aggregated ports
27 #define CSI_PORT_LANES 4 //4lane or 2lane
28 #define CSI_SERDES_TYPE 3 //0:m9296, 1:m96712, 2:m96724, 3:m96714
29 #if (CSI_SERDES_TYPE==0)
30 #define CSI_SERDES 2 // max cams connected to a dser
31 #elif (CSI_SERDES_TYPE==1)
32 #define CSI_SERDES 4 // max cams connected to a dser
33 #elif (CSI_SERDES_TYPE==2)
34 #define CSI_SERDES 4 // max cams connected to a dser
35 #elif (CSI_SERDES_TYPE==3)
36 #define CSI_SERDES 1 // max cams connected to a dser
37 #else
38 #define CSI_SERDES 2 // max cams connected to a dser
39 #endif
40 #define CSI_SERDES_GMSL
41
42 #define CAM_MODE isx021 //isx021, isx031, imx390isp

```

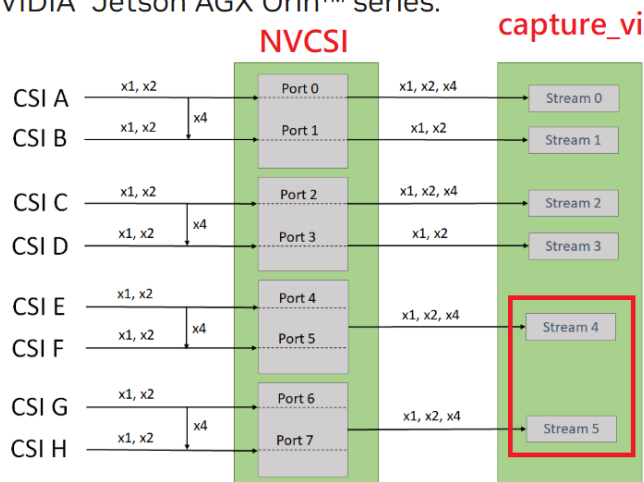
```

1 #define I2C_BUS i2c@3180000 // cam_i2c
2 // #define I2C_BUS cam_i2cmux
3 #if 0
4 #define I2C_MUX //tca9546@70
5 // #define I2C_MUX_TCA9546
6 #define I2C_BUS_CAM0 /bus@0/I2C_BUS/i2c@0
7 #define I2C_BUS_CAM1 /bus@0/I2C_BUS/i2c@0
8 #define I2C_BUS_CAM2 /bus@0/I2C_BUS/i2c@1
9 #define I2C_BUS_CAM3 /bus@0/I2C_BUS/i2c@1
10 #define I2C_BUS_CAM4 /bus@0/I2C_BUS/i2c@1
11 #define I2C_BUS_CAM5 /bus@0/I2C_BUS/i2c@1
12 #define I2C_BUS_CAM6 /bus@0/I2C_BUS/i2c@1
13 #define I2C_BUS_CAM7 /bus@0/I2C_BUS/i2c@1
14 #else
15 #define I2C_BUS_CAM0 /bus@0/I2C_BUS
16 #define I2C_BUS_CAM1 /bus@0/I2C_BUS
17 #define I2C_BUS_CAM2 /bus@0/I2C_BUS
18 #define I2C_BUS_CAM3 /bus@0/I2C_BUS
19 #define I2C_BUS_CAM4 /bus@0/I2C_BUS
20 #define I2C_BUS_CAM5 /bus@0/I2C_BUS
21 #define I2C_BUS_CAM6 /bus@0/I2C_BUS
22 #define I2C_BUS_CAM7 /bus@0/I2C_BUS
23 #endif
24
25 #define CSI_NUM_CHANS 1 //support max cams
26 #define CSI_NUM_PORTS 1 //Number of aggregated ports
27 #define CSI_PORT_LANES 2 //4lane or 2lane
28 #define CSI_SERDES_TYPE 3 //0:m9296, 1:m96712, 2:m96724, 3:m96714
29 #if (CSI_SERDES_TYPE==0)
30 #define CSI_SERDES 2 // max cams connected to a dser
31 #elif (CSI_SERDES_TYPE==1)
32 #define CSI_SERDES 4 // max cams connected to a dser
33 #elif (CSI_SERDES_TYPE==2)
34 #define CSI_SERDES 4 // max cams connected to a dser
35 #elif (CSI_SERDES_TYPE==3)
36 #define CSI_SERDES 1 // max cams connected to a dser
37 #else
38 #define CSI_SERDES 2 // max cams connected to a dser
39 #endif
40 #define CSI_SERDES_GMSL
41
42 #define CAM_MODE isx021 //isx021, isx031, imx390isp, i

```

M9296 4lane AGX Orin (8*otocam v.s. Orin NX(Nano) :

> NVIDIA® Jetson AGX Orin™ series:



文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	26 of 68	

tegra234-camera-config-m9296-4lane_8otocam222.h	tegra234-camera-config-m9296-4lane_2otocam222.h
<pre> 1 #define I2C_BUS i2c@3180000 // cam_i2c 2 // #define I2C_MUX TCA9546 3 #if 1 4 #define I2C_MUX //tca9546@70 5 #define I2C_MUX TCA9546 6 #define I2C_BUS_CAM0 /bus@0/I2C_BUS/i2c@0 7 #define I2C_BUS_CAM1 /bus@0/I2C_BUS/i2c@0 8 #define I2C_BUS_CAM2 /bus@0/I2C_BUS/i2c@1 9 #define I2C_BUS_CAM3 /bus@0/I2C_BUS/i2c@1 10 #define I2C_BUS_CAM4 /bus@0/I2C_BUS/i2c@1 11 #define I2C_BUS_CAM5 /bus@0/I2C_BUS/i2c@1 12 #define I2C_BUS_CAM6 /bus@0/I2C_BUS/i2c@1 13 #define I2C_BUS_CAM7 /bus@0/I2C_BUS/i2c@1 14 #else 15 #define I2C_BUS_CAM0 /bus@0/I2C_BUS 16 #define I2C_BUS_CAM1 /bus@0/I2C_BUS 17 #define I2C_BUS_CAM2 /bus@0/I2C_BUS 18 #define I2C_BUS_CAM3 /bus@0/I2C_BUS 19 #define I2C_BUS_CAM4 /bus@0/I2C_BUS 20 #define I2C_BUS_CAM5 /bus@0/I2C_BUS 21 #define I2C_BUS_CAM6 /bus@0/I2C_BUS 22 #define I2C_BUS_CAM7 /bus@0/I2C_BUS 23 #endif 24 25 #define CSI_NUM_CHANS 8 //support max cams 26 #define CSI_NUM_PORTS 4 //Number of aggregated ports 27 #define CSI_PORT_LANES 4 //4lane or 2lane 28 #define CSI_SERDES_TYPE 0 //0:m9296, 1:m96712, 2:m96724 29 #if (CSI_SERDES_TYPE==0) 30 #define CSI_SERDES 2 // max cams connected to a dser 31 #elif (CSI_SERDES_TYPE==1) 32 #define CSI_SERDES 4 // max cams connected to a dser 33 #elif (CSI_SERDES_TYPE==2) 34 #define CSI_SERDES 4 // max cams connected to a dser 35 #else 36 #define CSI_SERDES 2 // max cams connected to a dser 37 #endif 38 #define CSI_SERDES_GMSL 39 #define AGX_ORIN_M9296SENSING 40 41 #define CAM_MODE isx021 //isx021, isx031, imx390isp </pre>	<pre> 1 #define I2C_BUS i2c@3180000 // cam_i2c 2 // #define I2C_MUX TCA9546 3 #if 0 4 #define I2C_MUX //tca9546@70 5 // #define I2C_MUX TCA9546 6 #define I2C_BUS_CAM0 /bus@0/I2C_BUS/i2c@0 7 #define I2C_BUS_CAM1 /bus@0/I2C_BUS/i2c@0 8 #define I2C_BUS_CAM2 /bus@0/I2C_BUS/i2c@1 9 #define I2C_BUS_CAM3 /bus@0/I2C_BUS/i2c@1 10 #define I2C_BUS_CAM4 /bus@0/I2C_BUS/i2c@1 11 #define I2C_BUS_CAM5 /bus@0/I2C_BUS/i2c@1 12 #define I2C_BUS_CAM6 /bus@0/I2C_BUS/i2c@1 13 #define I2C_BUS_CAM7 /bus@0/I2C_BUS/i2c@1 14 #else 15 #define I2C_BUS_CAM0 /bus@0/I2C_BUS 16 #define I2C_BUS_CAM1 /bus@0/I2C_BUS 17 #define I2C_BUS_CAM2 /bus@0/I2C_BUS 18 #define I2C_BUS_CAM3 /bus@0/I2C_BUS 19 #define I2C_BUS_CAM4 /bus@0/I2C_BUS 20 #define I2C_BUS_CAM5 /bus@0/I2C_BUS 21 #define I2C_BUS_CAM6 /bus@0/I2C_BUS 22 #define I2C_BUS_CAM7 /bus@0/I2C_BUS 23 #endif 24 25 #define CSI_NUM_CHANS 2 //support max cams 26 #define CSI_NUM_PORTS 1 //Number of aggregated ports 27 #define CSI_PORT_LANES 4 //4lane or 2lane 28 #define CSI_SERDES_TYPE 0 //0:m9296, 1:m96712, 2:m96724 29 #if (CSI_SERDES_TYPE==0) 30 #define CSI_SERDES 2 // max cams connected to a dser 31 #elif (CSI_SERDES_TYPE==1) 32 #define CSI_SERDES 4 // max cams connected to a dser 33 #elif (CSI_SERDES_TYPE==2) 34 #define CSI_SERDES 4 // max cams connected to a dser 35 #else 36 #define CSI_SERDES 2 // max cams connected to a dser 37 #endif 38 #define CSI_SERDES_GMSL 39 40 #define CAM_MODE isx021 //isx021, isx031, imx390isp, 41 </pre>

2.1.2 Dynamic by dts generator tool

The .dts of dynamic_dts_generator is generated using a tool ([Excel on Windows/Linux](#)). Currently it's designed for NVIDIA Orin NX(Nano) EVK + Maxim EVK or [OrinNX\(AGX Orin\) EVK + oToAdapter](#).

Excel: Each yellow cell below contains a list of options. After selecting, press the "Generate DTS" button to generate the device tree. Insufficient bandwidth may occur with certain option combinations, resulting in a warning message, then please refer section 2.1.4 for the device tree installation.

There are three sheets : OrinNX(for OrinNX EVK+Maxim EVK), [oToAdapterNX\(for OrinNX EVK+oToAdapter\)](#), [oToAdapterAGX\(for AGX Orin EVK+oToAdapter\)](#)

CSI_Mode : 2x4 for 4lane, 4x2 for 2lane

oToCam :

otocam222/otocam223/otocam260isp/otocam264isp/otocam271isp/otocam274isp/otocam276isp

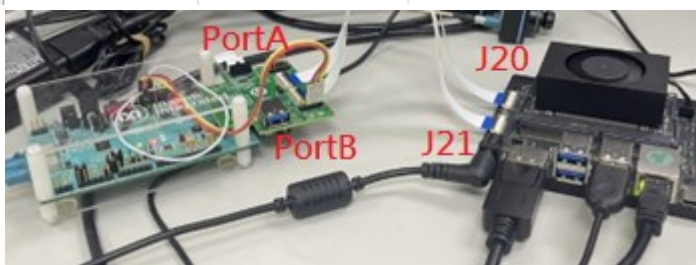
Deserializer : 9296/96712/96714/96724

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	27 of 68	

max lane speed : low / high

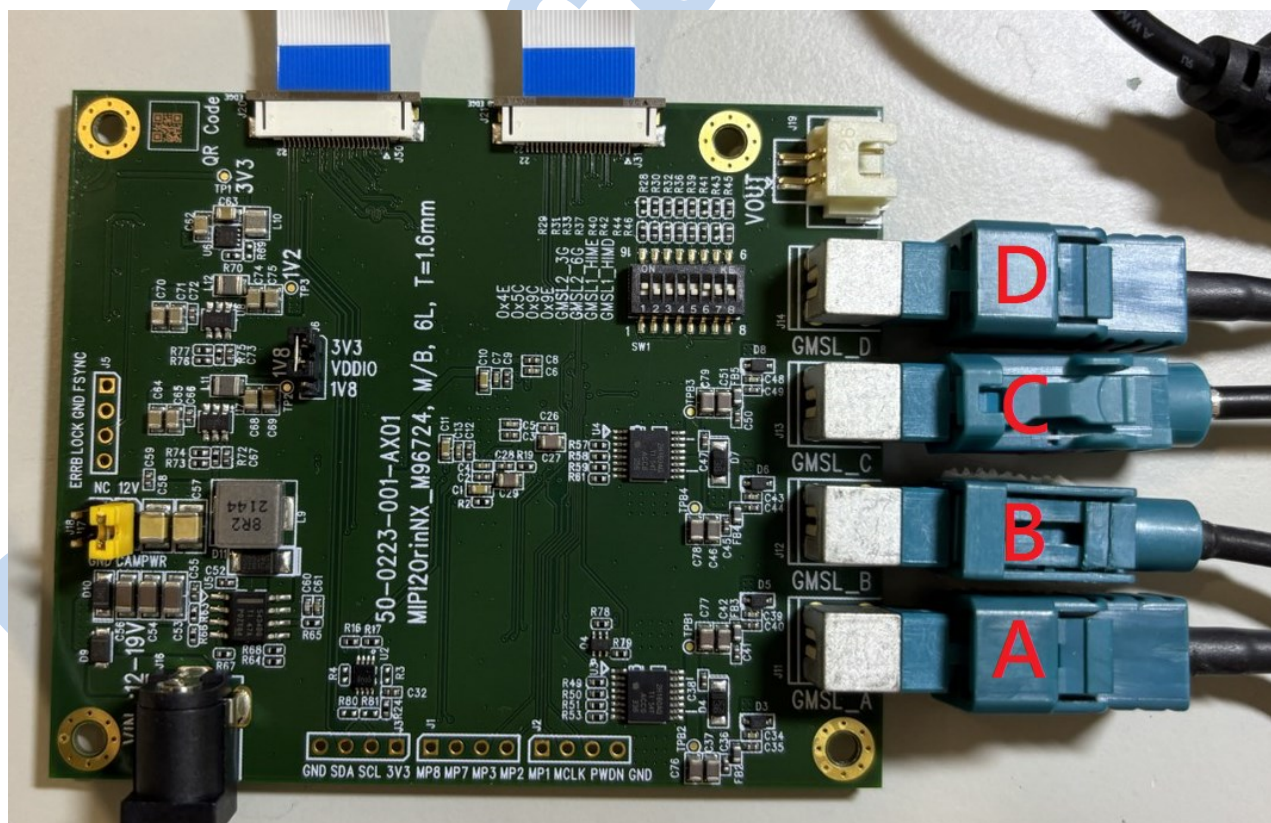
OrinNX EVK+Maxim EVK :

OrinNX	CSI_Mode	oToCam	Deserializer	lane speed	
Setting	2x4	otocam222	9296	high	Generate DTS
		otocam222			
		otocam222			
		otocam222			

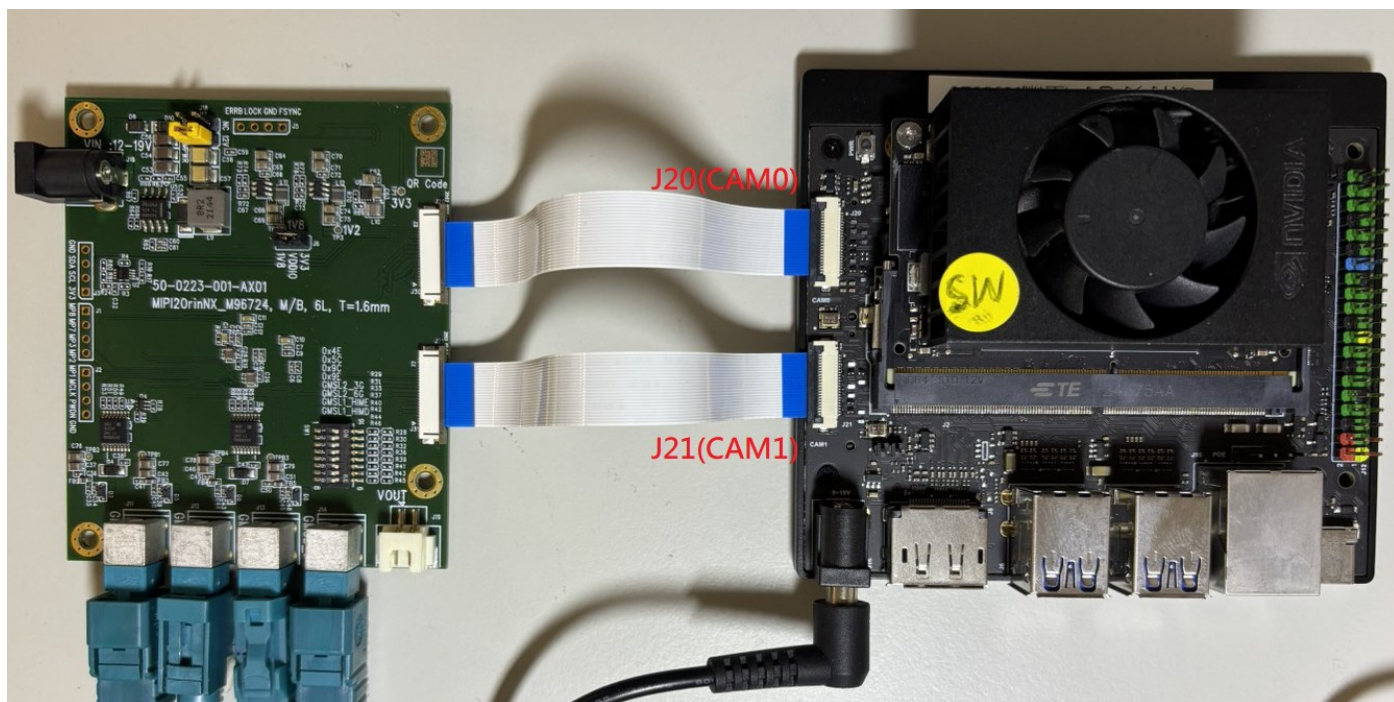


OrinNX EVK+oToAdapter :

OrinNX	CSI_Mode	oToCam	Deserializer	lane speed
Setting	2x4	otocam222 A	96724	high
		otocam222 B		
		otocam222 C		
		otocam222 D		



文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
文件名稱	oToCAM_nVidiaDriver_UserGuide-v2.14.docx	頁 數
		28 of 68



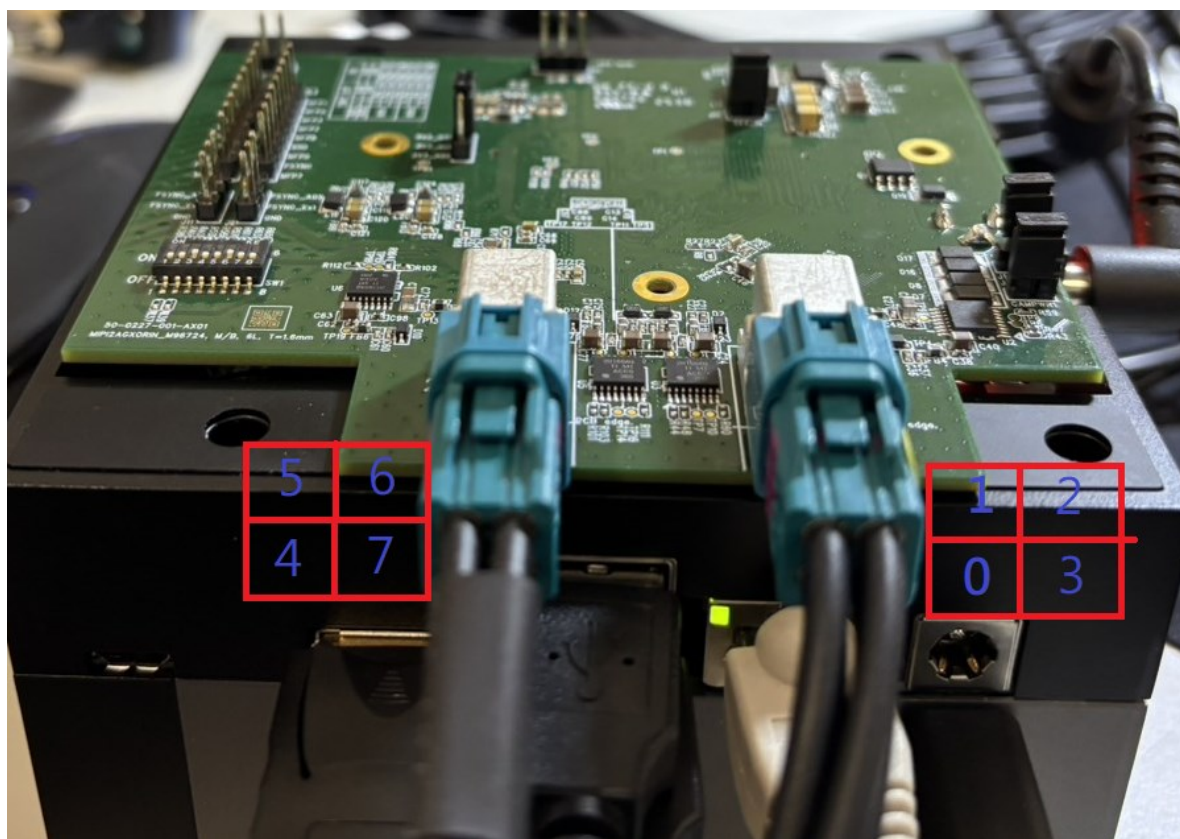
Frame sync Setting : (see section 2.1.7 for detail)

fsync_setting	internal fsync
Deserializer1	on
fsync_setting	external fsync
Deserializer1	off

AGX Orin EVK+oToAdapter :

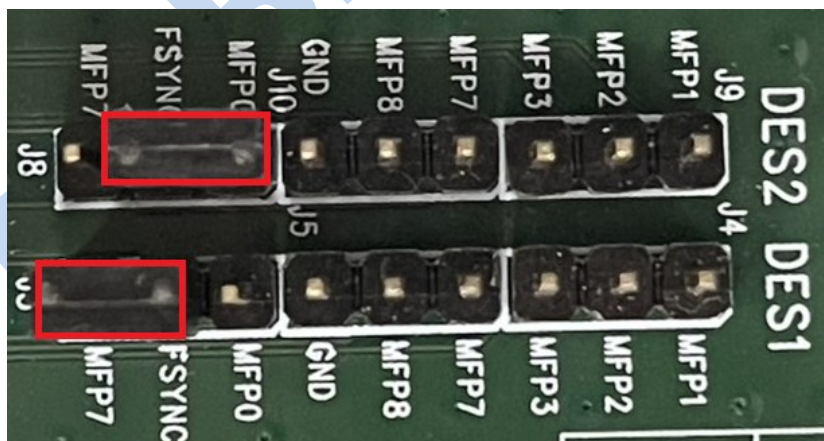
oToAdapterAGX	CSI_Mode	oToCam	Deserializer	lane speed
Setting	2x4	otocam27lisp 0	96724	low
		otocam27lisp 1	96724	
		otocam27lisp 2		
		otocam27lisp 3		
		otocam27lisp 4		
		otocam27lisp 5		
		otocam27lisp 6		
		otocam27lisp 7		

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	29 of 68	



Frame sync Setting : (see section 2.1.7 for detail)

fsync_setting	internal fsync
Deserializer1	off
Deserializer2	on
fsync_setting	external fsync
Deserializer1	on
Deserializer2	off



文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	30 of 68	

p.s.

- lane speed: **high(2.5gb)/low(1.5gb)**. Generally, high is selected. However, if there are issues with board design or cable quality, low should be selected. In this case, the amount of data that can be transmitted will be smaller, and fewer cameras will be supported.
- If the selected combination bandwidth is insufficient, a warning message like the one shown below will appear, and the generated .dts file will contain the "**BandwidthERR**" error message. The .dts file is still usable in this case, but you should reduce the number of connected cameras.

2x4:4lane, 4x2:2lane

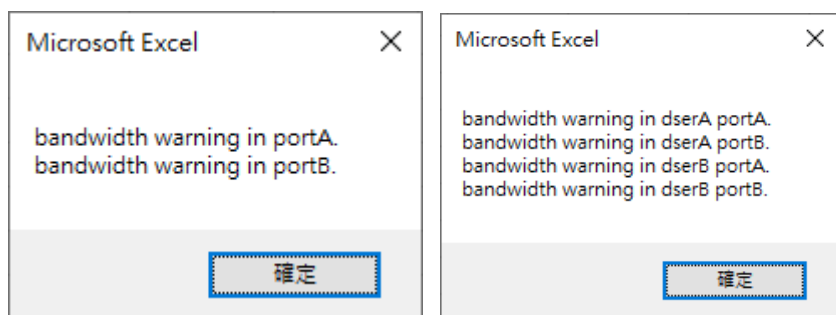
BandwidthERR: oToAdapterAGX-2x4-otocam271isp-otocam271isp-otocam271isp-otocam271isp-otocam271isp-otocam271isp-otocam271isp-otocam271isp-otocam271isp-low.dts

BandwidthERR: oToAdapterNX-2x4-otocam260isp-otocam222-otocam222-otocam222-low.dts

oToAdapterAGX-2x4-otocam223-otocam223-otocam223-otocam223-otocam223-otocam223-otocam223-otocam223-otocam223-high.dts

oToAdapterNX-2x4-otocam260isp-otocam222-otocam222-otocam222-high.dts

oToAdapter Type



3. Bandwidth calculation : $\text{Max_cam} * \text{cam_w} * \text{cam_h} * 8 * 2 * 1.25 * 30(\text{fps}) > \text{max_lane} * \text{lane_speed}(\text{high:2.5g, low 1.5g}) \rightarrow \text{BandwidthERR}$, for example:

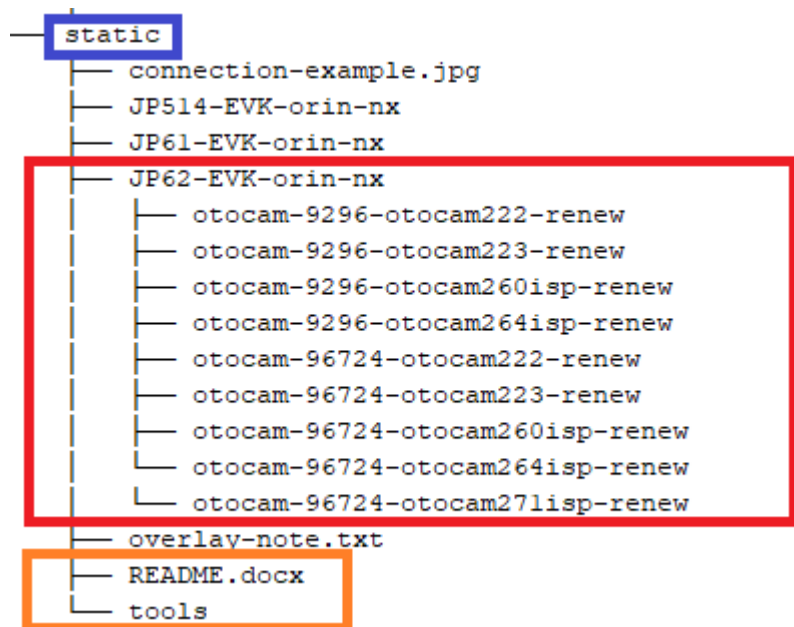
$\text{bw_low}(4\text{lane}) : 4 * 1920 * 1536 * 8 * 2 * 1.25 * 30 = 7\text{g} > 4 * 1.5 = 6\text{g} \rightarrow \text{BandwidthERR}$, in this case, if the

connected cameras are 3, then 5.3g is ok.

$\text{bw_high}(4\text{lane}) : 4 * 1920 * 1536 * 8 * 2 * 1.25 * 30 = 7\text{g} < 4 * 2.5 = 10\text{g} \rightarrow 4$ of the connected cameras is ok.

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	31 of 68	

2.1.3 The static device tree example



2.1.4 The installation of device tree

Please refer to the file README.docx of static folder, and the related tools are on the tools folder.

Step1. copy driver, tree file to tools folder.

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	32 of 68	

```
nvidia@ubuntu:~/tools_20260105$ ls
check_IQ
'GetVoltage&Tempure'
gst-showall2.sh
gst-showall_repeat.sh
gst-showall.sh
insmod-otocam.sh
oToAdapterAGX-2x4-2x4-otocam223-otocam223-otocam223-otocam223-otocam2
23-otocam223-otocam223-high.dts
oToAdapterAGX-2x4-2x4-otocam260isp-otocam260isp-otocam260isp-otocam260isp-otoca
m260isp-otocam260isp-otocam260isp-otocam260isp-high.dts
oToAdapterNX-2x4-otocam223-otocam223-otocam223-otocam223-high.dts
oToCAM.ko
oToSerDes.ko
rmmod.sh
run_replace-v2.sh
set-dtb-to-dts.sh
set-dts-to-dtb.sh
show-debug.sh
show-fps.sh
show-parse-tree.sh
test-nv16.sh
test-otocam276isp.sh
nvidia@ubuntu:~/tools_20260105$ ./run_replace-v2.sh
ls: cannot access '*.dtbo': No such file or directory
1: oToAdapterAGX-2x4-2x4-otocam223-otocam223-otocam223-otocam223-otoca
m223-otocam223-otocam223-high.dts
2: oToAdapterAGX-2x4-2x4-otocam260isp-otocam260isp-otocam260isp-otocam260isp-oto
cam260isp-otocam260isp-otocam260isp-otocam260isp-high.dts
3: oToAdapterNX-2x4-otocam223-otocam223-otocam223-otocam223-high.dts
Enter dts file index to apply: 
```

Step2. Run “[sh run_replace-v2.sh](#)”, then select the target .dts or .dtbo (this step will [auto reboot the system](#)).
If you want to apply run_replace-v2.sh to the dynamic overlay .dts, please change the content of xxx.dts
“overlay-name = "Jetson Camera IMX390";” to “overlay-name = "Camera xxx";”
for JP6.2.2 :

```
// tegra-camera-base-r36.dtsi
/ {
    under JP6.2.1
    overlay-name = "Camera oToAdapterNX-2x4-otocam222-otocam222-otocam222-high-fsync11-fsyncE0";
    jetson-header-name = "Jetson 24pin CSI Connector";
}

355 // tegra-camera-base-r36.dtsi
356
357 / {
    JP6.2.2
    overlay-name = "Camera oToAdapterNX-2x4-otocam222-otocam222-otocam222-high-fsync11-fsyncE0";
    jetson-header-name = "Jetson 24pin CSI Connector";
}
359
```

Step3. Run “[sh insmod-otocam.sh](#)”, then use “[ls /dev/video*](#)” to check video node.

Step4. Run “[sh gst-showall.sh](#)” or “[gst-showall_repeat.sh](#)” in GUI mode or Run “[show-fps.sh](#)” in console mode to show frame rate.

p.s. Please install gstreamer1.0-plugins-bad by “[sudo apt-get install gstreamer1.0-plugins-bad](#)” for [gst-showall*.sh](#) (after JP6.2.2)

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	33 of 68	

4-1. gst-showall2.sh for otocam276isp

gst-showall2.sh 0 -> for 3840x2160

gst-showall2.sh 1 -> for 512x288

4-2. test-otocam276isp.sh

test-otocam276isp.sh 0 -> save for 3840x2160 yuv file

test-otocam276isp.sh 1 -> save for 512x288 yuv file

4-3. gnome-test-nv16-multi.sh / gnome-test-nv16-multi-512k.sh(log size to 512 KB) for frame rate monitoring(output log to videoN.txt).

Step5. Check result

p.s. Please install v4l-utils by “sudo apt-get install v4l-utils -y” for gnome-test-nv16-multi.sh.

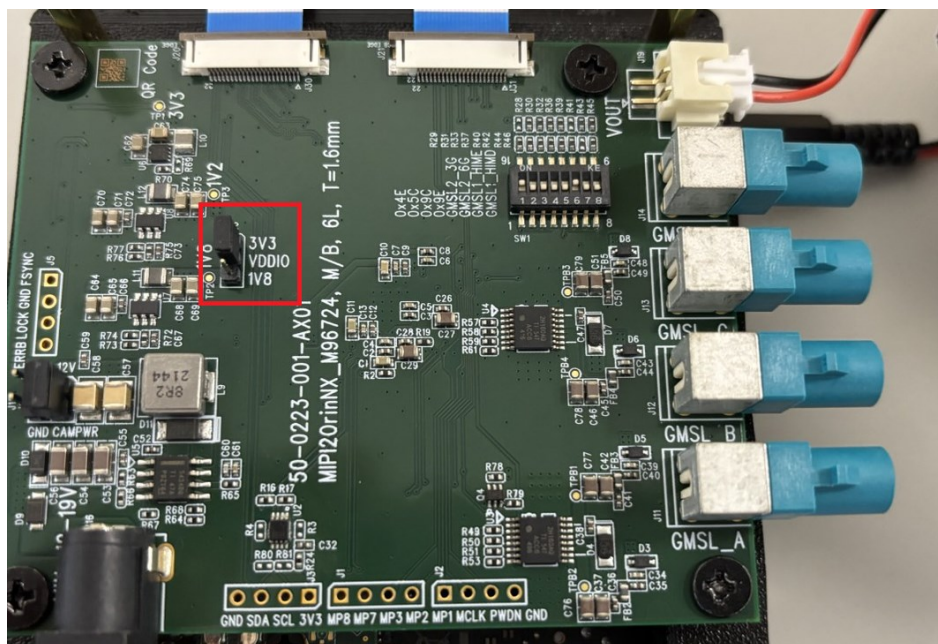
```
nvidia@ubuntu:~/tools_2.4.5$ sudo apt-get install v4l-utils -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libv4l2rds0
The following NEW packages will be installed:
  libv4l2rds0 v4l-utils
0 upgraded, 2 newly installed, 0 to remove and 404 not upgraded.
Need to get 669 kB of archives.
After this operation, 2,200 kB of additional disk space will be used.
Get:1 http://ports.ubuntu.com/ubuntu-ports jammy/main arm64 libv4l2rds0 arm64 1.22.1-2build1 [17.4 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports jammy/universe arm64 v4l-utils arm64 1.22.1-2build1 [652 kB]
Fetched 669 kB in 2s (363 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libv4l2rds0:arm64.
(Reading database ... 189164 files and directories currently installed.)
Preparing to unpack .../libv4l2rds0_1.22.1-2build1_arm64.deb ...
Unpacking libv4l2rds0:arm64 (1.22.1-2build1) ...
Selecting previously unselected package v4l-utils.
Preparing to unpack .../v4l-utils_1.22.1-2build1_arm64.deb ...
Unpacking v4l-utils (1.22.1-2build1) ...
Setting up libv4l2rds0:arm64 (1.22.1-2build1) ...
Setting up v4l-utils (1.22.1-2build1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
Processing triggers for man-db (2.10.2-1) ...
```

2.1.4.1 The device tree installation of oToAdapter

Orin Nano+oToAdapter :

(1) Set to 3.3v for Old HW, New HW : no need to set

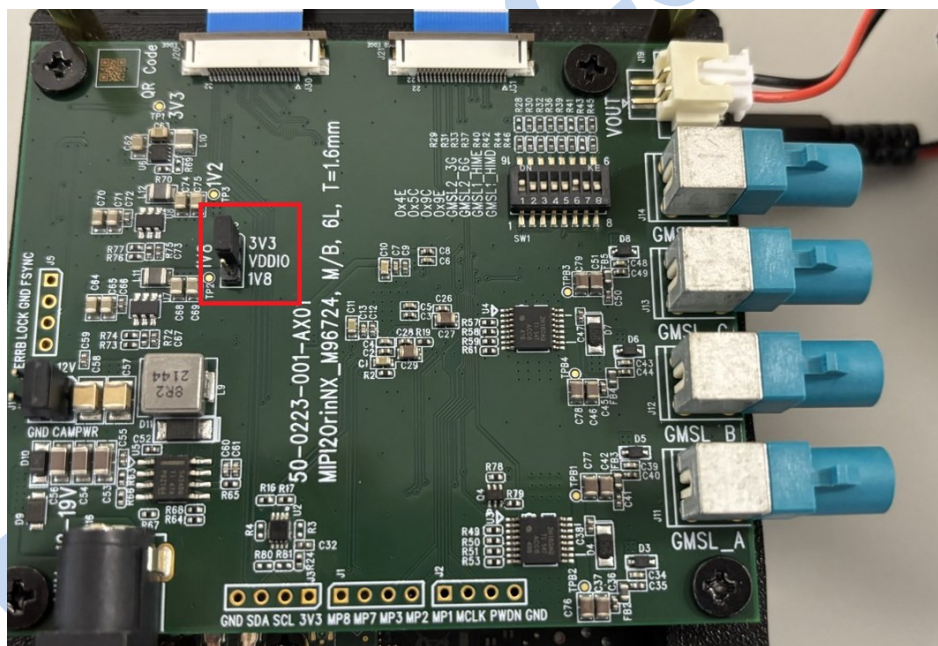
文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	34 of 68	



- (2) Insert a [sd card](#) ($\geq 64\text{GB}$) and flash Jetson Linux bsp image
- (3) Install device tree & display camera image by 2.1.4

Orin NX+oToAdapter :

- (1) Set to 3.3v for Old HW, New HW : no need to set



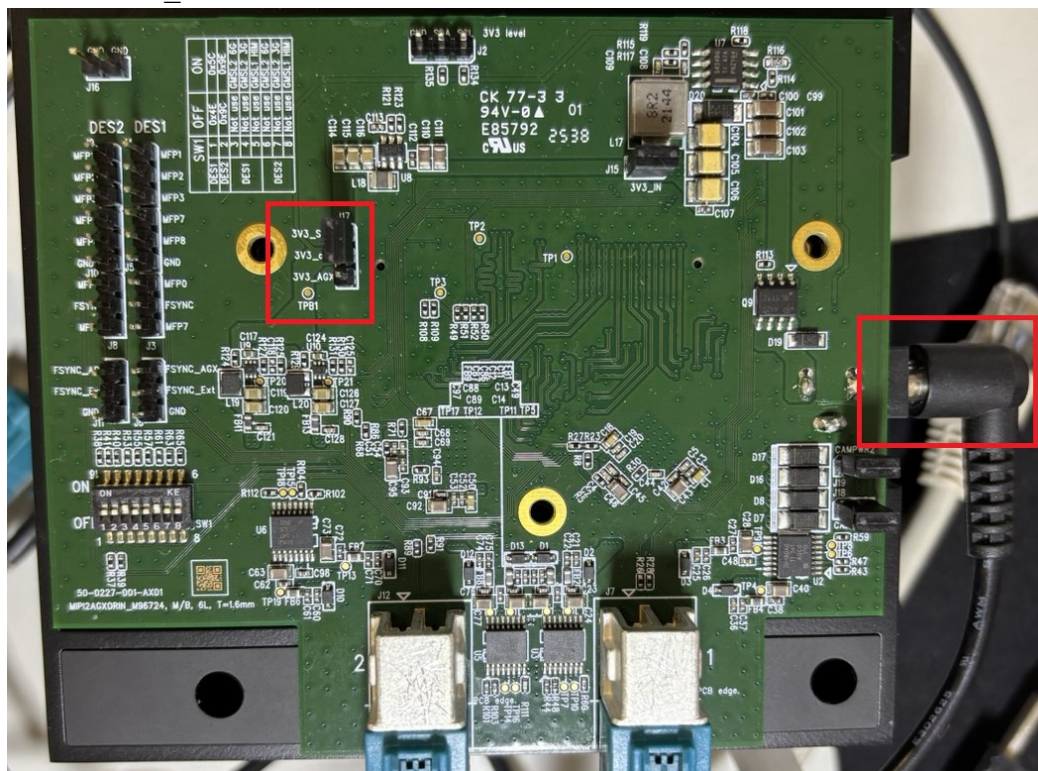
- (2) Insert a [SSD](#) ($\geq 128\text{GB}$) and flash Jetson Linux bsp image
- (3) Install device tree & display camera image by 2.1.4

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	35 of 68	

文件名稱	oToCAM_nVidiaDriver_UserGuide-v2.14.docx	頁 數	35 of 68
------	--	-----	----------

AGX Orin +oToAdapter :

- (1) Please use an external power supply(12V~19V), the jumper is configured to connect to 3V3_Sys and 3V3_out



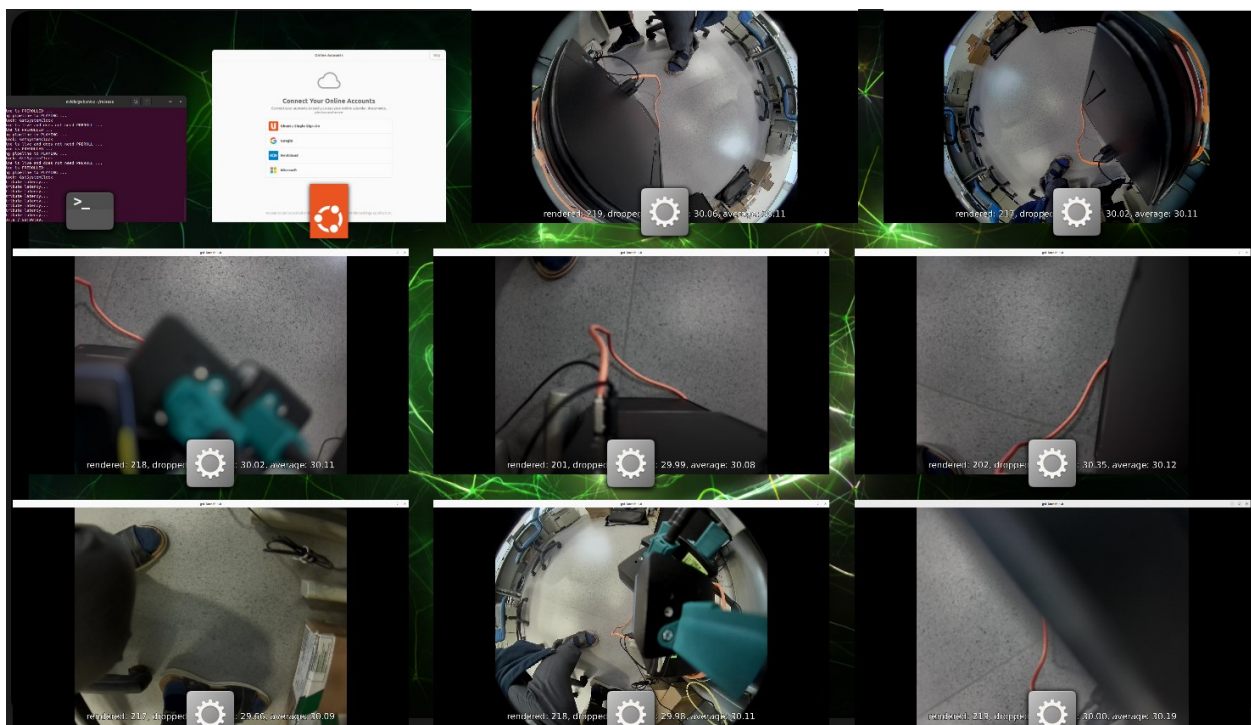
- ## (2) flash Jetson Linux bsp image by Nvidia SDK Manager

The original Jetson Linux is R35 with Ubuntu v20.04. Please install JP6.2 (Ubuntu v22.04).

```
nvidia@ubuntu:~$ more /etc/nv_tegra_release
# R36 (release), RIVISION: 4.3, GCID: 38968081, BOARD: generic, EABI: aarch64, D
ATE: Wed Jan  8 01:49:37 UTC 2025
# KERNEL_VARIANT: oot
TARGET_USERSPACE_LIB_DIR=nvidia
TARGET_USERSPACE_LIB_DIR_PATH=usr/lib/aarch64-linux-gnu/nvidia
nvidia@ubuntu:~$ more /etc/os-release
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
```

- ### (3) Install device tree & display camera image by session 2.1.4

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	36 of 68	



2.1.5 Device tree update for oToCAM.ko & oToSerDes.ko

device tree update for oToCAM.ko & oToSerDes.ko is as the following, please search “compatible” string and update the related string (red box) .

O:\nvidia\test\device_tree\vecow\device-tree_ori.dts	O:\nvidia\test\device_tree\vecow\device-tree_fix.dts
<pre> 241 }; 242 i2c@1 { 243 reg = <0x01>; 244 i2c-mux,deselect-on-exit; 245 i2c-mux-idle-disconnect; 246 #address-cells = <0x01>; 247 #size-cells = <0x00>; 248 max9296_c@48 { 249 compatible = "maxim,max9296"; 250 reg = <0x48>; 251 csi-mode = "2x4"; 252 max-src = <0x02>; 253 reset-gpios = <0xffffffff 0xa0 0x00>; 254 fasync-gpio = <0x06>; 255 phandle = <0x06>; 256 }; 257 max9295_prim@62 { 258 compatible = "maxim,max9295"; 259 reg = <0x62>; 260 is-prim-ser; 261 }; 262 max9295_c@40 { 263 compatible = "maxim,max9295"; 264 reg = <0x40>; 265 nvidia,gmsl-dser-device = <0x06>; 266 phandle = <0x07>; 267 }; 268 max9295_d@60 { 269 compatible = "maxim,max9295"; 270 reg = <0x60>; 271 nvidia,gmsl-dser-device = <0x06>; 272 phandle = <0x09>; 273 }; 274 otocamisp_c@1b { 275 status = "okay"; 276 reg = <0x1b>; 277 def-addr = <0x6d>; 278 clocks = <0xffffffff 0x24 0xffffffff 0x24>; 279 clock-names = "extperiph1@01lp_grtba"; 280 mclk = "extperiph1"; 281 nvidia,gmsl-dser-device = <0x07>; 282 nvidia,gmsl-dser-device = <0x06>; 283 devnode = "video2"; 284 compatible = "sony,imx398"; 285 physical_w = "15.0"; </pre>	<pre> 241 }; 242 i2c@1 { 243 reg = <0x01>; 244 i2c-mux,deselect-on-exit; 245 i2c-mux-idle-disconnect; 246 #address-cells = <0x01>; 247 #size-cells = <0x00>; 248 max9296_c@48 { 249 compatible = "otobrite,deserializer"; 250 reg = <0x48>; 251 csi-mode = "2x4"; 252 max-src = <0x02>; 253 reset-gpios = <0xffffffff 0xa0 0x00>; 254 fasync-gpio = <0x06>; 255 phandle = <0x06>; 256 }; 257 max9295_prim@62 { 258 compatible = "otobrite,serializer"; 259 reg = <0x62>; 260 is-prim-ser; 261 }; 262 max9295_c@40 { 263 compatible = "otobrite,serializer"; 264 reg = <0x40>; 265 nvidia,gmsl-dser-device = <0x06>; 266 phandle = <0x07>; 267 }; 268 max9295_d@60 { 269 compatible = "otobrite,serializer"; 270 reg = <0x60>; 271 nvidia,gmsl-dser-device = <0x06>; 272 phandle = <0x09>; 273 }; 274 otocamisp_c@1b { 275 status = "okay"; 276 reg = <0x1b>; 277 def-addr = <0x6d>; 278 clocks = <0xffffffff 0x24 0xffffffff 0x24>; 279 clock-names = "extperiph1@01lp_grtba"; 280 mclk = "extperiph1"; 281 nvidia,gmsl-dser-device = <0x07>; 282 nvidia,gmsl-dser-device = <0x06>; 283 devnode = "video2"; 284 compatible = "otobrite,otocam"; 285 physical_w = "15.0"; </pre>

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	37 of 68	

文件名稱	oToCAM_nVidiaDriver_UserGuide-v2.14.docx
------	--

2.1.6 Device tree update for sensor model

- This is the setting of otocam222.

```
mode0 {
    mclk_khz = "24000";
    num_lanes = "4";
    tegra_sinterface = "serial_a";
    vc_id = "0";
    discontinuous_clk = "no";
    dpcm_enable = "false";
    cil_settletime = "0";
    mode_type = "yuv";
    pixel_phase = "uyvy";
    csi_pixel_bit_depth = "16";
    dynamic_pixel_bit_depth = "16";
    active_w = "1920";
    active_h = "1280";
    readout_orientation = "0";
    line_length = "2200";
    inherent_gain = "1";
    pix_clk_hz = "375000000";
    gain_factor = "10";
    min_gain_val = "0";
    max_gain_val = "300";
    step_gain_val = "3";
    default_gain = "0";
    min_hdr_ratio = "1";
    max_hdr_ratio = "1";
    framerate_factor = "1000000";
    min_framerate = "30000000";
    max_framerate = "30000000";
    step_framerate = "1";
    default_framerate = "30000000";
    exposure_factor = "1000000";
    min_exp_time = "59";
    max_exp_time = "33333";
    step_exp_time = "1";
    default_exp_time = "33333";
    embedded_metadata_height = "0";
};
```

- Differences between otocam222 & otocam223/otocam274isp

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	38 of 68	

文件名稱 oToCAM_nVidiaDriver_UserGuide-v2.14.docx

<pre> 48 clocks = <0xffffffff 0x24 0xffffffff 0x24>; 49 clock-names = "extperiph1@p1lp_gtrba"; 50 mclk = "extperiph1"; 51 nvidia,gmsl-ser-device = <0x02>; 52 nvidia,gmsl-dser-device = <0x01>; 53 devmode = "video0"; 54 compatible = "otobrite,otocam"; 55 physical_w = "15.0"; 56 physical_h = "12.5"; 57 sensor_model = "isx001"; 58 post_crop_frame_drop = "0"; 59 use_decibel_gain = "true"; 60 use_sensor_mode_id = "true"; 61 phandle = <0x2b>; 62 63 mode0 { 64 mclk_khz = "24000"; 65 num_lanes = "4"; 66 tegra_sinterface = "serial_a"; 67 vc_id = "0"; 68 discontinuous_clk = "no"; 69 dpcn_enable = "false"; 70 cil_settletime = "0"; 71 mode_type = "yuv"; 72 pixel_phase = "uyvy"; 73 csi_pixel_bit_depth = "16"; 74 dynamic_pixel_bit_depth = "16"; 75 active_w = "1920"; 76 active_h = "1280"; 77 readout_orientation = "0"; 78 line_length = "2200"; 79 inherent_gain = "1"; 80 pix_clk_hz = "375000000"; 81 gain_factor = "10"; 82 min_gain_val = "0"; </pre>	<pre> 48 clocks = <0xffffffff 0x24 0xffffffff 0x24>; 49 clock-names = "extperiph1@p1lp_gtrba"; 50 mclk = "extperiph1"; 51 nvidia,gmsl-ser-device = <0x02>; 52 nvidia,gmsl-dser-device = <0x01>; 53 devmode = "video0"; 54 compatible = "otobrite,otocam"; 55 physical_w = "15.0"; 56 physical_h = "12.5"; 57 sensor_model = "isx001"; 58 post_crop_frame_drop = "0"; 59 use_decibel_gain = "true"; 60 use_sensor_mode_id = "true"; 61 phandle = <0x2b>; 62 63 mode0 { 64 mclk_khz = "24000"; 65 num_lanes = "4"; 66 tegra_sinterface = "serial_a"; 67 vc_id = "0"; 68 discontinuous_clk = "no"; 69 dpcn_enable = "false"; 70 cil_settletime = "0"; 71 mode_type = "yuv"; 72 pixel_phase = "uyvy"; 73 csi_pixel_bit_depth = "16"; 74 dynamic_pixel_bit_depth = "16"; 75 active_w = "1920"; 76 active_h = "1280"; 77 readout_orientation = "0"; 78 line_length = "2200"; 79 inherent_gain = "1"; 80 pix_clk_hz = "375000000"; 81 gain_factor = "10"; 82 min_gain_val = "0"; </pre>
--	--

● Differences between otocam222 & otocam260isp

<pre> 50 mclk = "extperiph1"; 51 nvidia,gmsl-ser-device = <0x02>; 52 nvidia,gmsl-dser-device = <0x01>; 53 devmode = "video0"; 54 compatible = "otobrite,otocam"; 55 physical_w = "15.0"; 56 physical_h = "12.5"; 57 sensor_model = "isx001"; 58 post_crop_frame_drop = "0"; 59 use_decibel_gain = "true"; 60 use_sensor_mode_id = "true"; 61 phandle = <0x2b>; 62 63 mode0 { 64 mclk_khz = "24000"; 65 num_lanes = "4"; 66 tegra_sinterface = "serial_a"; 67 vc_id = "0"; 68 discontinuous_clk = "no"; 69 dpcn_enable = "false"; 70 cil_settletime = "0"; 71 mode_type = "yuv"; 72 pixel_phase = "uyvy"; 73 csi_pixel_bit_depth = "16"; 74 dynamic_pixel_bit_depth = "16"; 75 active_w = "1920"; 76 active_h = "1280"; 77 readout_orientation = "0"; 78 line_length = "2200"; 79 inherent_gain = "1"; 80 pix_clk_hz = "375000000"; 81 gain_factor = "10"; 82 min_gain_val = "0"; 83 max_gain_val = "300"; 84 step_gain_val = "3"; 85 default_gain = "0"; 86 min_hdr_ratio = "1"; </pre>	<pre> 50 mclk = "extperiph1"; 51 nvidia,gmsl-ser-device = <0x02>; 52 nvidia,gmsl-dser-device = <0x01>; 53 devmode = "video0"; 54 compatible = "otobrite,otocam"; 55 physical_w = "15.0"; 56 physical_h = "12.5"; 57 sensor_model = "isx001isp"; 58 post_crop_frame_drop = "0"; 59 use_decibel_gain = "true"; 60 use_sensor_mode_id = "true"; 61 phandle = <0x2b>; 62 63 mode0 { 64 mclk_khz = "24000"; 65 num_lanes = "4"; 66 tegra_sinterface = "serial_a"; 67 vc_id = "0"; 68 discontinuous_clk = "no"; 69 dpcn_enable = "false"; 70 cil_settletime = "0"; 71 mode_type = "yuv"; 72 pixel_phase = "uyvy"; 73 csi_pixel_bit_depth = "16"; 74 dynamic_pixel_bit_depth = "16"; 75 active_w = "2880"; 76 active_h = "1800"; 77 readout_orientation = "0"; 78 line_length = "3200"; 79 inherent_gain = "1"; 80 pix_clk_hz = "375000000"; 81 gain_factor = "10"; 82 min_gain_val = "0"; 83 max_gain_val = "300"; 84 step_gain_val = "3"; 85 default_gain = "0"; 86 min_hdr_ratio = "1"; </pre>
---	--

● Differences between otocam222 & otocam264isp

<pre> 53 devmode = "video0"; 54 compatible = "otobrite,otocam"; 55 physical_w = "15.0"; 56 physical_h = "12.5"; 57 sensor_model = "isx001"; 58 post_crop_frame_drop = "0"; 59 use_decibel_gain = "true"; 60 use_sensor_mode_id = "true"; 61 phandle = <0x2b>; 62 63 mode0 { 64 mclk_khz = "24000"; 65 num_lanes = "4"; 66 tegra_sinterface = "serial_a"; 67 vc_id = "0"; 68 discontinuous_clk = "no"; 69 dpcn_enable = "false"; 70 cil_settletime = "0"; 71 mode_type = "yuv"; 72 pixel_phase = "uyvy"; 73 csi_pixel_bit_depth = "16"; 74 dynamic_pixel_bit_depth = "16"; 75 active_w = "1920"; 76 active_h = "1280"; 77 readout_orientation = "0"; 78 line_length = "2200"; 79 inherent_gain = "1"; 80 pix_clk_hz = "375000000"; </pre>	<pre> 53 devmode = "video0"; 54 compatible = "otobrite,otocam"; 55 physical_w = "15.0"; 56 physical_h = "12.5"; 57 sensor_model = "isx001isp"; 58 post_crop_frame_drop = "0"; 59 use_decibel_gain = "true"; 60 use_sensor_mode_id = "true"; 61 phandle = <0x2b>; 62 63 mode0 { 64 mclk_khz = "24000"; 65 num_lanes = "4"; 66 tegra_sinterface = "serial_a"; 67 vc_id = "0"; 68 discontinuous_clk = "no"; 69 dpcn_enable = "false"; 70 cil_settletime = "0"; 71 mode_type = "yuv"; 72 pixel_phase = "uyvy"; 73 csi_pixel_bit_depth = "16"; 74 dynamic_pixel_bit_depth = "16"; 75 active_w = "1920"; 76 active_h = "1800"; 77 readout_orientation = "0"; 78 line_length = "2200"; 79 inherent_gain = "1"; 80 pix_clk_hz = "375000000"; </pre>
--	---

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	39 of 68	

文件名稱 oToCAM_nVidiaDriver_UserGuide-v2.14.docx

● Differences between otocam222 & otocam271isp/otocam276isp(max_lane_speed = 2.5G)

R:\Release\Nvidia_Release\fix\static\UP62-EVK-orin-nx\otocam-9296-otocam222- new\device-tree.dts	R:\Release\Nvidia_Release\fix\static\UP62-EVK-orin-nx\otocam-9296-otocam271isp- new\device-tree.dts
<pre> 34 compatible = "nvidia, tegra-camera"; 55 physical_w = "15.0"; 56 physical_h = "12.5"; 57 sensor_model = "imx021"; 58 post_crop_frame_drop = "0"; 59 use_decibel_gain = "true"; 60 use_sensor_mode_id = "true"; 61 phandle = <0x2b>; 62 63 mode0 { 64 mclk_khz = "24000"; 65 num_lanes = "4"; 66 tegra_interface = "serial_a"; 67 vc_id = "0"; 68 discontinuous_clk = "no"; 69 dpcm_enable = "false"; 70 cll_settletime = "0"; 71 mode_type = "yuv"; 72 pixel_phase = "uyvy"; 73 csi_pixel_bit_depth = "16"; 74 dynamic_pixel_bit_depth = "16"; 75 active_w = "1920"; 76 active_h = "1280"; 77 readout_orientation = "0"; 78 line_length = "2200"; 79 inherent_gain = "1"; 80 pix_clk_hz = "375000000"; 81 gain_factor = "10"; 82 min_gain_val = "0"; </pre>	<pre> 34 compatible = "nvidia, tegra-camera"; 55 physical_w = "15.0"; 56 physical_h = "12.5"; 57 sensor_model = "imx728isp"; 58 post_crop_frame_drop = "0"; 59 use_decibel_gain = "true"; 60 use_sensor_mode_id = "true"; 61 phandle = <0x2b>; 62 63 mode0 { 64 mclk_khz = "24000"; 65 num_lanes = "4"; 66 tegra_interface = "serial_a"; 67 vc_id = "0"; 68 discontinuous_clk = "no"; 69 dpcm_enable = "false"; 70 cll_settletime = "0"; 71 mode_type = "yuv"; 72 pixel_phase = "uyvy"; 73 csi_pixel_bit_depth = "16"; 74 dynamic_pixel_bit_depth = "16"; 75 active_w = "3840"; 76 active_h = "2160"; 77 readout_orientation = "0"; 78 line_length = "4080"; 79 inherent_gain = "1"; 80 pix_clk_hz = "625000000"; 81 gain_factor = "10"; 82 min_gain_val = "0"; </pre>
<pre> 330 }; 331 332 fragment-camera@13 { 333 target-path = "/"; 334 335 __overlay__ { 336 337 tegra-camera-platform { 338 compatible = "nvidia, tegra-camera-platform"; 339 num_csi_lanes = <0x04>; 340 max_lane_speed = <0x16380>; 341 min_bits_per_pixel = <0x0a>; 342 vi_peak_byte_per_pixel = <0x02>; 343 vi_bw_margin_pct = <0x10>; 344 isp_peak_byte_per_pixel = <0x05>; 345 isp_bw_margin_pct = <0x19>; 346 phandle = <0x31>; 347 </pre>	<pre> 330 }; 331 332 fragment-camera@13 { 333 target-path = "/"; 334 335 __overlay__ { 336 337 tegra-camera-platform { 338 compatible = "nvidia, tegra-camera-platform"; 339 num_csi_lanes = <0x04>; 340 max_lane_speed = <0x262540>; 341 min_bits_per_pixel = <0x0a>; 342 vi_peak_byte_per_pixel = <0x02>; 343 vi_bw_margin_pct = <0x10>; 344 isp_peak_byte_per_pixel = <0x05>; 345 isp_bw_margin_pct = <0x19>; 346 phandle = <0x31>; 347 </pre>

● Differences between otocam222 & otocam264raw

O:\nvidia\test\device_tree\avermidia\device-tree-otocam222- dts	O:\nvidia\test\device_tree\avermidia\device-tree-otocam264raw- dts
<pre> 3258 physical_h = "12.5"; 3259 sensor_model = "imx021"; 3260 post_crop_frame_drop = "0"; 3261 use_decibel_gain = "true"; 3262 use_sensor_mode_id = "true"; 3263 phandle = <0x24f>; 3264 3265 mode0 { 3266 mclk_khz = "24000"; 3267 num_lanes = "4"; 3268 tegra_interface = "serial_a"; 3269 vc_id = "0"; 3270 discontinuous_clk = "no"; 3271 dpcm_enable = "false"; 3272 cll_settletime = "0"; 3273 mode_type = "yuv"; 3274 pixel_phase = "uyvy"; 3275 csi_pixel_bit_depth = "16"; 3276 dynamic_pixel_bit_depth = "16"; 3277 active_w = "1920"; 3278 active_h = "1280"; 3279 readout_orientation = "0"; 3280 line_length = "2200"; 3281 inherent_gain = "1"; 3282 pix_clk_hz = "375000000"; 3283 gain_factor = "10"; 3284 min_gain_val = "0"; 3285 max_gain_val = "300"; 3286 step_gain_val = "3"; 3287 default_gain = "0"; 3288 min_hdr_ratio = "1"; 3289 max_hdr_ratio = "1"; 3290 framerate_factor = "1000000"; 3291 min_framerate = "30000000"; 3292 max_framerate = "30000000"; 3293 step_framerate = "1"; 3294 default_framerate = "30000000"; 3295 exposure_factor = "1000000"; 3296 min_exp_time = "59"; 3297 max_exp_time = "33333"; 3298 step_exp_time = "1"; 3299 default_exp_time = "33333"; 3300 embedded_metadata_height = "0"; 3301 lane_polarity = "0"; 3302 }; 3303 3304 gmsl-link { 3305 streams = "yuv@0ued-ul"; 3306 src-csi-port = "b"; 3307 dst-csi-port = "a"; 3308 serdes-csi-link = "a"; </pre>	<pre> 3258 physical_h = "12.5"; 3259 sensor_model = "imx390"; 3260 post_crop_frame_drop = "0"; 3261 use_decibel_gain = "true"; 3262 use_sensor_mode_id = "true"; 3263 phandle = <0x24f>; 3264 3265 mode0 { 3266 mclk_khz = "24000"; 3267 num_lanes = "4"; 3268 tegra_interface = "serial_a"; 3269 vc_id = "0"; 3270 discontinuous_clk = "no"; 3271 dpcm_enable = "false"; 3272 cll_settletime = "0"; 3273 mode_type = "bayer"; 3274 pixel_phase = "rggb"; 3275 csi_pixel_bit_depth = "12"; 3276 dynamic_pixel_bit_depth = "12"; 3277 active_w = "1920"; 3278 active_h = "1080"; 3279 readout_orientation = "0"; 3280 line_length = "2200"; 3281 inherent_gain = "1"; 3282 pix_clk_hz = "625000000"; 3283 gain_factor = "10"; 3284 min_gain_val = "0"; 3285 max_gain_val = "300"; 3286 step_gain_val = "3"; 3287 default_gain = "0"; 3288 min_hdr_ratio = "1"; 3289 max_hdr_ratio = "1"; 3290 framerate_factor = "1000000"; 3291 min_framerate = "30000000"; 3292 max_framerate = "30000000"; 3293 step_framerate = "1"; 3294 default_framerate = "30000000"; 3295 exposure_factor = "1000000"; 3296 min_exp_time = "59"; 3297 max_exp_time = "33333"; 3298 step_exp_time = "1"; 3299 default_exp_time = "33333"; 3300 embedded_metadata_height = "0"; 3301 lane_polarity = "0"; 3302 }; 3303 3304 gmsl-link { 3305 streams = "raw12@ued-ul"; 3306 src-csi-port = "b"; 3307 dst-csi-port = "a"; 3308 serdes-csi-link = "a"; </pre>

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	40 of 68	

- Driver rename (nv_imx390.ko -> oToCAM.ko, max9296.ko -> oToSerDes.ko)

O:\nvidia\test\device_tree\vecow\device-tree_ori.dts	O:\nvidia\test\device_tree\vecow\device-tree_fix.dts
<pre> 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 </pre>	<pre> 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 </pre>

2.1.7 Frame sync

Internal Frame sync : The fsync signal is generated internally by deserializer. The red box below indicates that the 10Hz signal is generated internally by the deserializer.

```

dserb: max96724@4e {
    status = "okay";
    compatible = "otobrite,deserializer";
    reg = <0x4e>;
    force-clock = <0x1>;
    csi-mode = "2x4";
    max-src = <4>;
    fsync-inter-hz = <0x0a>;
};

```

External Frame sync : The fsync signal is generated externally, such as PWM. We need to configure the external signal and specify which GPIO input to use. The red box below indicates that the signal is input via gpio1.

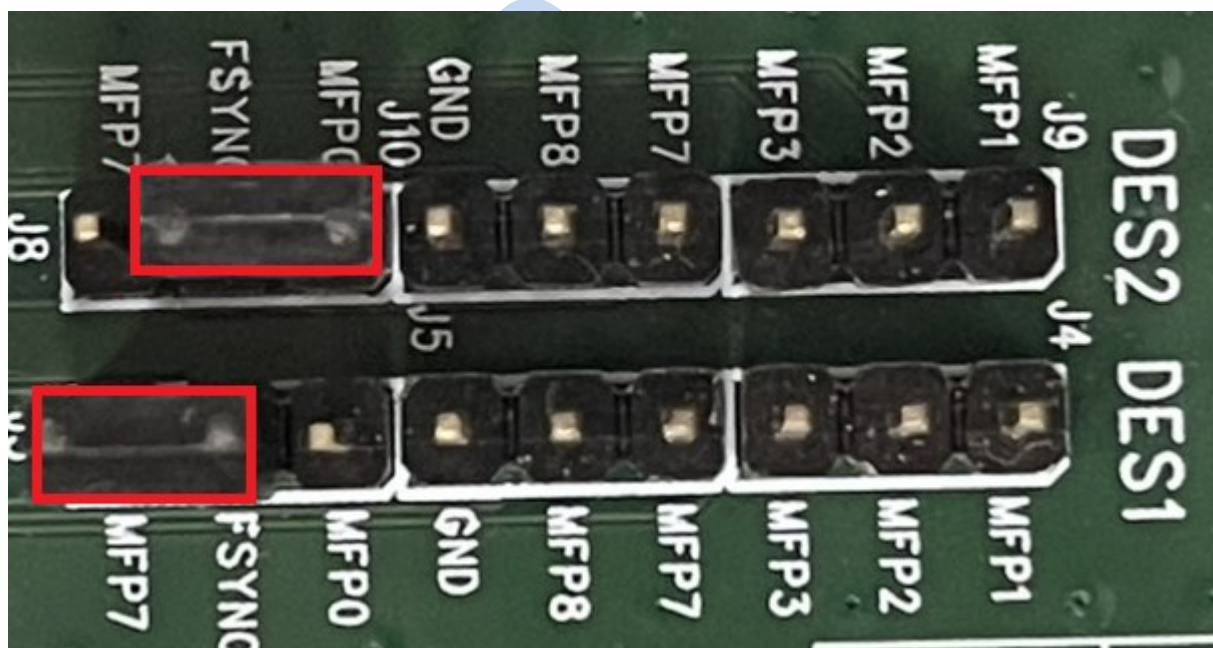
文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	41 of 68	

```
dsera: max96724@27 {
    status = "okay";
    compatible = "otobrite,deserializer";
    reg = <0x27>;
    force-clock = <0x1>;
    csi-mode = "2x4";
    max-src = <4>;
    fsync-gpio = <0x01>;
};
```

The available GPIOs for each deserializer can be found in the table below.

	internal	external
max96712 evk	2,5	0,2,4,5
max96724 evk	0	0,1,2,3,7,8
max96714 evk	0	0,1,2,3,4,6,7
max9296 evk	0	0,7,8
otoadapter NX	0	0,1,2,3
otoadapter AGX	0	0,1,2,3,7,8
TI interposer	3	1

The example of AGX Orin+oToAdapter :



DES2 is configured with internal fsync. The signal is generated from MFP0 and transmitted to FSYN. The

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	42 of 68	

DES2 internal line FSYNC of oToAdapter(m96724*2) is connected to DES1 FSYNC. Therefore, the fsync signal generated from DES2 will be transmitted to DES1's FSYNC, and then to DES1's MFP7 (gpio7). In the device tree, DES1 must be configured as external fsync (from gpio7), and DES2 as internal fsync, with a frequency of 30Hz. **DES2 must be connected to at least one camera (device tree ID: 4~7). Otherwise, internal fsync will fail to output.**

```

dsera: max96724@27 {
    status = "okay";
    compatible = "otobrite,deserializer";
    reg = <0x27>;
    fsync-gpio = <0x7>;
    force-clock = <0x1>;
    csi-mode = "2x4";
    max-src = <4>;
};

dserb: max96724@4e {
    status = "okay";
    compatible = "otobrite,deserializer";
    reg = <0x4e>;
    fsync-gpio = <0xd>;
    force-clock = <0x1>;
    csi-mode = "2x4";
    max-src = <4>;
    fsync-inter-hz = <0x1e>;
};

```

0xd=disable

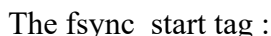
Fsync Start :

About deserializer m96724(oToAdapter)/m96712 , after capturing frames using gstreamer(gst-showall_XXX.sh) or v4l2-ctl(gnome-test-nv16-multi.sh), use "oToAdapter-fsync-internal-start.sh" to start fsync.

Check fsync by "gnome-test-nv16-multi.sh" :

If fsync start, the time stamp of video0.txt ~ videoN.txt will be < 1ms.

頁 數 | 43 of 68



```

496 cap dqbuf: 2 seq: 470 bytesused: 5898240 ts: 26313.047477 delta: 33.333 ms fps: 30.00 (ts-monotonic, ts-src-eof)
497 cap dqbuf: 3 seq: 471 bytesused: 5898240 ts: 26313.080810 delta: 33.333 ms fps: 30.00 (ts-monotonic, ts-src-eof)
498 cap dqbuf: 0 seq: 472 bytesused: 5898240 ts: 26313.114143 delta: 33.333 ms fps: 30.00 (ts-monotonic, ts-src-eof)
499 cap dqbuf: 1 seq: 473 bytesused: 5898240 ts: 26313.147476 delta: 33.333 ms fps: 30.00 (ts-monotonic, ts-src-eof)
500 #####fsync_start
501 cap dqbuf: 2 seq: 474 bytesused: 5898240 ts: 26313.180810 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof)
502 cap dqbuf: 3 seq: 475 bytesused: 5898240 ts: 26313.214143 delta: 33.333 ms fps: 30.00 (ts-monotonic, ts-src-eof)
503 cap dqbuf: 0 seq: 476 bytesused: 5898240 ts: 26313.270371 delta: 56.228 ms fps: 29.91 (ts-monotonic, ts-src-eof)
504 cap dqbuf: 1 seq: 477 bytesused: 5898240 ts: 26313.303704 delta: 33.333 ms fps: 29.91 (ts-monotonic, ts-src-eof)
505 cap dqbuf: 2 seq: 478 bytesused: 5898240 ts: 26313.337037 delta: 33.333 ms fps: 29.91 (ts-monotonic, ts-src-eof)
506 cap dqbuf: 3 seq: 479 bytesused: 5898240 ts: 26313.370370 delta: 33.333 ms fps: 29.91 (ts-monotonic, ts-src-eof)
507 cap dqbuf: 0 seq: 480 bytesused: 5898240 ts: 26313.403704 delta: 33.334 ms fps: 29.91 (ts-monotonic, ts-src-eof)

```

8otocam260isp:



歐特明電子股份有限公司

oToBrite Electronics, Inc.

文件名稱	oToCAM_nVidiaDriver_UserGuide-v2.14.docx	文件編號		版本
		生效日期	2026/3/20	V2.14
		制定部門	System-SW	
		頁數	44 of 68	

video0.txt - 記事本	video1.txt - 記事本	video2.txt - 記事本	video3.txt - 記事本
#####tsync_start##### cap dqbuf: 2 seq: 275 bytesused: 10713600 ts: 102.370185 delta: 33.334 ms fps: 29.92 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 276 bytesused: 10713600 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 277 bytesused: 10713600 ts: 102.436853 delta: 33.334 ms fps: 29.92 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 278 bytesused: 10713600 ts: 102.470187 delta: 33.334 ms fps: 29.92 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 279 bytesused: 10713600 ts: 102.503521 delta: 33.334 ms fps: 29.92 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 280 bytesused: 10713600 ts: 102.536855 delta: 33.334 ms fps: 29.92 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 281 bytesused: 10713600 ts: 102.570189 delta: 33.334 ms fps: 29.92 (ts-monotonic, ts-src-eof)	#####tsync_start##### cap dqbuf: 2 seq: 266 bytesused: 10713600 ts: 102.336829 delta: 33.334 ms fps: 29.85 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 267 bytesused: 10713600 ts: 102.370163 delta: 33.334 ms fps: 29.85 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 268 bytesused: 10713600 ts: 102.403497 delta: 33.334 ms fps: 29.85 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 269 bytesused: 10713600 ts: 102.436830 delta: 33.333 ms fps: 29.85 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 270 bytesused: 10713600 ts: 102.470164 delta: 33.334 ms fps: 29.85 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 271 bytesused: 10713600 ts: 102.503498 delta: 33.334 ms fps: 29.85 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 272 bytesused: 10713600 ts: 102.536832 delta: 33.334 ms fps: 29.85 (ts-monotonic, ts-src-eof)	#####tsync_start##### cap dqbuf: 2 seq: 258 bytesused: 10713600 ts: 102.303526 delta: 33.334 ms fps: 29.95 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 259 bytesused: 10713600 ts: 102.336860 delta: 33.334 ms fps: 29.95 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 260 bytesused: 10713600 ts: 102.370193 delta: 33.333 ms fps: 29.95 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 261 bytesused: 10713600 ts: 102.403527 delta: 33.334 ms fps: 29.95 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 262 bytesused: 10713600 ts: 102.436861 delta: 33.334 ms fps: 29.95 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 263 bytesused: 10713600 ts: 102.470195 delta: 33.334 ms fps: 29.95 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 264 bytesused: 10713600 ts: 102.503529 delta: 33.334 ms fps: 29.95 (ts-monotonic, ts-src-eof)	#####tsync_start##### cap dqbuf: 3 seq: 251 bytesused: 10713600 ts: 102.336851 delta: 33.334 ms fps: 29.95 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 252 bytesused: 10713600 ts: 102.370184 delta: 33.333 ms fps: 29.95 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 253 bytesused: 10713600 ts: 102.403518 delta: 33.334 ms fps: 29.95 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 254 bytesused: 10713600 ts: 102.436852 delta: 33.334 ms fps: 29.95 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 255 bytesused: 10713600 ts: 102.470186 delta: 33.334 ms fps: 29.95 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 256 bytesused: 10713600 ts: 102.503520 delta: 33.334 ms fps: 29.95 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 257 bytesused: 10713600 ts: 102.536854 delta: 33.334 ms fps: 29.95 (ts-monotonic, ts-src-eof)
video4.txt - 記事本	video5.txt - 記事本	video6.txt - 記事本	video7.txt - 記事本
#####tsync_start##### cap dqbuf: 2 seq: 242 bytesused: 10713600 ts: 102.303491 delta: 33.334 ms fps: 29.87 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 243 bytesused: 10713600 ts: 102.336824 delta: 33.333 ms fps: 29.87 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 244 bytesused: 10713600 ts: 102.370158 delta: 33.334 ms fps: 29.87 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 245 bytesused: 10713600 ts: 102.403492 delta: 33.334 ms fps: 29.87 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 246 bytesused: 10713600 ts: 102.436826 delta: 33.334 ms fps: 29.87 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 247 bytesused: 10713600 ts: 102.470160 delta: 33.334 ms fps: 29.87 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 248 bytesused: 10713600 ts: 102.503494 delta: 33.334 ms fps: 29.87 (ts-monotonic, ts-src-eof)	#####tsync_start##### cap dqbuf: 2 seq: 234 bytesused: 10713600 ts: 102.303496 delta: 33.334 ms fps: 29.84 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 235 bytesused: 10713600 ts: 102.336830 delta: 33.334 ms fps: 29.84 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 236 bytesused: 10713600 ts: 102.370164 delta: 33.334 ms fps: 29.84 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 237 bytesused: 10713600 ts: 102.403497 delta: 33.333 ms fps: 29.84 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 238 bytesused: 10713600 ts: 102.436831 delta: 33.334 ms fps: 29.84 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 239 bytesused: 10713600 ts: 102.470165 delta: 33.334 ms fps: 29.84 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 240 bytesused: 10713600 ts: 102.503499 delta: 33.334 ms fps: 29.84 (ts-monotonic, ts-src-eof)	#####tsync_start##### cap dqbuf: 2 seq: 226 bytesused: 10713600 ts: 102.303515 delta: 33.334 ms fps: 29.81 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 227 bytesused: 10713600 ts: 102.336849 delta: 33.334 ms fps: 29.81 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 228 bytesused: 10713600 ts: 102.370183 delta: 33.334 ms fps: 29.81 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 229 bytesused: 10713600 ts: 102.403517 delta: 33.334 ms fps: 29.81 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 230 bytesused: 10713600 ts: 102.436851 delta: 33.334 ms fps: 29.81 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 231 bytesused: 10713600 ts: 102.470184 delta: 33.333 ms fps: 29.81 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 232 bytesused: 10713600 ts: 102.503518 delta: 33.334 ms fps: 29.81 (ts-monotonic, ts-src-eof)	#####tsync_start##### cap dqbuf: 2 seq: 218 bytesused: 10713600 ts: 102.270169 delta: 33.334 ms fps: 29.81 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 219 bytesused: 10713600 ts: 102.303503 delta: 33.334 ms fps: 29.81 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 220 bytesused: 10713600 ts: 102.336837 delta: 33.334 ms fps: 29.81 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 221 bytesused: 10713600 ts: 102.370171 delta: 33.334 ms fps: 29.81 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 222 bytesused: 10713600 ts: 102.403505 delta: 33.334 ms fps: 29.81 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 223 bytesused: 10713600 ts: 102.436838 delta: 33.333 ms fps: 29.81 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 224 bytesused: 10713600 ts: 102.470172 delta: 33.334 ms fps: 29.81 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 225 bytesused: 10713600 ts: 102.503506 delta: 33.334 ms fps: 29.81 (ts-monotonic, ts-src-eof)

AGX Orin+oToAdatper+8otocam223 :

Non-Fsync :

video0.txt - 記事本	video1.txt - 記事本	video2.txt - 記事本	video3.txt - 記事本
(ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 271 bytesused: 5898240 ts: 293.911516 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 272 bytesused: 5898240 ts: 293.944855 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 273 bytesused: 5898240 ts: 293.978194 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 274 bytesused: 5898240 ts: 294.011534 delta: 33.340 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 275 bytesused: 5898240 ts: 294.044873 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 276 bytesused: 5898240 ts: 294.078212 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof)	(ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 269 bytesused: 5898240 ts: 293.931303 delta: 33.339 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 270 bytesused: 5898240 ts: 293.964641 delta: 33.338 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 271 bytesused: 5898240 ts: 293.997980 delta: 33.339 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 272 bytesused: 5898240 ts: 294.031318 delta: 33.338 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 273 bytesused: 5898240 ts: 294.064657 delta: 33.339 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 274 bytesused: 5898240 ts: 294.097996 delta: 33.339 ms fps: 30.00 (ts-monotonic, ts-src-eof)	(ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 263 bytesused: 5898240 ts: 293.923260 delta: 33.339 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 264 bytesused: 5898240 ts: 293.956598 delta: 33.338 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 265 bytesused: 5898240 ts: 293.989937 delta: 33.339 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 266 bytesused: 5898240 ts: 294.023275 delta: 33.338 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 267 bytesused: 5898240 ts: 294.056614 delta: 33.339 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 268 bytesused: 5898240 ts: 294.089953 delta: 33.339 ms fps: 30.00 (ts-monotonic, ts-src-eof)	(ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 254 bytesused: 5898240 ts: 293.907213 delta: 33.339 m (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 255 bytesused: 5898240 ts: 293.940552 delta: 33.339 m (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 256 bytesused: 5898240 ts: 293.973890 delta: 33.338 m (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 257 bytesused: 5898240 ts: 294.007229 delta: 33.339 m (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 258 bytesused: 5898240 ts: 294.040568 delta: 33.339 m (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 259 bytesused: 5898240 ts: 294.073906 delta: 33.338 m (ts-monotonic, ts-src-eof)
video4.txt - 記事本	video5.txt - 記事本	video6.txt - 記事本	video7.txt - 記事本
(ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 241 bytesused: 5898240 ts: 293.904926 delta: 33.339 ms fps: 28.84 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 242 bytesused: 5898240 ts: 293.938264 delta: 33.338 ms fps: 28.84 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 243 bytesused: 5898240 ts: 293.971603 delta: 33.339 ms fps: 28.84 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 244 bytesused: 5898240 ts: 294.004942 delta: 33.339 ms fps: 28.84 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 245 bytesused: 5898240 ts: 294.038280 delta: 33.338 ms fps: 28.84 (ts-monotonic, ts-src-eof)	(ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 235 bytesused: 5898240 ts: 293.925798 delta: 33.339 ms fps: 28.97 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 236 bytesused: 5898240 ts: 293.959137 delta: 33.339 ms fps: 28.97 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 237 bytesused: 5898240 ts: 293.992475 delta: 33.338 ms fps: 28.98 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 238 bytesused: 5898240 ts: 294.025814 delta: 33.339 ms fps: 28.98 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 239 bytesused: 5898240 ts: 294.059153 delta: 33.339 ms fps: 28.99 (ts-monotonic, ts-src-eof)	(ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 229 bytesused: 5898240 ts: 293.915071 delta: 33.338 ms fps: 29.31 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 230 bytesused: 5898240 ts: 293.948410 delta: 33.339 ms fps: 29.31 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 231 bytesused: 5898240 ts: 293.981749 delta: 33.339 ms fps: 29.31 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 232 bytesused: 5898240 ts: 294.015087 delta: 33.338 ms fps: 29.32 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 233 bytesused: 5898240 ts: 294.048426 delta: 33.339 ms fps: 29.32 (ts-monotonic, ts-src-eof)	(ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 220 bytesused: 5898240 ts: 293.903045 delta: 33.339 r (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 221 bytesused: 5898240 ts: 293.936384 delta: 33.339 r (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 222 bytesused: 5898240 ts: 293.969722 delta: 33.338 r (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 223 bytesused: 5898240 ts: 294.003061 delta: 33.339 r (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 224 bytesused: 5898240 ts: 294.036399 delta: 33.338 r (ts-monotonic, ts-src-eof)

Fsync :

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	45 of 68	

文件名稱 oToCAM_nVidiaDriver_UserGuide-v2.14.docx

video0.txt - 記事本 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 810 bytesused: 5898240 ts: 358.922382 delta: 33.340 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 811 bytesused: 5898240 ts: 358.955721 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 812 bytesused: 5898240 ts: 358.989060 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 813 bytesused: 5898240 ts: 359.022404 delta: 33.340 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 814 bytesused: 5898240 ts: 359.055739 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 815 bytesused: 5898240 ts: 359.089078 delta: 33.338 ms fps: 29.99 (ts-monotonic, ts-src-eof) 第 839 列, 第 52 行 100% Unix (LF) UTF-8	video1.txt - 記事本 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 805 bytesused: 5898240 ts: 358.922388 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 806 bytesused: 5898240 ts: 358.955726 delta: 33.338 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 807 bytesused: 5898240 ts: 358.989065 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 808 bytesused: 5898240 ts: 359.022404 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 809 bytesused: 5898240 ts: 359.055742 delta: 33.338 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 810 bytesused: 5898240 ts: 359.089082 delta: 33.340 ms fps: 29.99 (ts-monotonic, ts-src-eof) 第 834 列, 第 52 行 100% Unix (LF) UTF-8	video2.txt - 記事本 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 799 bytesused: 5898240 ts: 358.922397 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 800 bytesused: 5898240 ts: 358.955736 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 801 bytesused: 5898240 ts: 358.989074 delta: 33.338 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 802 bytesused: 5898240 ts: 359.022413 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 803 bytesused: 5898240 ts: 359.055752 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 804 bytesused: 5898240 ts: 359.089090 delta: 33.338 ms fps: 29.99 (ts-monotonic, ts-src-eof) 第 828 列, 第 52 行 100% Unix (LF) UTF-8	video3.txt - 記事本 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 790 bytesused: 5898240 ts: 358.922367 delta: 33.338 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 791 bytesused: 5898240 ts: 358.955706 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 792 bytesused: 5898240 ts: 358.989045 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 793 bytesused: 5898240 ts: 359.022383 delta: 33.338 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 794 bytesused: 5898240 ts: 359.055760 delta: 33.337 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 795 bytesused: 5898240 ts: 359.089099 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) 第 819 列, 第 52 行 100% Unix (LF) UTF-8
video4.txt - 記事本 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 776 bytesused: 5898240 ts: 358.922366 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 777 bytesused: 5898240 ts: 358.955704 delta: 33.338 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 778 bytesused: 5898240 ts: 358.989043 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 779 bytesused: 5898240 ts: 359.022381 delta: 33.338 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 780 bytesused: 5898240 ts: 359.055720 delta: 33.339 ms fps: 29.99 第 839 列, 第 52 行 100% Unix (LF) UTF-8	video5.txt - 記事本 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 772 bytesused: 5898240 ts: 358.922369 delta: 33.338 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 773 bytesused: 5898240 ts: 358.955708 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 774 bytesused: 5898240 ts: 358.989046 delta: 33.338 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 775 bytesused: 5898240 ts: 359.022385 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 776 bytesused: 5898240 ts: 359.055724 delta: 33.339 ms fps: 29.99 第 834 列, 第 52 行 100% Unix (LF) UTF-8	video6.txt - 記事本 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 767 bytesused: 5898240 ts: 358.922356 delta: 33.338 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 768 bytesused: 5898240 ts: 358.955695 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 769 bytesused: 5898240 ts: 358.989033 delta: 33.338 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 770 bytesused: 5898240 ts: 359.022372 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 771 bytesused: 5898240 ts: 359.055711 delta: 33.339 ms fps: 29.99 第 828 列, 第 52 行 100% Unix (LF) UTF-8	video7.txt - 記事本 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 759 bytesused: 5898240 ts: 358.922366 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 760 bytesused: 5898240 ts: 358.955704 delta: 33.338 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 761 bytesused: 5898240 ts: 358.989043 delta: 33.339 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 762 bytesused: 5898240 ts: 359.022381 delta: 33.338 ms fps: 29.99 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 763 bytesused: 5898240 ts: 359.055720 delta: 33.339 ms fps: 29.99 第 819 列, 第 52 行 100% Unix (LF) UTF-8

AGX Orin+oToAdatper+4otocam223(3M)+4otocam260isp(5M) :

Non-Fsync :

video0.txt - 記事本 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 278 bytesused: 10713600 ts: 877.910914 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 279 bytesused: 10713600 ts: 877.944249 delta: 33.335 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 280 bytesused: 10713600 ts: 877.977584 delta: 33.335 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 281 bytesused: 10713600 ts: 878.010918 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 282 bytesused: 10713600 ts: 878.044253 delta: 33.335 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 283 bytesused: 10713600 ts: 878.077587 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 284 bytesused: 10713600 ts: 878.110918 delta: 33.334 ms fps: 30.00 第 307 列, 第 53 行 100% Unix (LF) UTF-8	video1.txt - 記事本 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 271 bytesused: 10713600 ts: 877.929587 delta: 33.335 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 272 bytesused: 10713600 ts: 877.962922 delta: 33.335 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 273 bytesused: 10713600 ts: 877.996256 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 274 bytesused: 10713600 ts: 878.029591 delta: 33.335 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 275 bytesused: 10713600 ts: 878.062925 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 276 bytesused: 10713600 ts: 878.096260 delta: 33.335 ms fps: 30.00 (ts-monotonic, ts-src-eof) 第 300 列, 第 53 行 100% Unix (LF) UTF-8	video2.txt - 記事本 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 262 bytesused: 10713600 ts: 877.913593 delta: 33.335 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 263 bytesused: 10713600 ts: 877.946927 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 264 bytesused: 10713600 ts: 877.980262 delta: 33.335 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 265 bytesused: 10713600 ts: 878.013596 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 266 bytesused: 10713600 ts: 878.046931 delta: 33.335 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 267 bytesused: 10713600 ts: 878.080265 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) 第 291 列, 第 53 行 100% Unix (LF) UTF-8	video3.txt - 記事本 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 255 bytesused: 10713600 ts: 877.904266 delta: 33.335 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 256 bytesused: 10713600 ts: 877.937600 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 257 bytesused: 10713600 ts: 877.970935 delta: 33.335 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 258 bytesused: 10713600 ts: 878.004269 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 259 bytesused: 10713600 ts: 878.037604 delta: 33.335 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 260 bytesused: 10713600 ts: 878.070938 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) 第 284 列, 第 53 行 100% Unix (LF) UTF-8
video4.txt - 記事本 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 258 bytesused: 5898240 ts: 877.918133 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 259 bytesused: 5898240 ts: 877.951467 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 260 bytesused: 5898240 ts: 877.984801 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 261 bytesused: 5898240 ts: 878.018133 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 262 bytesused: 5898240 ts: 878.051469 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 263 bytesused: 5898240 ts: 878.084802 delta: 33.333 ms fps: 30.00 (ts-monotonic, ts-src-eof) 第 287 列, 第 52 行 100% Unix (LF) UTF-8	video5.txt - 記事本 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 253 bytesused: 5898240 ts: 877.913225 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 254 bytesused: 5898240 ts: 877.946559 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 255 bytesused: 5898240 ts: 877.979893 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 256 bytesused: 5898240 ts: 878.013227 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 257 bytesused: 5898240 ts: 878.046561 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 258 bytesused: 5898240 ts: 878.079894 delta: 33.333 ms fps: 30.00 (ts-monotonic, ts-src-eof) 第 282 列, 第 52 行 100% Unix (LF) UTF-8	video6.txt - 記事本 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 247 bytesused: 5898240 ts: 877.900820 delta: 33.333 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 248 bytesused: 5898240 ts: 877.934154 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 249 bytesused: 5898240 ts: 877.967488 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 250 bytesused: 5898240 ts: 878.000822 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 251 bytesused: 5898240 ts: 878.034156 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 252 bytesused: 5898240 ts: 878.067489 delta: 33.333 ms fps: 30.00 (ts-monotonic, ts-src-eof) 第 276 列, 第 52 行 100% Unix (LF) UTF-8	video7.txt - 記事本 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 240 bytesused: 5898240 ts: 877.923493 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 241 bytesused: 5898240 ts: 877.956827 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 2 seq: 242 bytesused: 5898240 ts: 877.990161 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 3 seq: 243 bytesused: 5898240 ts: 878.023495 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 0 seq: 244 bytesused: 5898240 ts: 878.056829 delta: 33.334 ms fps: 30.00 (ts-monotonic, ts-src-eof) cap dqbuf: 1 seq: 245 bytesused: 5898240 ts: 878.090162 delta: 33.333 ms fps: 30.00 (ts-monotonic, ts-src-eof) 第 269 列, 第 52 行 100% Unix (LF) UTF-8

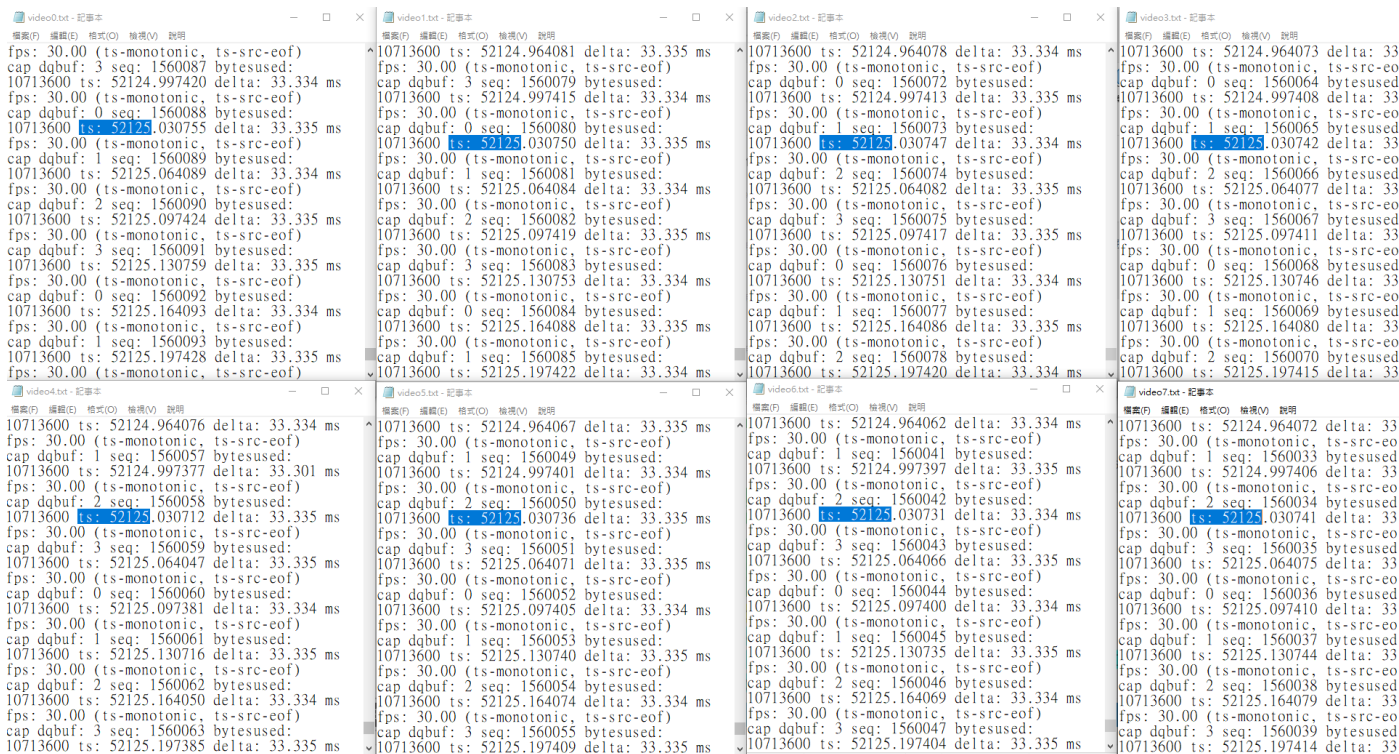
Fsync :

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	46 of 68	

文件名稱	oToCAM_nVidiaDriver_UserGuide-v2.14.docx
------	--

AGX Orin+oToAdatper+8otocam260isp(5M) 12hr. test:

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	47 of 68	



2.2. Reconfigure and Patch Linux Kernel

For JetPack 6.x, there's no need to modify kernel config since max9296, max9295, nv_imx390 are not built into the kernel. Simply replace max9296.ko, max9295.ko, nv_imx390.ko with oToBrite's version.

2.3. Update DTB

Install the DTB to target Jetson DevKit. Below are relevant path and files:

```

/boot/
├── Image
├── tegra234-p3737-0000+p3701-0004-nv.dtb    (JetPack6.0 AGX-Orin)
├── tegra234-p3701-0004-p3737-0000.dtb      (JetPack5.x AGX-Orin)
├── tegra194-p2888-0001-p2822-0000.dtb      (JetPack5.x AGX-Xavier)
└── extlinux/
    └── extlinux.conf

```


文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	48 of 68	

3. Driver Testing

3.1. Load Driver Kernel Module

Driver

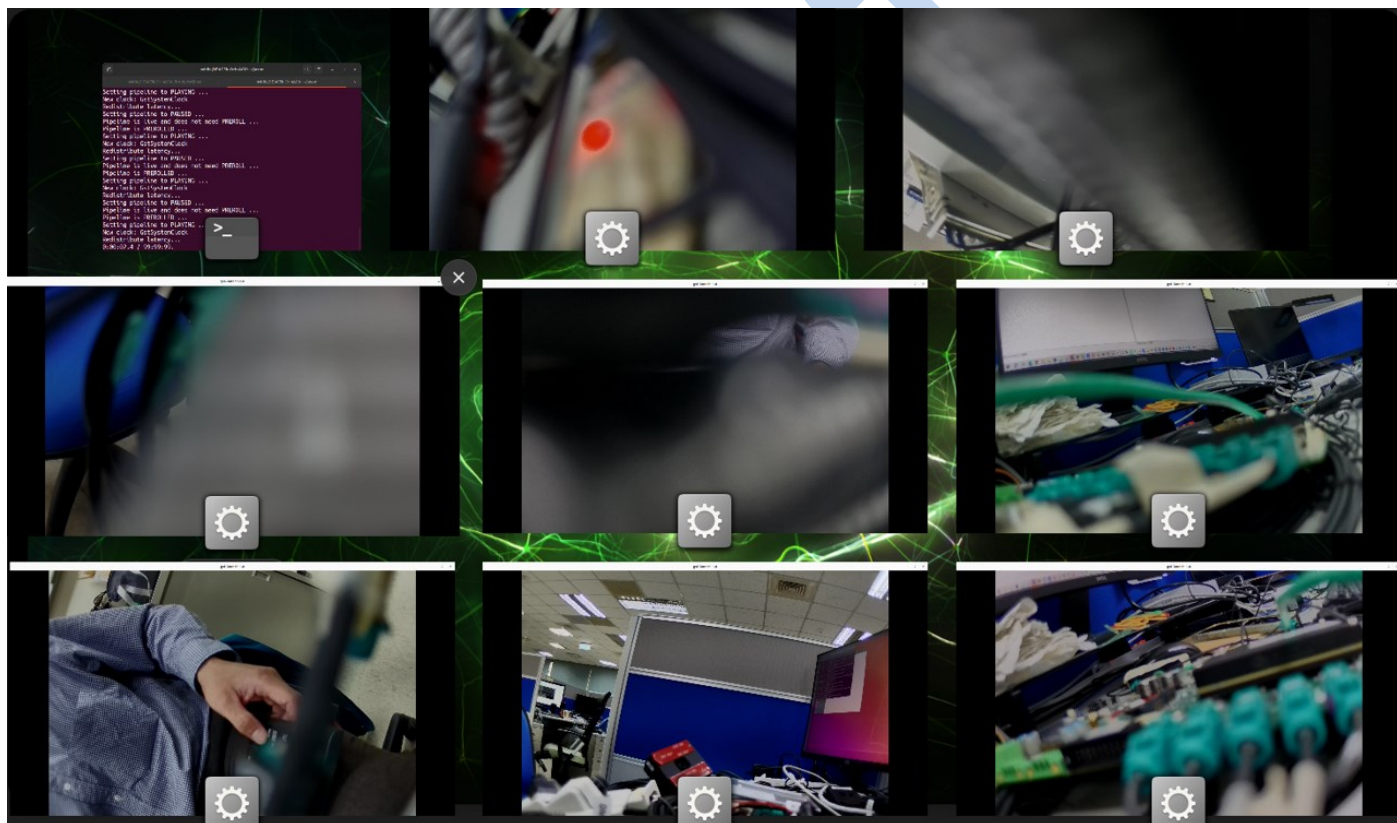
Kernel modules can be manually loaded as below:

```
$ sudo insmod oToSerDes.ko
$ sudo insmod oToCAM.ko
```

3.2. Run GStreamer to display

```
gst-launch-1.0 v4l2src device=/dev/video$1 ! videoconvert ! xvimagesink
```

```
gst-launch-1.0 -e v4l2src device=/dev/video$i ! fpsdisplaysink
```



文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	49 of 68	

3.3. Run V4L2-utils or other tool

1. v4l2-ctl command can activate video streaming, but don't display video on screen. Below is command example:

check frame rate :

```
v4l2-ctl -d /dev/video0 --set-fmt-video=pixelformat=NV16 --stream-mmap --stream-count=300
```

```
v4l2-ctl -d /dev/video0 --set-fmt-video=pixelformat=UYVY --stream-mmap --stream-count=300
```

get image/raw data:

```
v4l2-ctl -d /dev/video0 --set-fmt-video=pixelformat=NV16 --stream-mmap --stream-count=1 --stream-to=nv16-video0.raw
```

```
v4l2-ctl -d /dev/video0 --set-fmt-video=pixelformat=UYVY --stream-mmap --stream-count=1 --stream-to=uyvy-video0.raw
```

```
v4l2-ctl -d /dev/video0 --set-fmt-video=pixelformat=RG12 --stream-mmap --stream-count=1 --stream-to=rg12-video0.raw
```

```
$ v4l2-ctl -d /dev/video0 --set-fmt-video=pixelformat=NV16 --stream-mmap --stream-count=300  
<<<<<<<<<<<<<<<<<<<<<<<<< 30.00 fps  
<<<<<<<<<<<<<<<<<<<<<<<<< 29.80 fps  
<<<<<<<<
```

2. `gnome-test-nv16-multi.sh` / `gnome-test-nv16-multi-512k.sh`(log size to 512 KB) on tool folder for frame rate monitor (output log to `videoN.txt`).

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	50 of 68	

```

video0
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 1 seq: 1017 bytesused: 5898240 ts: 7993.432446 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 2 seq: 1018 bytesused: 5898240 ts: 7993.465778 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 3 seq: 1019 bytesused: 5898240 ts: 7993.499111 delta: 33.333 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 0 seq: 1020 bytesused: 5898240 ts: 7993.532443 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 1 seq: 1021 bytesused: 5898240 ts: 7993.565776 delta: 33.333 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 2 seq: 1022 bytesused: 5898240 ts: 7993.599108 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 3 seq: 1023 bytesused: 5898240 ts: 7993.632440 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 0 seq: 1024 bytesused: 5898240 ts: 7993.665773 delta: 33.333 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 1 seq: 1025 bytesused: 5898240 ts: 7993.699105 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 2 seq: 1026 bytesused: 5898240 ts: 7993.732437 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 3 seq: 1027 bytesused: 5898240 ts: 7993.765770 delta: 33.333 ms fps
: 30.00 (ts-monotonic, ts-src-eof)

video1
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 2 seq: 1006 bytesused: 5898240 ts: 7993.391156 delta: 33.333 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 3 seq: 1007 bytesused: 5898240 ts: 7993.424488 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 0 seq: 1008 bytesused: 5898240 ts: 7993.457820 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 1 seq: 1009 bytesused: 5898240 ts: 7993.491153 delta: 33.333 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 2 seq: 1010 bytesused: 5898240 ts: 7993.524485 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 3 seq: 1011 bytesused: 5898240 ts: 7993.557817 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 0 seq: 1012 bytesused: 5898240 ts: 7993.591150 delta: 33.333 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 1 seq: 1013 bytesused: 5898240 ts: 7993.624482 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 2 seq: 1014 bytesused: 5898240 ts: 7993.657815 delta: 33.333 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 3 seq: 1015 bytesused: 5898240 ts: 7993.691147 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 0 seq: 1016 bytesused: 5898240 ts: 7993.724479 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)

video2
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 3 seq: 999 bytesused: 5898240 ts: 7993.451660 delta: 33.333 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 0 seq: 1000 bytesused: 5898240 ts: 7993.484992 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 1 seq: 1001 bytesused: 5898240 ts: 7993.518324 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 2 seq: 1002 bytesused: 5898240 ts: 7993.551657 delta: 33.333 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 3 seq: 1003 bytesused: 5898240 ts: 7993.584989 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 0 seq: 1004 bytesused: 5898240 ts: 7993.618322 delta: 33.333 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 1 seq: 1005 bytesused: 5898240 ts: 7993.651654 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 2 seq: 1006 bytesused: 5898240 ts: 7993.684986 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 3 seq: 1007 bytesused: 5898240 ts: 7993.718319 delta: 33.333 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 0 seq: 1008 bytesused: 5898240 ts: 7993.751651 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 1 seq: 1009 bytesused: 5898240 ts: 7993.784983 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)

video3
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 2 seq: 986 bytesused: 5898240 ts: 7993.393441 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 3 seq: 987 bytesused: 5898240 ts: 7993.426774 delta: 33.333 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 0 seq: 988 bytesused: 5898240 ts: 7993.460106 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 1 seq: 989 bytesused: 5898240 ts: 7993.493438 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 2 seq: 990 bytesused: 5898240 ts: 7993.526771 delta: 33.333 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 3 seq: 991 bytesused: 5898240 ts: 7993.560103 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 0 seq: 992 bytesused: 5898240 ts: 7993.593436 delta: 33.333 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 1 seq: 993 bytesused: 5898240 ts: 7993.626768 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 2 seq: 994 bytesused: 5898240 ts: 7993.660100 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 3 seq: 995 bytesused: 5898240 ts: 7993.693433 delta: 33.333 ms fps
: 30.00 (ts-monotonic, ts-src-eof)
cap dqbuf: 0 seq: 996 bytesused: 5898240 ts: 7993.726765 delta: 33.332 ms fps
: 30.00 (ts-monotonic, ts-src-eof)

```

abnormal string of frame drop : “dropped”

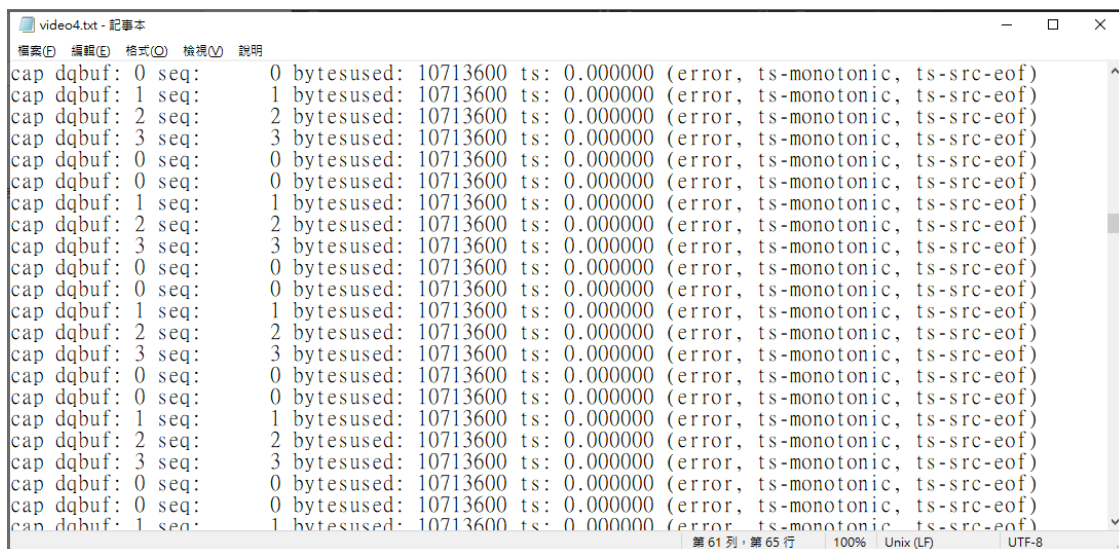
```

cap dqbuf: 0 seq: 0 bytesused: 10713600 ts: 156.734801 (ts-monotonic, ts-src-eof)
cap dqbuf: 1 seq: 1 bytesused: 10713600 ts: 156.768135 delta: 33.334 ms (error, ts-monotonic, ts-src-eof)
cap dqbuf: 2 seq: 2 bytesused: 10713600 ts: 156.868137 delta: 100.002 ms dropped: 1 (ts-monotonic, ts-src-eof)
cap dqbuf: 3 seq: 3 bytesused: 10713600 ts: 156.901471 delta: 33.334 ms (ts-monotonic, ts-src-eof)
cap dqbuf: 0 seq: 4 bytesused: 10713600 ts: 156.934804 delta: 33.333 ms (ts-monotonic, ts-src-eof)
cap dqbuf: 1 seq: 5 bytesused: 10713600 ts: 156.968138 delta: 33.334 ms fps: 21.43 (ts-monotonic, ts-src-eof)

```

abnormal string of streaming error :

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	51 of 68	



3. check bit loss by “tt-otoadapter_serdes_info.sh” or “tt-otoadapter_serdes_info_repeat.sh”.

normal :

```
nvidia@ubuntu:~/tools_2.4.5$ ./tt-otoadapter_serdes_info.sh
0x27 dser DEV_ID: 0xa2
linkA is link locked.
linkB is link locked.
linkC is link locked.
linkD is link locked.
linkA is video locked.
linkB is video locked.
linkC is video locked.
linkD is video locked.
0x4e dser DEV_ID: 0xa2
linkA is link locked.
linkB is link locked.
linkC is link locked.
linkD is link locked.
linkA is video locked.
linkB is video locked.
linkC is video locked.
linkD is video locked.
```

abnormal :

(1)error of linkB unlocked

```
0x27 dser DEV_ID: 0xa2
linkA is link locked.
linkB is link locked.
linkC is link locked.
linkD is link locked.
0x4e dser DEV_ID: 0xa2
linkA is link locked.
linkC is link locked.
linkD is link locked.
```

(2)error of linkB video streaming

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	52 of 68	

文件名稱	oToCAM_nVidiaDriver_UserGuide-v2.14.docx
------	--

```
0x27 dser DEV_ID: 0xa2
linkA is link locked.
linkB is link locked.
linkC is link locked.
linkD is link locked.
linkA is video locked.
linkB is video locked.
linkC is video locked.
linkD is video locked.
0x4e dser DEV_ID: 0xa2
linkA is link locked.
linkB is link locked.
linkC is link locked.
linkD is link locked.
linkA is video locked.
linkC is video locked.
linkD is video locked.
```

OToBrite Confidential

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	53 of 68	

4. Feedback

When issues arise during integration and testing, please feedback oToBrite with below information:

- dmesg logs for module insertion, and video streaming.
- check camera PoC/power supply, FAKRA/SerDes cable connection.
- check whether video nodes exist:
\$ ls /dev/video*
- check v4l2 sub-device nodes binding:
\$ media-ctl -p -d /dev/media0
- check video settings:
\$ gst-device-monitor-1.0 /dev/videoN

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	54 of 68	

5. Reference

5.1. Nvidia Development Documents

<https://developer.nvidia.com/embedded/jetson-linux> Jetson Linux Driver Package

<https://docs.nvidia.com/jetson/archives/r35.4.1/DeveloperGuide/index.html> Camera Development

5.2. Nvidia Carrier Board Specification

https://developer.nvidia.com/embedded/dlc/Jetson_AGX_Xavier_Developer_Kit_Carrier_Board_Specification

5.3. Nvidia Xavier PinMux Table

<https://developer.nvidia.com/jetson-xavier-nx-pinmux-configuration-template-v106>

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	55 of 68	

Appendix A. Device tree programming

device tree programming example (AGX Orin):

driver version: 0.04.05 ◦

Dser use 2x4 config ◦

Sensor is otocam222 ◦

Step 1. Dser : check i2c address to fill the following block.

```
bus@0 {
    i2c@3180000 {
        max9296_a@48 {
            status = "okay";
            compatible = "maxim,max9296";
            reg = <0x6b>;
            csi-mode = "2x4";
            max-src = <0x04>;
            reset-gpios = <0xffffffff 0x3b 0x00>;
            fsync-gpio = <0x00>;
            phandle = <0x01>;
        };
    };
};
```

- two Dser, one for 3180000 , another for 31e0000 ◦
- “reg” : Dser i2c address ◦
- “csi-mode” 、 “ max-src” : set Dser config , it is 2x4 ◦
- “fsync-gpio” : for fsync , not used ◦

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	56 of 68	

Step 2. Ser : write to the same area with Dser.

```
max9295_a@40 {
    status = "okay";
    compatible = "maxim,max9295";
    reg = <0x18>;
    nvidia,gmsl-dser-device = <0x01>;
    phandle = <0x02>;
};
```

- The default i2c address of 9295A is 0x62 , after insert driver, it will be remapped to the assigned address of reg .
- The tree will describe which Dser the Ser is connected to. The value of "nvidia,gmsl-dser-device" is a pointer to the "phandle" of the Dser. .

Step 3. Cam : Usually written in the same section as the corresponding Ser and Dser.

```
imx390@10 {
    status = "okay";
    reg = <0x10>;
    def-addr = <0x1a>;
    clocks = <0xffffffff 0x24 0xffffffff 0x24>;
    clock-names = "extperiph1\0pll_p_grtba";
    mclk = "extperiph1";
    nvidia,gmsl-ser-device = <0x02>;
    nvidia,gmsl-dser-device = <0x01>;
    devnode = "video0";
    compatible = "sony,imx390";
    physical_w = "15.0";
    physical_h = "12.5";
    sensor_model = "isx021";
    post_crop_frame_drop = "0";
    use_decibel_gain = "true";
    use_sensor_mode_id = "true";
    phandle = <0x2b>;
};
```

- The default i2c address of Isx021 is 0x1a, and it will be remapped to the address specified by "reg" after the driver is installed. .
- The tree will describe which Ser and Dser the Cam is connected to, "nvidia,gmsl-ser-device", "nvidia,gmsl-dser-device".
- "Sensor_model" refers to the sensor model used.

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	57 of 68	

- Basic information of Cam, which is used by the mainboard program (v4l2, gstreamer) to parse the received data.
- This is the setting of otocam222.

```
mode0 {
    mclk_khz = "24000";
    num_lanes = "4";
    tegra_sinterface = "serial_a";
    vc_id = "0";
    discontinuous_clk = "no";
    dpcm_enable = "false";
    cil_settletime = "0";
    mode_type = "yuv";
    pixel_phase = "uyvy";
    csi_pixel_bit_depth = "16";
    dynamic_pixel_bit_depth = "16";
    active_w = "1920";
    active_h = "1280";
    readout_orientation = "0";
    line_length = "2200";
    inherent_gain = "1";
    pix_clk_hz = "375000000";
    gain_factor = "10";
    min_gain_val = "0";
    max_gain_val = "300";
    step_gain_val = "3";
    default_gain = "0";
    min_hdr_ratio = "1";
    max_hdr_ratio = "1";
    framerate_factor = "1000000";
    min_framerate = "30000000";
    max_framerate = "30000000";
    step_framerate = "1";
    default_framerate = "30000000";
    exposure_factor = "1000000";
    min_exp_time = "59";
    max_exp_time = "33333";
    step_exp_time = "1";
    default_exp_time = "33333";
    embedded_metadata_height = "0";
};
```

- Differences between otocam222 & otocam223

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	58 of 68	

文件名稱 oToCAM_nVidiaDriver_UserGuide-v2.14.docx

<pre> 48 clocks = <0xffffffff 0x24 0xffffffff 0x24>; 49 clock-names = "extperiph1@p1lp_gtrba"; 50 mclk = "extperiph1"; 51 nvidia,gsml-ser-device = <0x02>; 52 nvidia,gsml-dser-device = <0x01>; 53 devmode = "video0"; 54 compatible = "otobrite,otocam"; 55 physical_w = "15.0"; 56 physical_h = "12.5"; 57 sensor_model = "isx081"; 58 post_crop_frame_drop = "0"; 59 use_decibel_gain = "true"; 60 use_sensor_mode_id = "true"; 61 phandle = <0x2b>; 62 63 mode0 { 64 mclk_khz = "24000"; 65 num_lanes = "4"; 66 tegra_sinterface = "serial_a"; 67 vc_id = "0"; 68 discontinuous_clk = "no"; 69 dpcn_enable = "false"; 70 cll_settletime = "0"; 71 mode_type = "yuv"; 72 pixel_phase = "uyvy"; 73 csi_pixel_bit_depth = "16"; 74 dynamic_pixel_bit_depth = "16"; 75 active_w = "1920"; 76 active_h = "1280"; 77 readout_orientation = "0"; 78 line_length = "2200"; 79 inherent_gain = "1"; 80 pix_clk_hz = "375000000"; 81 gain_factor = "10"; 82 min_gain_val = "0"; </pre>	<pre> 48 clocks = <0xffffffff 0x24 0xffffffff 0x24>; 49 clock-names = "extperiph1@p1lp_gtrba"; 50 mclk = "extperiph1"; 51 nvidia,gsml-ser-device = <0x02>; 52 nvidia,gsml-dser-device = <0x01>; 53 devmode = "video0"; 54 compatible = "otobrite,otocam"; 55 physical_w = "15.0"; 56 physical_h = "12.5"; 57 sensor_model = "isx081"; 58 post_crop_frame_drop = "0"; 59 use_decibel_gain = "true"; 60 use_sensor_mode_id = "true"; 61 phandle = <0x2b>; 62 63 mode0 { 64 mclk_khz = "24000"; 65 num_lanes = "4"; 66 tegra_sinterface = "serial_a"; 67 vc_id = "0"; 68 discontinuous_clk = "no"; 69 dpcn_enable = "false"; 70 cll_settletime = "0"; 71 mode_type = "yuv"; 72 pixel_phase = "uyvy"; 73 csi_pixel_bit_depth = "16"; 74 dynamic_pixel_bit_depth = "16"; 75 active_w = "1920"; 76 active_h = "1280"; 77 readout_orientation = "0"; 78 line_length = "2200"; 79 inherent_gain = "1"; 80 pix_clk_hz = "375000000"; 81 gain_factor = "10"; 82 min_gain_val = "0"; </pre>
--	--

● Differences between otocam222 & otocam260isp

<pre> 50 mclk = "extperiph1"; 51 nvidia,gsml-ser-device = <0x02>; 52 nvidia,gsml-dser-device = <0x01>; 53 devmode = "video0"; 54 compatible = "otobrite,otocam"; 55 physical_w = "15.0"; 56 physical_h = "12.5"; 57 sensor_model = "isx081"; 58 post_crop_frame_drop = "0"; 59 use_decibel_gain = "true"; 60 use_sensor_mode_id = "true"; 61 phandle = <0x2b>; 62 63 mode0 { 64 mclk_khz = "24000"; 65 num_lanes = "4"; 66 tegra_sinterface = "serial_a"; 67 vc_id = "0"; 68 discontinuous_clk = "no"; 69 dpcn_enable = "false"; 70 cll_settletime = "0"; 71 mode_type = "yuv"; 72 pixel_phase = "uyvy"; 73 csi_pixel_bit_depth = "16"; 74 dynamic_pixel_bit_depth = "16"; 75 active_w = "1920"; 76 active_h = "1280"; 77 readout_orientation = "0"; 78 line_length = "2200"; 79 inherent_gain = "1"; 80 pix_clk_hz = "375000000"; 81 gain_factor = "10"; 82 min_gain_val = "0"; 83 max_gain_val = "300"; 84 step_gain_val = "3"; 85 default_gain = "0"; 86 min_hdr_ratio = "1"; </pre>	<pre> 50 mclk = "extperiph1"; 51 nvidia,gsml-ser-device = <0x02>; 52 nvidia,gsml-dser-device = <0x01>; 53 devmode = "video0"; 54 compatible = "otobrite,otocam"; 55 physical_w = "15.0"; 56 physical_h = "12.5"; 57 sensor_model = "isx081isp"; 58 post_crop_frame_drop = "0"; 59 use_decibel_gain = "true"; 60 use_sensor_mode_id = "true"; 61 phandle = <0x2b>; 62 63 mode0 { 64 mclk_khz = "24000"; 65 num_lanes = "4"; 66 tegra_sinterface = "serial_a"; 67 vc_id = "0"; 68 discontinuous_clk = "no"; 69 dpcn_enable = "false"; 70 cll_settletime = "0"; 71 mode_type = "yuv"; 72 pixel_phase = "uyvy"; 73 csi_pixel_bit_depth = "16"; 74 dynamic_pixel_bit_depth = "16"; 75 active_w = "2880"; 76 active_h = "1800"; 77 readout_orientation = "0"; 78 line_length = "3200"; 79 inherent_gain = "1"; 80 pix_clk_hz = "375000000"; 81 gain_factor = "10"; 82 min_gain_val = "0"; 83 max_gain_val = "300"; 84 step_gain_val = "3"; 85 default_gain = "0"; 86 min_hdr_ratio = "1"; </pre>
---	--

● Differences between otocam222 & otocam264isp

<pre> 53 devmode = "video0"; 54 compatible = "otobrite,otocam"; 55 physical_w = "15.0"; 56 physical_h = "12.5"; 57 sensor_model = "isx081"; 58 post_crop_frame_drop = "0"; 59 use_decibel_gain = "true"; 60 use_sensor_mode_id = "true"; 61 phandle = <0x2b>; 62 63 mode0 { 64 mclk_khz = "24000"; 65 num_lanes = "4"; 66 tegra_sinterface = "serial_a"; 67 vc_id = "0"; 68 discontinuous_clk = "no"; 69 dpcn_enable = "false"; 70 cll_settletime = "0"; 71 mode_type = "yuv"; 72 pixel_phase = "uyvy"; 73 csi_pixel_bit_depth = "16"; 74 dynamic_pixel_bit_depth = "16"; 75 active_w = "1920"; 76 active_h = "1280"; 77 readout_orientation = "0"; 78 line_length = "2200"; 79 inherent_gain = "1"; 80 pix_clk_hz = "375000000"; </pre>	<pre> 53 devmode = "video0"; 54 compatible = "otobrite,otocam"; 55 physical_w = "15.0"; 56 physical_h = "12.5"; 57 sensor_model = "isx081isp"; 58 post_crop_frame_drop = "0"; 59 use_decibel_gain = "true"; 60 use_sensor_mode_id = "true"; 61 phandle = <0x2b>; 62 63 mode0 { 64 mclk_khz = "24000"; 65 num_lanes = "4"; 66 tegra_sinterface = "serial_a"; 67 vc_id = "0"; 68 discontinuous_clk = "no"; 69 dpcn_enable = "false"; 70 cll_settletime = "0"; 71 mode_type = "yuv"; 72 pixel_phase = "uyvy"; 73 csi_pixel_bit_depth = "16"; 74 dynamic_pixel_bit_depth = "16"; 75 active_w = "1920"; 76 active_h = "1280"; 77 readout_orientation = "0"; 78 line_length = "2200"; 79 inherent_gain = "1"; 80 pix_clk_hz = "375000000"; </pre>
--	---

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	59 of 68	

文件名稱 oToCAM_nVidiaDriver_UserGuide-v2.14.docx

● Differences between otocam222 & otocam271isp(max_lane_speed = 2.5G)

<pre> 34 compatible = "uc100rite,otocam"; 55 physical_w = "15.0"; 56 physical_h = "12.5"; 57 sensor_model = "isx021"; 58 post_crop_frame_drop = "0"; 59 use_decibel_gain = "true"; 60 use_sensor_mode_id = "true"; 61 phandle = <0x2b>; 62 63 mode0 { 64 mclk_khz = "24000"; 65 num_lanes = "4"; 66 tegra_sinterface = "serial_a"; 67 vc_id = "0"; 68 discontinuous_clk = "no"; 69 dpcm_enable = "false"; 70 cil_settletime = "0"; 71 mode_type = "yuv"; 72 pixel_phase = "uyvy"; 73 csi_pixel_bit_depth = "16"; 74 dynamic_pixel_bit_depth = "16"; 75 active_w = "1920"; 76 active_h = "1280"; 77 readout_orientation = "0"; 78 line_length = "2200"; 79 inherent_gain = "1"; 80 pix_clk_hz = "375000000"; 81 gain_factor = "10"; 82 min_gain_val = "0"; </pre>	<pre> 34 compatible = "uc100rite,otocam"; 55 physical_w = "15.0"; 56 physical_h = "12.5"; 57 sensor_model = "imx728isp"; 58 post_crop_frame_drop = "0"; 59 use_decibel_gain = "true"; 60 use_sensor_mode_id = "true"; 61 phandle = <0x2b>; 62 63 mode0 { 64 mclk_khz = "24000"; 65 num_lanes = "4"; 66 tegra_sinterface = "serial_a"; 67 vc_id = "0"; 68 discontinuous_clk = "no"; 69 dpcm_enable = "false"; 70 cil_settletime = "0"; 71 mode_type = "yuv"; 72 pixel_phase = "uyvy"; 73 csi_pixel_bit_depth = "16"; 74 dynamic_pixel_bit_depth = "16"; 75 active_w = "3840"; 76 active_h = "2160"; 77 readout_orientation = "0"; 78 line_length = "4080"; 79 inherent_gain = "1"; 80 pix_clk_hz = "625000000"; 81 gain_factor = "10"; 82 min_gain_val = "0"; </pre>
<pre> 330 }; 331 332 fragment-camera@13 { 333 target-path = "/"; 334 335 __overlay__ { 336 337 tegra-camera-platform { 338 compatible = "nvidia, tegra-camera-platform"; 339 num_csi_lanes = <0x04>; 340 max_lane_speed = <0x166360>; 341 min_bits_per_pixel = <0x0a>; 342 vi_peak_byte_per_pixel = <0x02>; 343 vi_bw_margin_pct = <0x10>; 344 isp_peak_byte_per_pixel = <0x05>; 345 isp_bw_margin_pct = <0x19>; 346 phandle = <0x31>; 347 </pre>	<pre> 330 }; 331 332 fragment-camera@13 { 333 target-path = "/"; 334 335 __overlay__ { 336 337 tegra-camera-platform { 338 compatible = "nvidia, tegra-camera-platform"; 339 num_csi_lanes = <0x04>; 340 max_lane_speed = <0x262540>; 341 min_bits_per_pixel = <0x0a>; 342 vi_peak_byte_per_pixel = <0x02>; 343 vi_bw_margin_pct = <0x10>; 344 isp_peak_byte_per_pixel = <0x05>; 345 isp_bw_margin_pct = <0x19>; 346 phandle = <0x31>; 347 </pre>

● Differences between otocam222 & otocam264raw

<pre> 3258 physical_h = "12.5"; 3259 sensor_model = "isx021"; 3260 post_crop_frame_drop = "0"; 3261 use_decibel_gain = "true"; 3262 use_sensor_mode_id = "true"; 3263 phandle = <0x24f>; 3264 3265 mode0 { 3266 mclk_khz = "24000"; 3267 num_lanes = "4"; 3268 tegra_sinterface = "serial_a"; 3269 vc_id = "0"; 3270 discontinuous_clk = "no"; 3271 dpcm_enable = "false"; 3272 cil_settletime = "0"; 3273 mode_type = "yuv"; 3274 pixel_phase = "uyvy"; 3275 csi_pixel_bit_depth = "16"; 3276 dynamic_pixel_bit_depth = "16"; 3277 active_w = "1920"; 3278 active_h = "1280"; 3279 readout_orientation = "0"; 3280 line_length = "2200"; 3281 inherent_gain = "1"; 3282 pix_clk_hz = "375000000"; 3283 gain_factor = "10"; 3284 min_gain_val = "0"; 3285 max_gain_val = "300"; 3286 step_gain_val = "3"; 3287 default_gain = "0"; 3288 min_hdr_ratio = "1"; 3289 max_hdr_ratio = "1"; 3290 framerate_factor = "1000000"; 3291 min_framerate = "30000000"; 3292 max_framerate = "30000000"; 3293 step_framerate = "1"; 3294 default_framerate = "30000000"; 3295 exposure_factor = "1000000"; 3296 min_exp_time = "59"; 3297 max_exp_time = "33333"; 3298 step_exp_time = "1"; 3299 default_exp_time = "33333"; 3300 embedded_metadata_height = "0"; 3301 lane_polarity = "6"; 3302 }; 3303 3304 gmsl-link { 3305 streams = "yuv0@ued-ul"; 3306 src-csi-port = "b"; 3307 dst-csi-port = "a"; 3308 serdes-csi-link = "a"; </pre>	<pre> 3258 physical_h = "12.5"; 3259 sensor_model = "imx390"; 3260 post_crop_frame_drop = "0"; 3261 use_decibel_gain = "true"; 3262 use_sensor_mode_id = "true"; 3263 phandle = <0x24f>; 3264 3265 mode0 { 3266 mclk_khz = "24000"; 3267 num_lanes = "4"; 3268 tegra_sinterface = "serial_a"; 3269 vc_id = "0"; 3270 discontinuous_clk = "no"; 3271 dpcm_enable = "false"; 3272 cil_settletime = "0"; 3273 mode_type = "bayer"; 3274 pixel_phase = "rggb"; 3275 csi_pixel_bit_depth = "12"; 3276 dynamic_pixel_bit_depth = "12"; 3277 active_w = "1920"; 3278 active_h = "1080"; 3279 readout_orientation = "0"; 3280 line_length = "2200"; 3281 inherent_gain = "1"; 3282 pix_clk_hz = "625000000"; 3283 gain_factor = "10"; 3284 min_gain_val = "0"; 3285 max_gain_val = "300"; 3286 step_gain_val = "3"; 3287 default_gain = "0"; 3288 min_hdr_ratio = "1"; 3289 max_hdr_ratio = "1"; 3290 framerate_factor = "1000000"; 3291 min_framerate = "30000000"; 3292 max_framerate = "30000000"; 3293 step_framerate = "1"; 3294 default_framerate = "30000000"; 3295 exposure_factor = "1000000"; 3296 min_exp_time = "59"; 3297 max_exp_time = "33333"; 3298 step_exp_time = "1"; 3299 default_exp_time = "33333"; 3300 embedded_metadata_height = "0"; 3301 lane_polarity = "6"; 3302 }; 3303 3304 gmsl-link { 3305 streams = "raw12@ued-ul"; 3306 src-csi-port = "b"; 3307 dst-csi-port = "a"; 3308 serdes-csi-link = "a"; </pre>
--	---

		文件編號		版本
		生效日期	2026/3/20	V2.14
		制定部門	System-SW	
文件名稱	oToCAM_nVidiaDriver_UserGuide-v2.14.docx		頁 數	60 of 68

gmsl-link describes that Cam enters Ser through port B (src-csi-port) in csi-lane 1x4 (csi-mode), and Ser then enters Dser through gmsl LinkA (serdes-csi-link), the packet format (streams), Virtual-channel (st-vc, vc-id) along the way, and then goes out from Port A (dst-csi-port) of Dser to the main board.

```
gmsl-link {  
    streams = "yuv8\0ued-u1";  
    src-csi-port = "b";  
    dst-csi-port = "a";  
    serdes-csi-link = "a";  
    csi-mode = "1x4";  
    st-vc = <0x00>;  
    vc-id = <0x00>;  
    num-lanes = <0x04>;  
};
```

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	61 of 68	

文件名稱 oToCAM_nVidiaDriver_UserGuide-v2.14.docx

```
ports {  
    #address-cells = <0x01>;  
    #size-cells = <0x00>;  
  
    port@0 {  
        reg = <0x00>;  
  
        endpoint {  
            vc-id = <0x00>;  
            port-index = <0x04>;  
            bus-width = <0x04>;  
            remote-endpoint = <0x03>;  
            phandle = <0x0e>;  
        };  
    };  
};
```

1

OToBrite Confidential

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	62 of 68	

```
bus@0 {  
  
    host1x@13e00000 {  
  
        nvcsi@15a00000 {  
            num-channels = <0x08>;  
            #address-cells = <0x01>;  
            #size-cells = <0x00>;  
            num-tpg-channels = <0x24>;  
            phandle = <0x34>;  
  
            channel@0 {  
                reg = <0x00>;  
                status = "okay";  
                phandle = <0x35>;  
  
                ports {  
                    #address-cells = <0x01>;  
                    #size-cells = <0x00>;  
  
                    port@0 {  
                        reg = <0x00>;  
                        status = "okay";  
                        phandle = <0x36>;  
  
                        endpoint@0 {  
                            port-index = <0x04>;  
                            bus-width = <0x04>;  
                            remote-endpoint = <0x0e>;  
                            status = "okay";  
                            phandle = <0x03>;  
                        };  
                    };  
                };  
            };  
        };  
    };  
};
```


文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	63 of 68	

```
tegra-capture-vi {
    num-channels = <0x08>;
    phandle = <0x2f>;

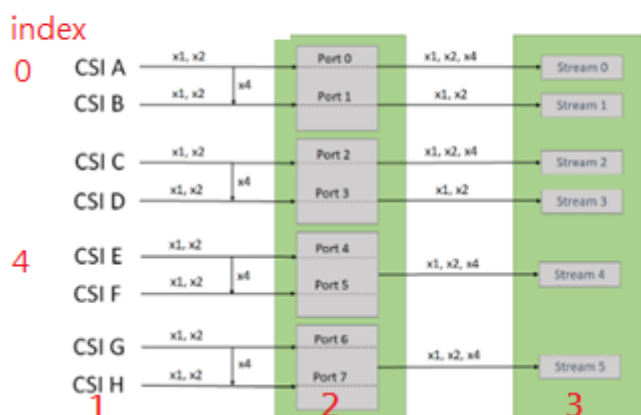
    ports {
        #address-cells = <0x01>;
        #size-cells = <0x00>;

        port@0 {
            reg = <0x00>;
            status = "okay";
            phandle = <0x30>;

            endpoint {
                vc-id = <0x00>;
                port-index = <0x04>;
                bus-width = <0x04>;
                remote-endpoint = <0x0a>;
                status = "okay";
                phandle = <0x0f>;
            };
        };
    };
};
```

The following graphic illustrates the port index mapping for different Jetson platforms.

> NVIDIA® Jetson AGX Orin™ series:



- The ports of Cam, NVCSI, and capture-vi describe together where Linux will receive the data. The Dser setting is 4 lines and both are output from port A. The circuit diagram shows that they are connected to the CSI (A/B) and CSI (E/F) of the motherboard respectively, so the port-index is 0

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	64 of 68	

文件名稱	oToCAM_nVidiaDriver_UserGuide-v2.14.docx
------	--

and 4 respectively.

oToBrite Confidential

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	65 of 68	

```
tegra-camera-platform {
    compatible = "nvidia, tegra-camera-platform";
    num_csi_lanes = <0x08>;
    max_lane_speed = <0x16E360>;
    min_bits_per_pixel = <0x0a>;
    vi_peak_byte_per_pixel = <0x02>;
    vi_bw_margin_pct = <0x19>;
    isp_peak_byte_per_pixel = <0x05>;
    isp_bw_margin_pct = <0x19>;
    tpg_max_iso = <0x3bc400>;
    phandle = <0x41>;

    modules {

        module0 {
            badge = "imx390_rear";
            position = "rear";
            orientation = "1";
            status = "okay";
            phandle = <0x42>;

            drivernode0 {
                pcl_id = "v4l2_sensor";
                sysfs-device-tree = "/sys/firmware/devicetree/base/bus@0/i2c@3180000/imx390@10";
                status = "okay";
                phandle = <0x43>;
            };

            drivernode1 {
                status = "disabled";
                pcl_id = "v4l2_lens";
                phandle = <0x44>;
            };
        };
    };
};
```

- When installing the driver, the main board will search for the camera data in the "sysfs-device-tree" to match.
- max_lane_speed will affect the output frequency of Dser.

p.s. 1. Driver rename (nv_imx390.ko -> oToCAM.ko, max9296.ko -> oToSerDes.ko)

O:\nvidia\test\device_tree\vecow\device-tree_ori.dts	O:\nvidia\test\device_tree\vecow\device-tree_fix.dts
241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291	241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291
};	};
i2c@1 {	i2c@1 {
reg = <0x01>;	reg = <0x01>;
i2c-mux,disable-on-exit;	i2c-mux,disable-on-exit;
i2c-mux-idle-disconnect;	i2c-mux-idle-disconnect;
#address-cells = <0x01>;	#address-cells = <0x01>;
#size-cells = <0x00>;	#size-cells = <0x00>;
max9296_c@40 {	max9296_c@40 {
compatible = "maxim,max9296";	compatible = "otobrite,deserializer";
reg = <0x48>;	reg = <0x48>;
csi-mode = "2x4";	csi-mode = "2x4";
max-src = <0x02>;	max-src = <0x02>;
reset-gpios = <0xffffffff 0xa0 0x00>;	reset-gpios = <0xffffffff 0xa0 0x00>;
fsync-gpio = <0x06>;	fsync-gpio = <0x06>;
phandle = <0x00>;	phandle = <0x06>;
};	};
max9295_prio@62 {	max9295_prio@62 {
compatible = "maxim,max9295";	compatible = "otobrite,serializer";
reg = <0x62>;	reg = <0x62>;
is-prim-ser;	is-prim-ser;
};	};
max9295_c@40 {	max9295_c@40 {
compatible = "maxim,max9295";	compatible = "otobrite,serializer";
reg = <0x40>;	reg = <0x40>;
nvidia,gmsl-dser-device = <0x06>;	nvidia,gmsl-dser-device = <0x06>;
phandle = <0x07>;	phandle = <0x07>;
};	};
max9295_d@60 {	max9295_d@60 {
compatible = "maxim,max9295";	compatible = "otobrite,serializer";
reg = <0x60>;	reg = <0x60>;
nvidia,gmsl-dser-device = <0x06>;	nvidia,gmsl-dser-device = <0x06>;
phandle = <0x09>;	phandle = <0x09>;
};	};
otocamisp_c@1b {	otocamisp_c@1b {
status = "okay";	status = "okay";
reg = <0x1b>;	reg = <0x1b>;
def-addr = <0x0d>;	def-addr = <0x0d>;
clocks = <0xffffffff 0x24 0xffffffff 0x24>;	clocks = <0xffffffff 0x24 0xffffffff 0x24>;
clock-names = "extperiph10p1p_grtba";	clock-names = "extperiph10p1p_grtba";
mc1k = "extperiph1";	mc1k = "extperiph1";
nvidia,gmsl-ser-device = <0x07>;	nvidia,gmsl-ser-device = <0x07>;
nvidia,gmsl-dser-device = <0x06>;	nvidia,gmsl-dser-device = <0x06>;
devnode = "video2";	devnode = "video2";
compatible = "nvidia,imx390";	compatible = "otobrite,otocam";
physical_w = "15.0";	physical_w = "15.0";

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁 數	66 of 68	

2. i2c mux: 2-channel i2c-mux(nxp,pca9543), 4-channel i2c-mux(nxp,pca9546)

```

fragment@10 {
    target-path = "/";
    __overlay__ {
        bus@0 {
            i2c@3100000 {
                tca9546@70 {
                    status = "okay";
                    compatible = "nxp,pca9546";
                    reg = <0x70>;
                    #address-cells = <0x01>;
                    #size-cells = <0x00>;
                    skip_mux_detect = "yes";

                    i2c@0 {
                        reg = <0x00>;
                        i2c-mux,deselect-on-exit;
                        #address-cells = <0x01>;
                        #size-cells = <0x00>;
                        status = "okay";

                        max9296_a@48 {
                            status = "okay";
                            compatible = "maxim,max9296";
                            reg = <0x48>;
                            csi-mode = "2x4";
                            max-src = <0x02>;
                            reset-gpios = <0xffffffff 0x3b 0x00>;
                            fsync-gpio = <0x00>;
                            phandle = <0x01>;
                        };
                    };
                };
            };
        };
    };
}

fragment@10 {
    target-path = "/";
    __overlay__ {
        bus@0 {
            i2c@3100000 {
                max9296_a@48 {
                    status = "okay";
                    compatible = "maxim,max9296";
                    reg = <0x48>;
                    csi-mode = "2x4";
                    max-src = <0x04>;
                    reset-gpios = <0xffffffff 0x3b 0x00>;
                    fsync-gpio = <0x00>;
                    phandle = <0x01>;
                };
                max9295_a@40 {
                    status = "okay";
                    compatible = "maxim,max9295";
                    reg = <0x40>;
                    mvidia,gmsl-dser-device = <0x01>;
                    phandle = <0x02>;
                };
                max9295_b@60 {
                    status = "okay";
                    compatible = "maxim,max9295";
                    reg = <0x60>;
                    mvidia,gmsl-dser-device = <0x01>;
                    phandle = <0x04>;
                };
            };
        };
    };
}

```

Ser i2c remap addr:

文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	67 of 68	

文件名稱	oToCAM_nVidiaDriver_UserGuide-v2.14.docx
------	--

```

max9295_a@40 {
    status = "okay";
    compatible = "maxim,max9295";
    reg = <0x18>;
    nvidia,gmsl-dser-device = <0x01>;
    phandle = <0x02>;
};

max9295_b@60 {
    status = "okay";
    compatible = "maxim,max9295";
    reg = <0x19>;
    nvidia,gmsl-dser-device = <0x01>;
    phandle = <0x04>;
};

max9295_c@40 {
    compatible = "maxim,max9295";
    reg = <0x1a>;
    nvidia,gmsl-dser-device = <0x01>;
    phandle = <0x06>;
};

max9295_d@60 {
    compatible = "maxim,max9295";
    reg = <0x1b>;
    nvidia,gmsl-dser-device = <0x01>;
    phandle = <0x08>;
};

max9295_e@40 {
    status = "okay";
    compatible = "maxim,max9295";
    reg = <0x1c>;
    nvidia,gmsl-dser-device = <0x16>;
    phandle = <0x17>;
};

max9295_f@60 {
    status = "okay";
    compatible = "maxim,max9295";
    reg = <0x1d>;
    nvidia,gmsl-dser-device = <0x16>;
    phandle = <0x19>;
};

max9295_g@40 {
    compatible = "maxim,max9295";
    reg = <0x1e>;
    nvidia,gmsl-dser-device = <0x16>;
    phandle = <0x1b>;
};

max9295_h@60 {
    compatible = "maxim,max9295";
    reg = <0x1f>;
    nvidia,gmsl-dser-device = <0x16>;
    phandle = <0x1d>;
};

```


文件編號		版本
生效日期	2026/3/20	V2.14
制定部門	System-SW	
頁數	68 of 68	

文件名稱	oToCAM_nVidiaDriver_UserGuide-v2.14.docx
------	--

oToBrite Confidential